

Bd. 22 - Endliche Wörter und Klammerungsbäume

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Kapitel 1

Einführung

Dieser Band stellt die kombinatorischen Werkzeuge bereit, mit denen später Produkte beliebiger endlicher Länge behandelt werden. Endliche Wörter werden als spezielle endliche Folgen konstruiert. Volle planare Binärbäume werden anschließend durch eindeutig zerlegbare Wörter codiert. Insbesondere werden weder eine Wortstruktur noch eine Baumstruktur axiomatisch vorausgesetzt. Induktion und Rekursion auf diesen Mengen sind Sätze, die auf der natürlichen Induktion und dem Dedekindschen Rekursionssatz beruhen.

Die Trennung ist für die folgenden Bände wesentlich: Erst Band 23 führt mit der Halbgruppe wieder eine axiomatisch definierte algebraische Struktur ein. Assoziativität wird dort genau einmal im Blockgesetz benutzt; die übrige Umklammerungsaussage folgt aus den hier konstruierten Werkzeugen.

Kapitel 2

Endliche Wörter

2.1 Wortmenge und Länge

Ein Wort über A ist der Funktionsgraph einer nullbasierten endlichen Folge mit Werten in A . Die folgende Definition ist durch $\mathcal{P}(\mathbb{N} \times A)$ gebunden und bezeichnet daher eine Menge.

Definition 22.2.1.1 (Menge der endlichen Wörter).

Mengen: A, w, n, A^*

Neues Symbol:

Endliche Wörter: $\triangleright A^*$

$$A^* := \{w \in \mathcal{P}(\mathbb{N} \times A) \mid \exists n \in \mathbb{N} (w: \mathbb{N}_{<n} \rightarrow A)\}$$

Theorem 22.2.1.1 (Typisierte endliche Folgen sind endliche Wörter).

Mengen: A, a, i, n, w, z

$$n \in \mathbb{N}, w: \mathbb{N}_{<n} \rightarrow A \vdash w \in A^*$$

$$1 \quad 1 \quad n \in \mathbb{N} \quad A$$

$$2 \quad 2 \quad w: \mathbb{N}_{<n} \rightarrow A \quad A$$

$$2 \quad 3 \quad w \subseteq \mathbb{N}_{<n} \times A \quad \text{Def. 5.2.1.1(2)}$$

$$1 \quad 4 \quad \mathbb{N}_{<n} \subseteq \mathbb{N} \quad \text{Th. 10.4.6.2(1)}$$

$$1,2 \quad 5 \quad w \subseteq \mathbb{N} \times A$$

$$\text{Th. 3.5.2.6, Th. 3.16.5.4, Th. 3.16.5.5(3, 4)}$$

$$1,2 \quad 6 \quad w \in \mathcal{P}(\mathbb{N} \times A) \quad \text{Th. 3.16.2.1(5)}$$

$$1,2 \quad 7 \quad w \in A^*$$

$$\text{Def. 22.2.1.1(6, 1, 2)}$$

Theorem 22.2.1.2 (Eindeutige Länge eines endlichen Wortes).

Mengen: A, w, m, n

$$w \in A^* \vdash \exists! n \in \mathbb{N} (w : \mathbb{N}_{<n} \rightarrow A)$$

1	1	$w \in A^*$	A
1	2	$\exists n \in \mathbb{N} (w : \mathbb{N}_{<n} \rightarrow A)$	<i>Def.</i> 22.2.1.1(1)
3	3	$m \in \mathbb{N} \wedge w : \mathbb{N}_{<m} \rightarrow A$	A
4	4	$n \in \mathbb{N} \wedge w : \mathbb{N}_{<n} \rightarrow A$	A
3,4	5	$\mathbb{N}_{<m} = \mathbb{N}_{<n}$	<i>Def.</i> 5.2.1.1(3,4)
3,4	6	$\mathbb{N}_{<m} \subseteq \mathbb{N}_{<n}$	<i>Th.</i> 3.5.3.3(5)
3,4	7	$\mathbb{N}_{<n} \subseteq \mathbb{N}_{<m}$	<i>Th.</i> 3.5.3.4(5)
3,4	8	$m \leq n$	<i>Th.</i> 10.4.6.22(3,4,6)
3,4	9	$n \leq m$	<i>Th.</i> 10.4.6.22(3,4,7)
3,4	10	$m = n$	<i>Th.</i> 10.4.5.25(3,4,8,9)
11	$\forall m \forall n$	$((m \in \mathbb{N} \wedge w : \mathbb{N}_{<m} \rightarrow A) \wedge (n \in \mathbb{N} \wedge w : \mathbb{N}_{<n} \rightarrow A) \rightarrow m = n)$	$\forall I(\forall I(\rightarrow I(\wedge I(3,4), 10)))$
12	$\exists^{\leq 1} n$	$(n \in \mathbb{N} \wedge w : \mathbb{N}_{<n} \rightarrow A)$	<i>Th.</i> 2.10.8.1(11)
1	13	$\exists! n \in \mathbb{N} (w : \mathbb{N}_{<n} \rightarrow A)$	<i>Th.</i> 2.10.9.4(2,12)

Definition 22.2.1.2 (Länge eines endlichen Wortes).

Mengen: A, w, n

Neues Symbol:

Wortlänge: $\triangleright |w|$

$$w \in A^* \vdash |w| := \iota n \in \mathbb{N} (w : \mathbb{N}_{<n} \rightarrow A)$$

Theorem 22.2.1.3 (Typisierung durch die Wortlänge).

Mengen: A, w

$$w \in A^* \vdash |w| \in \mathbb{N}, w : \mathbb{N}_{<|w|} \rightarrow A$$

1	1	$w \in A^*$	A
1	2	$\exists! n (n \in \mathbb{N} \wedge w : \mathbb{N}_{<n} \rightarrow A)$	<i>Th.</i> 22.2.1.2(1)
1	3	$ w \in \mathbb{N} \wedge w : \mathbb{N}_{< w } \rightarrow A$	<i>Th.</i> 2.10.9.7(2), <i>Def.</i> 22.2.1.2

$$\begin{array}{ll}
1 & 4 \quad |w| \in \mathbb{N} \quad \wedge E1(3) \\
1 & 5 \quad w: \mathbb{N}_{<|w|} \rightarrow A \quad \wedge E2(3)
\end{array}$$

Theorem 22.2.1.4 (Länge aus einer gegebenen Worttypisierung).

Mengen: A, w, n, k

$$w \in A^*, n \in \mathbb{N}, w: \mathbb{N}_{<n} \rightarrow A \vdash |w| = n$$

$$\begin{array}{ll}
1 & 1 \quad w \in A^* \quad A \\
2 & 2 \quad n \in \mathbb{N} \quad A \\
3 & 3 \quad w: \mathbb{N}_{<n} \rightarrow A \quad A \\
1 & 4 \quad |w| \in \mathbb{N} \quad Th. 22.2.1.3(1) \\
1 & 5 \quad w: \mathbb{N}_{<|w|} \rightarrow A \quad Th. 22.2.1.3(1) \\
1 & 6 \quad \exists! k \in \mathbb{N} (w: \mathbb{N}_{<k} \rightarrow A) \quad Th. 22.2.1.2(1) \\
1 & 7 \quad |w| \in \mathbb{N} \wedge w: \mathbb{N}_{<|w|} \rightarrow A \wedge I(4, 5) \\
2, 3 & 8 \quad n \in \mathbb{N} \wedge w: \mathbb{N}_{<n} \rightarrow A \quad \wedge I(2, 3) \\
1, 2, 3 & 9 \quad |w| = n \quad Th. 2.10.9.1(6, 7, 8)
\end{array}$$

2.2 Leerwort, Buchstabenwörter und Nichtleerwörter

Definition 22.2.2.1 (Leerwort).

Mengen: A, ε_A
Neues Symbol:
Leerwort: $\triangleright \varepsilon_A$

$$\varepsilon_A := \emptyset$$

Theorem 22.2.2.1 (Das Leerwort hat Länge null).

Mengen: A

$$\varepsilon_A \in A^*, |\varepsilon_A| = 0$$

$$\begin{array}{ll}
1 & 0 \in \mathbb{N} \quad Ax. 10.3.1 \\
2 & \mathbb{N}_{<0} = \emptyset \quad Th. 10.4.6.5 \\
3 & \varepsilon_A = \emptyset \quad Def. 22.2.2.1 \\
4 & \emptyset: \emptyset \rightarrow A \quad Th. 5.3.4.7 \\
5 & \varepsilon_A: \mathbb{N}_{<0} \rightarrow A = E(2, 3, 4)
\end{array}$$

$$\begin{array}{ll} 6 \ \varepsilon_A \in A^* & Th. \ 22.2.1.1(1,5) \\ 7 \ |\varepsilon_A| = 0 & Th. \ 22.2.1.4(6,1,5) \end{array}$$

Definition 22.2.2.2 (Menge der nichtleeren Wörter).

Mengen: A, A^*, A^+
Neues Symbol:
Nichtleere Wörter: $\triangleright A^+$

$$A^+ := A^* \setminus \{\varepsilon_A\}$$

Theorem 22.2.2.2 (Elementkriterium für nichtleere Wörter).

Mengen: A, w

$$w \in A^+ \dashv\vdash w \in A^*, |w| \neq 0$$

Beweis.

$\vdash$:

$$\begin{array}{ll} 1 \quad 1 \ w \in A^+ & A \\ 1 \quad 2 \ w \in A^* \wedge w \notin \{\varepsilon_A\} & \\ & Def. \ 22.2.2.2(1), Th. \ 3.12.1 \\ 1 \quad 3 \ w \in A^* & \wedge E1(2) \\ 1 \quad 4 \ w \notin \{\varepsilon_A\} & \wedge E2(2) \\ 1 \quad 5 \ w \neq \varepsilon_A & Th. \ 3.11.3.8(4) \\ 6 \quad 6 \ |w| = 0 & A \\ 1 \quad 7 \ w: \mathbb{N}_{<|w|} \rightarrow A & Th. \ 22.2.1.3(3) \\ 1,6 \quad 8 \ w: \emptyset \rightarrow A & Th. \ 10.4.6.5(6,7) \\ 1,6 \quad 9 \ w = \emptyset & Th. \ 5.3.4.6(8) \\ 1,6 \quad 10 \ w = \varepsilon_A & Def. \ 22.2.2.1(9) \\ 1,6 \quad 11 \ \perp & \perp I(5,10) \\ 1 \quad 12 \ |w| \neq 0 & \neg I(6,11) \end{array}$$

$\dashv\vdash$:

$$\begin{array}{ll} 1 \quad 1 \ w \in A^* & A \\ 2 \quad 2 \ |w| \neq 0 & A \\ 3 \quad 3 \ w = \varepsilon_A & A \\ 1,3 \quad 4 \ |w| = 0 & Th. \ 22.2.2.1(3) \\ 1,2,3 \quad 5 \ \perp & \perp I(2,4) \\ 1,2 \quad 6 \ w \neq \varepsilon_A & \neg I(3,5) \\ 1,2 \quad 7 \ w \notin \{\varepsilon_A\} & Th. \ 3.11.3.8(6) \end{array}$$

$$1,2 \quad w \in A^+$$

Def. 22.2.2.2, Th. 3.12.1(1, 7)

□

Theorem 22.2.2.3 (Typisierung nichtleerer Wörter).

Mengen: A, w

$$w \in A^+ \vdash w \in A^*, \quad |w| \in \mathbb{N}, \quad w: \mathbb{N}_{<|w|} \rightarrow A$$

$$\begin{array}{ll} 1 \quad 1 \quad w \in A^+ & A \\ 1 \quad 2 \quad w \in A^* & \text{Th. 22.2.2.2(1)} \\ 1 \quad 3 \quad |w| \in \mathbb{N} & \text{Th. 22.2.1.3(2)} \\ 1 \quad 4 \quad w: \mathbb{N}_{<|w|} \rightarrow A & \text{Th. 22.2.1.3(2)} \end{array}$$

Definition 22.2.2.3 (Buchstabenwort).

Mengen: $A, a, \langle a \rangle$

Neues Symbol:

Buchstabenwort: $\triangleright \langle a \rangle$

$$a \in A \vdash \langle a \rangle := \{(0, a)\}$$

Theorem 22.2.2.4 (Grundgesetze der Buchstabenwörter).

Mengen: A, a

$$a \in A \vdash \langle a \rangle \in A^+, \quad |\langle a \rangle| = 1, \quad \langle a \rangle(0) = a$$

$$\begin{array}{ll} 1 \quad 1 \quad a \in A & A \\ \quad 2 \quad 1 \in \mathbb{N} & \text{Th. 10.4.4.11} \\ 1 \quad 3 \quad \langle a \rangle \subseteq \mathbb{N}_{<1} \times A & \\ & \text{Def. 22.2.2.3(1), Th. 10.5.1.12, Th. 3.16.5.5} \\ 1 \quad 4 \quad \forall i \in \mathbb{N}_{<1} \exists! b \in A ((i, b) \in \langle a \rangle) & \\ & \text{Def. 22.2.2.3(1), Th. 10.5.1.12, Th. 3.11.3.8, Th. 3.11.4.1} \\ 1 \quad 5 \quad \langle a \rangle: \mathbb{N}_{<1} \rightarrow A & \text{Th. 5.2.6.1(3, 4)} \\ 1 \quad 6 \quad \langle a \rangle \in A^* & \text{Th. 22.2.1.1(2, 5)} \\ 1 \quad 7 \quad |\langle a \rangle| = 1 & \text{Th. 22.2.1.4(6, 2, 5)} \\ \quad 8 \quad 1 \neq 0 & \\ & \text{Def. 10.4.4.2, Ax. 10.3.5} \\ 1 \quad 9 \quad \langle a \rangle \in A^+ & \text{Th. 22.2.2.2(6, 7, 8)} \end{array}$$

10 $0 \in \mathbb{N}_{<1}$ *Th.* 10.5.1.12
 1 11 $(0, \langle a \rangle(0)) \in \langle a \rangle Th.$ 5.2.3.2(5, 10)
 1 12 $\langle a \rangle(0) = a$

Def. 22.2.2.3(11), *Th.* 3.11.3.8, *Th.* 3.11.4.1

2.3 Konkatenation

Definition 22.2.3.1 (Mitgliedschaft in verschobenen Bildmengen).

Mengen: J, j, r, x

$$x \in \{r + j \mid j \in J\} \dashv\vdash \exists j (j \in J \wedge x = r + j)$$

Definition 22.2.3.2 (Konkatenation endlicher Wörter).

Mengen: A, u, v, j

Neues Symbol:

Wortkonkatenation: $\triangleright u \frown v$

$$u, v \in A^* \vdash u \frown v := u \cup \{(|u| + j, v(j)) \mid j \in \mathbb{N}_{<|v|}\}$$

Theorem 22.2.3.1 (Blockzerlegung eines Anfangsabschnitts).

Mengen: j, r, s

$$r, s \in \mathbb{N} \vdash \mathbb{N}_{<r+s} = \mathbb{N}_{<r} \cup \{r + j \mid j \in \mathbb{N}_{<s}\}$$

Beweis.

Induktionsanfang

Th. 22.2.3.1(i)

$$r \in \mathbb{N} \vdash \mathbb{N}_{<r+0} = \mathbb{N}_{<r} \cup \{r + j \mid j \in \mathbb{N}_{<0}\}$$

1 1 $r \in \mathbb{N}$ A
 2 $\mathbb{N}_{<0} = \emptyset$ *Th.* 10.4.6.5
 3 $\forall x (x \in \{r + j \mid j \in \mathbb{N}_{<0}\} \leftrightarrow \exists j (j \in \mathbb{N}_{<0} \wedge x = r + j))$

Def. 22.2.3.1

4 $\forall x x \notin \{r + j \mid j \in \mathbb{N}_{<0}\}$ *Th.* 3.6.2.2(2, 3)

5 $\{r + j \mid j \in \mathbb{N}_{<0}\} = \emptyset$

Ax. 3.4.1.1(4, *Th.* 3.6.2.2)

1 6 $r + 0 = r$ *Th.* 10.4.4.2(1)

1 7 $\mathbb{N}_{<r+0} = \mathbb{N}_{<r} \cup \{r + j \mid j \in \mathbb{N}_{<0}\}$ *Th.* 3.13.3.23(5, 6)

Induktionsschritt

Th. 22.2.3.1(ii)

$$\begin{aligned} r, s \in \mathbb{N}, \mathbb{N}_{<r+s} &= \mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<s}\} \vdash \\ \mathbb{N}_{<r+\text{succ}(s)} &= \mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<\text{succ}(s)}\} \end{aligned}$$

$$\begin{array}{ll} 1 & 1 \ r \in \mathbb{N} \quad A \\ 2 & 2 \ s \in \mathbb{N} \quad A \\ 3 & 3 \ \mathbb{N}_{<r+s} = \mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<s}\} \quad A \\ 1,2 & 4 \ r+s \in \mathbb{N} \quad \text{Th. 10.4.4.1(1,2)} \\ 1,2 & 5 \ r+\text{succ}(s) = \text{succ}(r+s) \quad \text{Th. 10.4.4.3(1,2)} \\ 1,2 & 6 \ \mathbb{N}_{<\text{succ}(r+s)} = \mathbb{N}_{<r+s} \cup \{r+s\} \quad \text{Th. 10.4.6.8(4)} \\ 2 & 7 \ \mathbb{N}_{<\text{succ}(s)} = \mathbb{N}_{<s} \cup \{s\} \quad \text{Th. 10.4.6.8(2)} \\ 1,2 & 8 \ \forall x \left(x \in \{r+j \mid j \in \mathbb{N}_{<\text{succ}(s)}\} \leftrightarrow x \in \{r+j \mid j \in \mathbb{N}_{<s}\} \vee x = r+s \right) \\ & \quad \text{Def. 22.2.3.1, Th. 3.13.2.8(7), Th. 3.11.3.8} \\ 1,2 & 9 \ \forall x \left(x \in \{r+j \mid j \in \mathbb{N}_{<\text{succ}(s)}\} \leftrightarrow x \in \{r+j \mid j \in \mathbb{N}_{<s}\} \cup \{r+s\} \right) \\ & \quad \text{Th. 3.13.2.8(8), Th. 3.11.3.8} \\ 1,2 & 10 \ \{r+j \mid j \in \mathbb{N}_{<\text{succ}(s)}\} = \{r+j \mid j \in \mathbb{N}_{<s}\} \cup \{r+s\} \text{Ax. 3.4.1.1(9)} \\ 1,2 & 11 \ \mathbb{N}_{<r+\text{succ}(s)} = \mathbb{N}_{<\text{succ}(r+s)} \quad = E(5) \\ 1,2,3 & 12 \ \mathbb{N}_{<\text{succ}(r+s)} = (\mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<s}\}) \cup \{r+s\} \quad = E(3,6) \\ 1,2,3 & 13 \ (\mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<s}\}) \cup \{r+s\} = \mathbb{N}_{<r} \cup (\{r+j \mid j \in \mathbb{N}_{<s}\} \cup \{r+s\}) \\ & \quad \text{Th. 3.13.3.33} \\ 1,2 & 14 \ \mathbb{N}_{<r} \cup (\{r+j \mid j \in \mathbb{N}_{<s}\} \cup \{r+s\}) = \mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<\text{succ}(s)}\} \\ & \quad \text{Th. 2.9.1.1(10)} \\ 1,2,3 & 15 \ \mathbb{N}_{<r+\text{succ}(s)} = \mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<\text{succ}(s)}\} \quad \text{Tr. (11, 12, 13, 14)} \end{array}$$

Induktionsschluss:

$$\begin{array}{ll} 1 & 1 \ r \in \mathbb{N}A \\ 2 & 2 \ s \in \mathbb{N}A \\ 1,2 & 3 \ \mathbb{N}_{<r+s} = \mathbb{N}_{<r} \cup \{r+j \mid j \in \mathbb{N}_{<s}\} \\ & \quad \text{Ax. 10.3.9(IA, IS, 1, 2)} \end{array}$$

□

Theorem 22.2.3.2 (Typisierung und Koordinaten der Konkatenation).

Mengen: A, u, v, i, j

$$\begin{aligned} u, v \in A^* \vdash u \frown v &: \mathbb{N}_{<|u|+|v|} \rightarrow A, \\ u, v \in A^*, i \in \mathbb{N}_{<|u|} \vdash u \frown v(i) &= u(i), \\ u, v \in A^*, j \in \mathbb{N}_{<|v|} \vdash u \frown v(|u|+j) &= v(j) \end{aligned}$$

Beweis.

Typisierung

Th. 22.2.3.2(i)

$$u, v \in A^* \vdash u \smallfrown v : \mathbb{N}_{<|u|+|v|} \rightarrow A$$

1	1	$u \in A^*$	A
2	2	$v \in A^*$	A
1	3	$ u \in \mathbb{N}$	Th. 22.2.1.3(1)
2	4	$ v \in \mathbb{N}$	Th. 22.2.1.3(2)
1	5	$u : \mathbb{N}_{< u } \rightarrow A$	Th. 22.2.1.3(1)
2	6	$v : \mathbb{N}_{< v } \rightarrow A$	Th. 22.2.1.3(2)
1,2	7	$\mathbb{N}_{< u + v } = \mathbb{N}_{< u } \cup \{ u + j \mid j \in \mathbb{N}_{< v }\}$	Th. 22.2.3.1(3, 4)
1,2	8	$\forall j \in \mathbb{N}_{< v } (u + j \notin \mathbb{N}_{< u })$	Th. 10.4.6.2(3, 4), Th. 10.4.4.8, Th. 10.4.6.15
1,2	9	$\forall j, k \in \mathbb{N}_{< v } (u + j = u + k \rightarrow j = k)$	Th. 10.4.6.2(3, 4), Th. 10.4.5.29
1,2	10	$\forall p \left(p \in u \smallfrown v \leftrightarrow p \in u \vee \right.$ $\left. \exists j \in \mathbb{N}_{< v } p = (u + j, v(j)) \right)$	Def. 22.2.3.2
1,2	11	$u \smallfrown v \subseteq \mathbb{N}_{< u + v } \times A$	Def. 5.2.1.1 (5, 6) Th. 5.2.3.8 (6) Th. 3.16.5.5 (7, 10) Def. 3.5.1.1 (5, 6, 7, 10)
12	12	$q \in \mathbb{N}_{< u + v }$	A
1,2,12	13	$q \in \mathbb{N}_{< u } \vee \exists j \in \mathbb{N}_{< v } q = u + j$	Th. 22.2.3.1(3, 4, 7, 12)
1,2,12	14	$\exists a \in A ((q, a) \in u \smallfrown v)$	Th. 5.2.3.2(5, 6, 10, 13)
1,2,12	15	$\forall a, b \in A \left((q, a) \in u \smallfrown v \wedge (q, b) \in u \smallfrown v \right.$ $\left. \rightarrow a = b \right)$	Def. 5.2.1.1(5, 6, 8, 9, 10)
1,2,12	16	$\exists^{\leq 1} a (a \in A \wedge (q, a) \in u \smallfrown v)$	Th. 2.10.8.1(15)
1,2,12	17	$\exists! a \in A ((q, a) \in u \smallfrown v)$	Th. 2.10.9.4(14, 16)
1,2	18	$\forall q \in \mathbb{N}_{< u + v } \exists! a \in A ((q, a) \in u \smallfrown v)$	$\forall I(\rightarrow I(12, 17))$
1,2	19	$u \smallfrown v : \mathbb{N}_{< u + v } \rightarrow A$	Th. 5.2.6.1(11, 18)

Anfangsblock:

1	1	$u, v \in A^*$	A
2	2	$i \in \mathbb{N}_{< u }$	A
1	3	$u \smallfrown v : \mathbb{N}_{< u + v } \rightarrow A$	Th. 22.2.3.2(i)(1)
1	4	$u : \mathbb{N}_{< u } \rightarrow A$	Th. 22.2.1.3(1)
1,2	5	$(i, u(i)) \in u$	Th. 5.2.3.2(4, 2)

$$1,2 \quad 6 \quad (i, u(i)) \in u \smallfrown v$$

Def. 22.2.3.2(5), Th. 3.13.3.1

$$1,2 \quad 7 \quad u \smallfrown v(i) = u(i) \quad \text{Th. 5.2.3.10(3, 6)}$$

Verschobener Block:

$$\begin{array}{ll} 1 \quad 1 \quad u, v \in A^* & A \\ 2 \quad 2 \quad j \in \mathbb{N}_{<|v|} & A \\ 1 \quad 3 \quad u \smallfrown v: \mathbb{N}_{<|u|+|v|} \rightarrow A & \text{Th. 22.2.3.2(i)(1)} \\ 1 \quad 4 \quad v: \mathbb{N}_{<|v|} \rightarrow A & \text{Th. 22.2.1.3(1)} \\ 1,2 \quad 5 \quad (j, v(j)) \in v & \text{Th. 5.2.3.2(4, 2)} \\ 1,2 \quad 6 \quad (|u| + j, v(j)) \in \{(|u| + k, v(k)) \mid k \in \mathbb{N}_{<|v|}\} \exists I(2, 5) \\ 1,2 \quad 7 \quad (|u| + j, v(j)) \in u \smallfrown v \end{array}$$

Def. 22.2.3.2(6), Th. 3.13.3.2

$$1,2 \quad 8 \quad u \smallfrown v(|u| + j) = v(j) \quad \text{Th. 5.2.3.10(3, 7)}$$

□

Theorem 22.2.3.3 (Länge einer Konkatination).

Mengen: A, u, v

$$u, v \in A^* \vdash |u \smallfrown v| = |u| + |v|$$

$$\begin{array}{ll} 1 \quad 1 \quad u, v \in A^* & A \\ 1 \quad 2 \quad u \smallfrown v: \mathbb{N}_{<|u|+|v|} \rightarrow A & \text{Th. 22.2.3.2(1)} \\ 1 \quad 3 \quad |u| + |v| \in \mathbb{N} & \text{Th. 10.4.4.1(Th. 22.2.1.3(1), Th. 22.2.1.3(1))} \\ 1 \quad 4 \quad u \smallfrown v \in A^* & \text{Th. 22.2.1.1(3, 2)} \\ 1 \quad 5 \quad |u \smallfrown v| = |u| + |v| & \text{Th. 22.2.1.4(4, 3, 2)} \end{array}$$

Theorem 22.2.3.4 (Kürzung bei festem Konkatinationsschnitt).

Mengen: A, u, v, u', v', i, j

$$\begin{array}{l} u, v, u', v' \in A^*, \quad |u| = |u'|, \quad u \smallfrown v = u' \smallfrown v' \\ \vdash u = u', \quad v = v' \end{array}$$

$$\begin{array}{ll} 1 \quad 1 \quad u, v, u', v' \in A^* & A \\ 2 \quad 2 \quad |u| = |u'| & A \\ 3 \quad 3 \quad u \smallfrown v = u' \smallfrown v' & A \end{array}$$

1,3	4	$ u + v = u' + v' $		$= E(3, Th. 22.2.3.3(1))$
1,2,3	5	$ u + v = u + v' $	$= E(2, 4)$	
1	6	$ u , v , v' \in \mathbb{N}$	$Th. 22.2.1.3(1)$	
1,2,3	7	$ v = v' $	$Th. 10.4.5.29(6, 5)$	
1,2	8	$u, u': \mathbb{N}_{< u } \rightarrow A$		$= E(2, Th. 22.2.1.3(1))$
9	9	$i \in \mathbb{N}_{< u }$	A	
1,9	10	$u \frown v(i) = u(i)$	$Th. 22.2.3.2(1, 9)$	
1,2,9	11	$u' \frown v'(i) = u'(i)$		$= E(2, Th. 22.2.3.2(1, 9))$
1,2,3,9	12	$u(i) = u'(i)$	$= E(3, 10, 11)$	
1,2,3	13	$\forall i \in \mathbb{N}_{< u } u(i) = u'(i) \forall I(\rightarrow I(9, 12))$		
1,2,3	14	$u = u'$	$Th. 5.2.3.16(8, 13)$	
1,2,3	15	$v, v': \mathbb{N}_{< v } \rightarrow A$		$= E(7, Th. 22.2.1.3(1))$
16	16	$j \in \mathbb{N}_{< v }$	A	
1,16	17	$u \frown v(u + j) = v(j)$	$Th. 22.2.3.2(1, 16)$	
1,2,7,16	18	$u' \frown v'(u + j) = v'(j)$		$= E(2, 7, Th. 22.2.3.2(1, 16))$
1,2,3,7,16	19	$v(j) = v'(j)$	$= E(3, 17, 18)$	
1,2,3,7	20	$\forall j \in \mathbb{N}_{< v } v(j) = v'(j) \forall I(\rightarrow I(16, 19))$		
1,2,3	21	$v = v'$	$Th. 5.2.3.16(15, 20)$	

Theorem 22.2.3.5 (Leerwortgesetze der Konkatination).

Mengen: A, w

$$w \in A^* \vdash \varepsilon_A \frown w = w, w \frown \varepsilon_A = w$$

Beweis.

Linkes Leerwortgesetz:

1	1	$w \in A^*$	A
	2	$\varepsilon_A \in A^*$	$Th. 22.2.2.1$
	3	$ \varepsilon_A = 0$	$Th. 22.2.2.1$
1	4	$\varepsilon_A \frown w: \mathbb{N}_{< \varepsilon_A + w } \rightarrow A$	$Th. 22.2.3.2(2, 1)$
1	5	$\varepsilon_A \frown w: \mathbb{N}_{< w } \rightarrow A$	$Th. 10.4.4.4(3, 4)$
1	6	$w: \mathbb{N}_{< w } \rightarrow A$	$Th. 22.2.1.3(1)$
7	7	$i \in \mathbb{N}_{< w }$	A
1,7	8	$\varepsilon_A \frown w(\varepsilon_A + i) = w(i)$	$Th. 22.2.3.2(2, 1, 7)$
1,7	9	$\varepsilon_A \frown w(i) = w(i)$	$Th. 10.4.4.4(3, 8)$
1	10	$\forall i \in \mathbb{N}_{< w } \varepsilon_A \frown w(i) = w(i) \forall I(\rightarrow I(7, 9))$	

$$1 \quad 11 \quad \varepsilon_A \frown w = w$$

Th. 5.2.3.16(5, 6, 10)

Rechtes Leerwortgesetz:

$$\begin{array}{ll}
1 \quad 1 \quad w \in A^* & A \\
2 \quad \varepsilon_A \in A^* & \text{Th. 22.2.2.1} \\
3 \quad |\varepsilon_A| = 0 & \text{Th. 22.2.2.1} \\
1 \quad 4 \quad w \frown \varepsilon_A : \mathbb{N}_{<|w|+|\varepsilon_A|} \rightarrow A & \text{Th. 22.2.3.2(1, 2)} \\
1 \quad 5 \quad w \frown \varepsilon_A : \mathbb{N}_{<|w|} \rightarrow A & \text{Th. 10.4.4.2(3, 4)} \\
1 \quad 6 \quad w : \mathbb{N}_{<|w|} \rightarrow A & \text{Th. 22.2.1.3(1)} \\
7 \quad 7 \quad i \in \mathbb{N}_{<|w|} & A \\
1,7 \quad 8 \quad w \frown \varepsilon_A(i) = w(i) & \text{Th. 22.2.3.2(1, 2, 7)} \\
1 \quad 9 \quad \forall i \in \mathbb{N}_{<|w|} \quad w \frown \varepsilon_A(i) = w(i) \forall I(\rightarrow I(7, 8)) & \\
1 \quad 10 \quad w \frown \varepsilon_A = w & \text{Th. 5.2.3.16(5, 6, 9)}
\end{array}$$

□

Theorem 22.2.3.6 (Assoziativität der Wortkonkatenation).

Mengen: A, u, v, w

$$u, v, w \in A^* \vdash u \frown v \frown w = u \frown v \frown w$$

Beweis.

Gemeinsame Typisierung

Th. 22.2.3.6(i)

$$u, v, w \in A^* \vdash u \frown v \frown w, u \frown v \frown w : \mathbb{N}_{<(|u|+|v|)+|w|} \rightarrow A$$

$$\begin{array}{ll}
1 \quad 1 \quad u, v, w \in A^* & A \\
1 \quad 2 \quad u \frown v \in A^* & \\
& \text{Th. 22.2.1.1(Th. 10.4.4.1(Th. 22.2.1.3(1), Th. 22.2.1.3(1)), Th. 22.2.3.2(1))} \\
1 \quad 3 \quad v \frown w \in A^* & \\
& \text{Th. 22.2.1.1(Th. 10.4.4.1(Th. 22.2.1.3(1), Th. 22.2.1.3(1)), Th. 22.2.3.2(1))} \\
1 \quad 4 \quad |u \frown v| = |u| + |v| & \text{Th. 22.2.3.3(1)} \\
1 \quad 5 \quad |v \frown w| = |v| + |w| & \text{Th. 22.2.3.3(1)} \\
1 \quad 6 \quad u \frown v \frown w : \mathbb{N}_{<(|u|+|v|)+|w|} \rightarrow A & \text{Th. 22.2.3.2(2, 1, 4)} \\
1 \quad 7 \quad u \frown v \frown w : \mathbb{N}_{<|u|+(|v|+|w|)} \rightarrow A & \text{Th. 22.2.3.2(1, 3, 5)} \\
1 \quad 8 \quad (|u| + |v|) + |w| = |u| + (|v| + |w|) & \\
& \text{Th. 10.4.4.7(Th. 22.2.1.3(1), Th. 22.2.1.3(1), Th. 22.2.1.3(1))} \\
1 \quad 9 \quad u \frown v \frown w : \mathbb{N}_{<(|u|+|v|)+|w|} \rightarrow A = E(8, 7) &
\end{array}$$

Dreifachzerlegung

Th. 22.2.3.6(ii)

$$\begin{aligned} u, v, w \in A^*, \quad k \in \mathbb{N}_{<(|u|+|v|)+|w|} \vdash \\ k \in \mathbb{N}_{<|u|} \vee \\ (\exists j \in \mathbb{N}_{<|v|} k = |u| + j) \vee \\ (\exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h) \end{aligned}$$

$$\begin{aligned} 1 \quad 1 \quad u, v, w \in A^* & A \\ 2 \quad 2 \quad k \in \mathbb{N}_{<(|u|+|v|)+|w|} & A \\ 1,2 \quad 3 \quad k \in \mathbb{N}_{<|u|+|v|} \vee \exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h & \\ & Th. 22.2.3.1(Th. 22.2.1.3(1), Th. 10.4.4.1(Th. 22.2.1.3(1), Th. 22.2.1.3(1)), 2) \\ 4 \quad 4 \quad k \in \mathbb{N}_{<|u|+|v|} & A \\ 1,4 \quad 5 \quad k \in \mathbb{N}_{<|u|} \vee \exists j \in \mathbb{N}_{<|v|} k = |u| + j & \\ & Th. 22.2.3.1(Th. 22.2.1.3(1), 4) \\ 6 \quad 6 \quad k \in \mathbb{N}_{<|u|} & A \\ 6 \quad 7 \quad k \in \mathbb{N}_{<|u|} \vee (\exists j \in \mathbb{N}_{<|v|} k = |u| + j) \vee \vee I1(6) & \\ & (\exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h) \\ 8 \quad 8 \quad \exists j \in \mathbb{N}_{<|v|} k = |u| + j & A \\ 8 \quad 9 \quad k \in \mathbb{N}_{<|u|} \vee (\exists j \in \mathbb{N}_{<|v|} k = |u| + j) \vee \vee I2(\vee I1(8)) & \\ & (\exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h) \\ 1,4 \quad 10 \quad k \in \mathbb{N}_{<|u|} \vee (\exists j \in \mathbb{N}_{<|v|} k = |u| + j) \vee \vee E(5, 6, 7, 8, 9) & \\ & (\exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h) \\ 11 \quad 11 \quad \exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h & A \\ 11 \quad 12 \quad k \in \mathbb{N}_{<|u|} \vee (\exists j \in \mathbb{N}_{<|v|} k = |u| + j) \vee \vee I2(\vee I2(11)) & \\ & (\exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h) \\ 1,2 \quad 13 \quad k \in \mathbb{N}_{<|u|} \vee (\exists j \in \mathbb{N}_{<|v|} k = |u| + j) \vee \vee E(3, 4, 10, 11, 12) & \\ & (\exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h) \end{aligned}$$

Punktweise Gleichheit:

$$\begin{aligned} 1 \quad 1 \quad u, v, w \in A^* & A \\ 1 \quad 2 \quad u \frown v \frown w, u \frown v \frown w: \mathbb{N}_{<(|u|+|v|)+|w|} \rightarrow A & Th. 22.2.3.6(i)(1) \\ 1 \quad 3 \quad u \frown v, v \frown w \in A^* & \\ & Th. 22.2.1.1(Th. 10.4.4.1(Th. 22.2.1.3(1), Th. 22.2.1.3(1)), Th. 22.2.3.2(1)) \\ 1 \quad 4 \quad |u \frown v| = |u| + |v| \wedge |v \frown w| = |v| + |w| & Th. 22.2.3.3(1) \\ 5 \quad 5 \quad k \in \mathbb{N}_{<(|u|+|v|)+|w|} & A \\ 1,5 \quad 6 \quad k \in \mathbb{N}_{<|u|} \vee & \\ & (\exists j \in \mathbb{N}_{<|v|} k = |u| + j) \vee \\ & (\exists h \in \mathbb{N}_{<|w|} k = (|u| + |v|) + h) & \\ & Th. 22.2.3.6(ii)(1, 5) \\ 7 \quad 7 \quad k \in \mathbb{N}_{<|u|} & A \\ 1,7 \quad 8 \quad k \in \mathbb{N}_{<|u|+|v|} & \\ & Th. 22.2.3.1(Th. 22.2.1.3(1), 7), Th. 3.13.3.1 \\ 1,7 \quad 9 \quad u \frown v \frown w(k) = u \frown v(k) & \\ & Th. 22.2.3.2(3, 1, 4, 8) \end{aligned}$$

1,7 10	$u \frown v(k) = u(k)$	<i>Th.</i> 22.2.3.2(1, 7)
1,7 11	$u \frown v \frown w(k) = u(k)$	<i>Th.</i> 22.2.3.2(1, 3, 7)
1,7 12	$u \frown v \frown w(k) = u \frown v \frown w(k)$	<i>Th.</i> 2.9.1.3(9, 10, 11)
13 13	$\exists j \in \mathbb{N}_{< v } k = u + j$	<i>A</i>
14 14	$j \in \mathbb{N}_{< v } \wedge k = u + j$	<i>A</i>
1,14 15	$k \in \mathbb{N}_{< u + v }$	<i>Th.</i> 22.2.3.1(<i>Th.</i> 22.2.1.3(1), 14)
1,14 16	$u \frown v \frown w(k) = u \frown v(k)$	<i>Th.</i> 22.2.3.2(3, 1, 4, 15)
1,14 17	$u \frown v(k) = v(j)$	<i>Th.</i> 22.2.3.2(1, 14)
1,14 18	$j \in \mathbb{N}_{< v + w }$	<i>Th.</i> 22.2.3.1(<i>Th.</i> 22.2.1.3(1), 14), <i>Th.</i> 3.13.3.1
1,14 19	$v \frown w(j) = v(j)$	<i>Th.</i> 22.2.3.2(1, 14)
1,14 20	$u \frown v \frown w(k) = v \frown w(j)$	<i>Th.</i> 22.2.3.2(1, 3, 4, 14, 18)
1,14 21	$u \frown v \frown w(k) = u \frown v \frown w(k)$	<i>Tr.</i> (16, 17, <i>Th.</i> 2.9.1.1(19), <i>Th.</i> 2.9.1.1(20))
1,13 22	$u \frown v \frown w(k) = u \frown v \frown w(k)$	$\exists E(13, 14, 21)$
23 23	$\exists h \in \mathbb{N}_{< w } k = (u + v) + h$	<i>A</i>
24 24	$h \in \mathbb{N}_{< w } \wedge k = (u + v) + h$	<i>A</i>
1,24 25	$u \frown v \frown w(k) = w(h)$	<i>Th.</i> 22.2.3.2(3, 1, 4, 24)
1,24 26	$ v + h \in \mathbb{N}_{< v + w }$	<i>Th.</i> 22.2.3.1(<i>Th.</i> 22.2.1.3(1), 24)
1,24 27	$v \frown w(v + h) = w(h)$	<i>Th.</i> 22.2.3.2(1, 24)
1,24 28	$k = u + (v + h)$	<i>Th.</i> 10.4.4.7(<i>Th.</i> 22.2.1.3(1), 24)
1,24 29	$u \frown v \frown w(k) = v \frown w(v + h)$	<i>Th.</i> 22.2.3.2(1, 3, 4, 26, 28)
1,24 30	$u \frown v \frown w(k) = u \frown v \frown w(k)$	<i>Tr.</i> (25, <i>Th.</i> 2.9.1.1(27), <i>Th.</i> 2.9.1.1(29))
1,23 31	$u \frown v \frown w(k) = u \frown v \frown w(k)$	$\exists E(23, 24, 30)$
1,5 32	$u \frown v \frown w(k) = u \frown v \frown w(k)$	$\vee E(6, 7, 12, 13, 22, 23, 31)$
1 33	$\forall k \in \mathbb{N}_{<(u + v)+ w } u \frown v \frown w(k) = u \frown v \frown w(k) \forall I(\rightarrow I(5, 32))$	
1 34	$u \frown v \frown w = u \frown v \frown w$	<i>Th.</i> 5.2.3.16(2, 33)

□

Theorem 22.2.3.7 (Abgeschlossenheit nichtleerer Konkatenationen).

Mengen: A, u, v

$$u, v \in A^+ \vdash u \frown v \in A^+$$

1	1	$u, v \in A^+$	A
1	2	$u, v \in A^*$	$Th. 22.2.2.3(1)$
1	3	$ u , v \in \mathbb{N}$	$Th. 22.2.2.3(1)$
1	4	$ u \neq 0$	$Th. 22.2.2.2(1)$
1	5	$\exists k \in \mathbb{N} u = \text{succ}(k)$	$Ax. 10.3.6(3, 4)$
6	6	$k \in \mathbb{N} \wedge u = \text{succ}(k)$	A
1,6	7	$ u + v = \text{succ}(k + v)$	$Th. 10.4.4.6(3, 6)$
1,6	8	$k + v \in \mathbb{N}$	$Th. 10.4.4.1(3, 6)$
1,6	9	$ u + v \neq 0$	$Ax. 10.3.5(7, 8)$
1	10	$u \frown v \in A^*$	
			$Th. 22.2.1.1(Th. 10.4.4.1(3), Th. 22.2.3.2(2))$
1,6	11	$ u \frown v \neq 0$	$Th. 22.2.3.3(2, 9)$
1,6	12	$u \frown v \in A^+$	$Th. 22.2.2.2(10, 11)$
1	13	$u \frown v \in A^+$	$\exists E(5, 6, 12)$

2.4 Endzerlegung, Induktion und Rekursion

Theorem 22.2.4.1 (Eindeutige Rechtszerlegung eines Nichtleerwortes).

Mengen: A, w, u, a

$$w \in A^+ \vdash \exists! u \in A^* \exists! a \in A (w = u \frown \langle a \rangle)$$

Beweis.

Existenz Th. 22.2.4.1(i)

$$w \in A^+ \vdash \exists u \in A^* \exists a \in A w = u \frown \langle a \rangle$$

1	1	$w \in A^+$	A
1	2	$w \in A^*$	$Th. 22.2.2.3(1)$
1	3	$ w \in \mathbb{N}$	$Th. 22.2.2.3(1)$
1	4	$ w \neq 0$	$Th. 22.2.2.2(1)$
1	5	$\text{pred}(w) \in \mathbb{N}$	$Th. 10.4.2.16(3)$
1	6	$ w = \text{succ}(\text{pred}(w))$	$Th. 10.4.2.19(3, 4)$
1	7	$\mathbb{N}_{<\text{pred}(w)} \subseteq \mathbb{N}_{< w }$	$Th. 10.4.6.10(5, 6)$
1	8	$\text{pred}(w) \in \mathbb{N}_{< w }$	$Th. 10.4.6.18(5, 6)$
1	9	$w \upharpoonright_{\mathbb{N}_{<\text{pred}(w)}} : \mathbb{N}_{<\text{pred}(w)} \rightarrow A$	

		<i>Th.</i> 5.3.2.12(<i>Th.</i> 22.2.1.3(2), 7)
1	10	$w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \in A^*$
		<i>Th.</i> 22.2.1.1(5, 9)
1	11	$ w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} = \text{pred}(w)$
		<i>Th.</i> 22.2.1.4(10, 5, 9)
1	12	$w(\text{pred}(w)) \in A$
		<i>Th.</i> 5.2.3.8(<i>Th.</i> 22.2.1.3(2), 8)
1	13	$\langle w(\text{pred}(w)) \rangle \in A^* \wedge \langle w(\text{pred}(w)) \rangle = 1$
		<i>Th.</i> 22.2.2.4(12), <i>Th.</i> 22.2.2.3
1	14	$w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \neg \langle w(\text{pred}(w)) \rangle : \mathbb{N}_{< w } \rightarrow A$
		<i>Th.</i> 22.2.3.2(10, 11, 13), <i>Th.</i> 10.4.4.12(5, 6)
1	15	$w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \neg \langle w(\text{pred}(w)) \rangle (\text{pred}(w)) = w(\text{pred}(w))$
		<i>Th.</i> 22.2.3.2(10, 11, 13), <i>Th.</i> 22.2.2.4(12), <i>Th.</i> 10.5.1.12, <i>Th.</i> 10.4.4.2
16	16	$i \in \mathbb{N}_{< w }$
		<i>A</i>
1,16	17	$i \in \mathbb{N}_{<\text{pred}(w)} \vee i = \text{pred}(w)$
		<i>Th.</i> 10.4.6.9(5, 6, 16)
18	18	$i \in \mathbb{N}_{<\text{pred}(w)}$
		<i>A</i>
1,18	19	$w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \neg \langle w(\text{pred}(w)) \rangle (i) = w(i)$
		<i>Th.</i> 22.2.3.2(10, 13, 18), <i>Th.</i> 5.3.2.14(7, 18)
20	20	$i = \text{pred}(w)$
		<i>A</i>
1,20	21	$w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \neg \langle w(\text{pred}(w)) \rangle (i) = w(i) = E(20, 15)$
1,16	22	$w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \neg \langle w(\text{pred}(w)) \rangle (i) = w(i)$
		$\vee E(17, 18, 19, 20, 21)$
1	23	$\forall i \in \mathbb{N}_{< w } w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \neg \langle w(\text{pred}(w)) \rangle (i) = w(i)$
		$\forall I(\rightarrow I(16, 22))$
1	24	$w = w \Vdash_{\mathbb{N}_{<\text{pred}(w)}} \neg \langle w(\text{pred}(w)) \rangle$
		<i>Th.</i> 2.9.1.1(<i>Th.</i> 5.2.3.16(<i>Th.</i> 22.2.1.3(2), 14, 23))
1	25	$\exists u \in A^* \exists a \in A w = u \neg \langle a \rangle$
		$\exists I(\wedge I(10, \exists I(\wedge I(12, 24))))$

Eindeutigkeit

Th. 22.2.4.1(ii)

$$w \in A^+ \vdash \forall u \in A^* \forall v \in A^* \forall a \in A \forall b \in A \left((w = u \neg \langle a \rangle \wedge w = v \neg \langle b \rangle) \rightarrow (u = v \wedge a = b) \right)$$

1	1	$w \in A^+$	<i>A</i>
2	2	$u \in A^*$	<i>A</i>
3	3	$v \in A^*$	<i>A</i>
4	4	$a \in A$	<i>A</i>
5	5	$b \in A$	<i>A</i>
6	6	$w = u \neg \langle a \rangle \wedge w = v \neg \langle b \rangle$	<i>A</i>
6	7	$w = u \neg \langle a \rangle$	$\wedge E1(6)$
6	8	$w = v \neg \langle b \rangle$	$\wedge E2(6)$
4,5	9	$\langle a \rangle, \langle b \rangle \in A^* \wedge \langle a \rangle = \langle b \rangle = 1$	

$$\begin{array}{ll}
& Th. 22.2.2.4(4, 5), Th. 22.2.2.3 \\
2,3,4,5,6 \ 10 \ u \frown \langle a \rangle = v \frown \langle b \rangle & Th. 2.9.1.4(7, 8) \\
2,3,4,5,6 \ 11 \ \text{succ}(|u|) = \text{succ}(|v|) & \\
& Th. 22.2.3.3(2, 3, 9, 10), Th. 10.4.4.12 \\
2,3,4,5,6 \ 12 \ |u| = |v| & \\
& Ax. 10.3.7(Th. 22.2.1.3(2, 3), 11) \\
2,3,4,5,6 \ 13 \ u = v & \\
& Th. 22.2.3.4(2, 3, 9, 12, 10) \\
2,3,4,5,6 \ 14 \ \langle a \rangle = \langle b \rangle & \\
& Th. 22.2.3.4(2, 3, 9, 12, 10) \\
2,3,4,5,6 \ 15 \ \langle a \rangle(0) = \langle b \rangle(0) & = E(14) \\
2,3,4,5,6 \ 16 \ a = b & \\
& = E(Th. 22.2.2.4(4), Th. 22.2.2.4(5), 15) \\
1,2,3,4,5,6 \ 17 \ u = v \wedge a = b & \wedge I(13, 16) \\
1 \ 18 \ \forall u \in A^* \forall v \in A^* \forall a \in A \forall b \in A \Big(& \\
\quad (w = u \frown \langle a \rangle \wedge w = v \frown \langle b \rangle) \rightarrow (u = v \wedge a = b) \Big) & \\
\quad \forall I(\rightarrow I(2, \forall I(\rightarrow I(3, \forall I(\rightarrow I(4, \forall I(\rightarrow I(5, \rightarrow I(6, 17)))))))) &
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 \ 1 \ w \in A^+ & A \\
1 \ 2 \ \exists u \in A^* \exists a \in A \ w = u \frown \langle a \rangle & Th. 22.2.4.1(i)(1) \\
1 \ 3 \ \forall u \in A^* \forall v \in A^* \forall a \in A \forall b \in A \Big(& Th. 22.2.4.1(ii)(1) \\
\quad (w = u \frown \langle a \rangle \wedge w = v \frown \langle b \rangle) \rightarrow (u = v \wedge a = b) \Big) & \\
1 \ 4 \ \exists! u \in A^* \exists! a \in A \ w = u \frown \langle a \rangle & Th. 2.10.9.6(2, 3)
\end{array}$$

□

Theorem 22.2.4.2 (Induktion über nichtleere Wörter).

Mengen: A, a, u, w

Prädikate: P

$$\begin{array}{l}
\forall a \in A \ P(\langle a \rangle), \\
\forall u \in A^+ \forall a \in A \ (P(u) \rightarrow P(u \frown \langle a \rangle)) \\
\vdash \forall w \in A^+ \ P(w)
\end{array}$$

Beweis.

Induktionsanfang

Th. 22.2.4.2(i)

$$\forall w \in A^+ \ (|w| = \text{succ}(0) \rightarrow P(w))$$

$$1 \ 1 \ \forall a \in A \ P(\langle a \rangle) \quad A$$

2	2	$w \in A^+$	A	
3	3	$ w = \text{succ}(0)$	A	
2,3	4	$w: \mathbb{N}_{<1} \rightarrow A$		<i>Th.</i> 22.2.2.3(2,3), <i>Def.</i> 10.4.4.2
2,3	5	$w(0) \in A$		<i>Th.</i> 5.2.3.8(4, <i>Th.</i> 10.5.1.12)
2,3	6	$\langle w(0) \rangle: \mathbb{N}_{<1} \rightarrow A \wedge \langle w(0) \rangle(0) = w(0)$		<i>Th.</i> 22.2.2.4(5), <i>Th.</i> 22.2.2.3
2,3	7	$\forall i \in \mathbb{N}_{<1} w(i) = \langle w(0) \rangle(i)$	<i>Th.</i> 10.5.1.12(4,6)	
2,3	8	$w = \langle w(0) \rangle$	<i>Th.</i> 5.2.3.16(4,6,7)	
1,2,3	9	$P(w)$		$= E(8, \rightarrow E(\forall E(1), 5))$
1	10	$\forall w \in A^+ (w = \text{succ}(0) \rightarrow P(w))$		$\forall I(\rightarrow I(2, \rightarrow I(3, 9)))$

Induktionsschritt

Th. 22.2.4.2(ii)

$$n \in \mathbb{N}, \forall u \in A^+ (|u| = \text{succ}(n) \rightarrow P(u)) \vdash \\ \forall w \in A^+ (|w| = \text{succ}(\text{succ}(n)) \rightarrow P(w))$$

1	1	$n \in \mathbb{N}$	A	
2	2	$\forall u \in A^+ (u = \text{succ}(n) \rightarrow P(u))$	A	
3	3	$\forall u \in A^+ \forall a \in A (P(u) \rightarrow P(u \smallfrown \langle a \rangle))$	A	
4	4	$w \in A^+$	A	
5	5	$ w = \text{succ}(\text{succ}(n))$	A	
4	6	$\exists u \in A^* \exists a \in A w = u \smallfrown \langle a \rangle$	<i>Th.</i> 22.2.4.1(i)(4)	
7	7	$u \in A^* \wedge a \in A \wedge w = u \smallfrown \langle a \rangle$	A	
1,5,7	8	$ u = \text{succ}(n)$		<i>Th.</i> 22.2.3.3(7), <i>Th.</i> 22.2.2.4(7), <i>Th.</i> 10.4.4.12, <i>Ax.</i> 10.3.7(1,5)
1,5,7	9	$u \in A^+$		<i>Th.</i> 22.2.2.2(7,8), <i>Ax.</i> 10.3.5(1)
1,2,5,7	10	$P(u)$		$\rightarrow E(\forall E(2), \wedge I(9,8))$
1,2,3,5,7	11	$P(u \smallfrown \langle a \rangle)$		$\rightarrow E(\rightarrow E(\rightarrow E(\forall E(3), 9), \wedge E1(\wedge E2(7))), 10)$
1,2,3,5,7	12	$P(w)$	$= E(7, 11)$	
1,2,3,4,5	13	$P(w)$	$\exists E(6, 7, 12)$	
1,2,3	14	$\forall w \in A^+ (w = \text{succ}(\text{succ}(n)) \rightarrow P(w))$		$\forall I(\rightarrow I(4, \rightarrow I(5, 13)))$

Induktionsschluss:

1	1	$\forall a \in A P(\langle a \rangle)$	A
2	2	$\forall u \in A^+ \forall a \in A (P(u) \rightarrow P(u \smallfrown \langle a \rangle))$	A

3	3	$w \in A^+$	A
3	4	$ w \in \mathbb{N}$	$Th. 22.2.2.3(3)$
3	5	$ w \neq 0$	$Th. 22.2.2.2(3)$
3	6	$\exists n \in \mathbb{N} w = \text{succ}(n)$	$Ax. 10.3.6(4, 5)$
7	7	$n \in \mathbb{N} \wedge w = \text{succ}(n)$	A
1,2,7	8	$\forall u \in A^+ (u = \text{succ}(n) \rightarrow P(u))$	$Ax. 10.3.9(IA, IS, 7)$
1,2,3,7	9	$P(w)$	$\rightarrow E(\forall E(8), \wedge I(3, \wedge E2(7)))$
1,2,3	10	$P(w)$	$\exists E(6, 7, 9)$
1,2	11	$\forall w \in A^+ P(w)$	$\forall I(\rightarrow I(3, 10))$

□

Theorem 22.2.4.3 (Rekursion über nichtleere Wörter).

Mengen: A, X, a, u

Funktionen: f, s, F

$$\begin{aligned}
 &f: A \rightarrow X, \quad s: X \times A \rightarrow X \vdash \\
 &\exists! F: A^+ \rightarrow X \left(\forall a \in A F(\langle a \rangle) = f(a) \wedge \right. \\
 &\quad \left. \forall u \in A^+ \forall a \in A F(u \frown \langle a \rangle) = s(F(u), a) \right)
 \end{aligned}$$

Beweis.

Rekursionsdaten

Th. 22.2.4.3(i)

$$\begin{aligned}
 &f: A \rightarrow X, \quad s: X \times A \rightarrow X \vdash G_0 \in M, \\
 &\forall w \forall y \left((w, y) \in G_0 \leftrightarrow w \in W \wedge y \in X \wedge \right. \\
 &\quad \left. |w| = 1 \wedge y = f(w(0)) \right), \\
 &\forall G \in M \forall w \forall y \left((w, y) \in T(G) \leftrightarrow \right. \\
 &\quad \left. w \in W \wedge y \in X \wedge \exists u \in W \exists x \in X \exists a \in A \left(\right. \right. \\
 &\quad \left. \left. (u, x) \in G \wedge w = u \frown \langle a \rangle \wedge y = s(x, a) \right) \right), \\
 &\mathcal{E}: M \rightarrow M, \quad \forall G \in M \mathcal{E}(G) = T(G), \\
 &H: \mathbb{N} \rightarrow M, \quad H(0) = G_0, \\
 &\forall n \in \mathbb{N} H(\text{succ}(n)) = \mathcal{E}(H(n)), \\
 &\forall n \in \mathbb{N} \forall w (w \in W_n \leftrightarrow w \in W \wedge |w| = \text{succ}(n)), \\
 &D: \mathbb{N} \rightarrow \mathcal{P}(W), \quad \forall n \in \mathbb{N} D(n) = W_n
 \end{aligned}$$

$$1 \quad 1 \quad f: A \rightarrow X$$

$$A$$

2	$2\ s: X \times A \rightarrow X$	A
	$3\ W := A^+$	$= I$
	$4\ M := \mathcal{P}(W \times X)$	$= I$
	$5\ G_0 := \{p \in W \times X \mid \pi_1(p) = 1 \wedge \pi_2(p) = f(\pi_1(p)(0))\}$	$= I$
1	$6\ \forall w \forall y \left((w, y) \in G_0 \leftrightarrow w \in W \wedge y \in X \wedge w = 1 \wedge y = f(w(0)) \right)$	$Th. 3.7.2.5(5), Th. 3.16.5.5, Th. 7.3.1.3, Th. 7.3.1.7$
1	$7\ G_0 \subseteq W \times X$	$Def. 3.5.1.1(5)$
1	$8\ G_0 \in M$	$Th. 3.16.2.1(4, 7)$
	$9\ T(G) := \{p \in W \times X \mid \exists u \in W \exists x \in X \exists a \in A ((u, x) \in G \wedge \pi_1(p) = u \smallfrown \langle a \rangle \wedge \pi_2(p) = s(x, a))\}$	$= I$
10	$10\ G \in M$	A
10	$11\ G \subseteq W \times X$	$Th. 3.16.2.1(4, 10)$
2,10	$12\ \forall w \forall y \left((w, y) \in T(G) \leftrightarrow w \in W \wedge y \in X \wedge \exists u \in W \exists x \in X \exists a \in A ((u, x) \in G \wedge w = u \smallfrown \langle a \rangle \wedge y = s(x, a)) \right)$	$Th. 3.7.2.5(9), Th. 3.16.5.5, Th. 7.3.1.3, Th. 7.3.1.7$
2,10	$13\ T(G) \subseteq W \times X$	$Def. 3.5.1.1(9)$
2,10	$14\ T(G) \in M$	$Th. 3.16.2.1(4, 13)$
	$15\ \forall G \in M\ T(G) \in M$	$\forall I(\rightarrow I(10, 14))$
2	$16\ \forall G \in M\ \forall w \forall y \left((w, y) \in T(G) \leftrightarrow w \in W \wedge y \in X \wedge \exists u \in W \exists x \in X \exists a \in A ((u, x) \in G \wedge w = u \smallfrown \langle a \rangle \wedge y = s(x, a)) \right)$	$\forall I(\rightarrow I(10, 12))$
2	$17\ \exists! E: M \rightarrow M\ \forall G \in M\ E(G) = T(G)$	$Th. 5.2.6.9(15)$
	$18\ \mathcal{E} := \iota E: M \rightarrow M$	$= I$
	$\forall G \in M\ E(G) = T(G)$	
2	$19\ \mathcal{E}: M \rightarrow M\ \forall G \in M\ \mathcal{E}(G) = T(G)$	$Th. 2.10.9.7(17, 18)$
1,2	$20\ \exists! H: \mathbb{N} \rightarrow M \left(H(0) = G_0 \wedge \forall n \in \mathbb{N}\ H(\text{succ}(n)) = \mathcal{E}(H(n)) \right)$	$Th. 10.4.3.20(8, 19)$
	$21\ H := \iota K: \mathbb{N} \rightarrow M \left(K(0) = G_0 \wedge \forall n \in \mathbb{N}\ K(\text{succ}(n)) = \mathcal{E}(K(n)) \right)$	$= I$
1,2	$22\ H: \mathbb{N} \rightarrow M\ H(0) = G_0 \wedge \forall n \in \mathbb{N}\ H(\text{succ}(n)) = \mathcal{E}(H(n))$	$Th. 2.10.9.7(20, 21)$
	$23\ W_n := \{w \in W \mid w = \text{succ}(n)\}$	$= I$
	$24\ \forall n \in \mathbb{N}\ \forall w (w \in W_n \leftrightarrow w \in W \wedge w = \text{succ}(n))$	$Th. 3.7.2.5(23)$
	$25\ \forall n \in \mathbb{N}\ W_n \subseteq W$	$Def. 3.5.1.1(24)$
	$26\ \forall n \in \mathbb{N}\ W_n \in \mathcal{P}(W)$	$Th. 3.16.2.1(25)$
	$27\ \exists! D: \mathbb{N} \rightarrow \mathcal{P}(W)$	$Th. 5.2.6.9(26)$
	$\forall n \in \mathbb{N}\ D(n) = W_n$	
	$28\ D := \iota J: \mathbb{N} \rightarrow \mathcal{P}(W)$	$= I$
	$\forall n \in \mathbb{N}\ J(n) = W_n$	

$$29 \ D: \mathbb{N} \rightarrow \mathcal{P}(W) \wedge \\ \forall n \in \mathbb{N} D(n) = W_n$$

Th. 2.10.9.7(27, 28)

Stufengerechte Rechtszerlegung

Th. 22.2.4.3(ii)

$$f: A \rightarrow X, \ s: X \times A \rightarrow X, \ n \in \mathbb{N} \vdash \\ \forall w \in W_{\text{succ}(n)} \exists! u \in W_n \exists! a \in A \ w = u \smallfrown \langle a \rangle$$

1	1 $f: A \rightarrow X$	A
2	2 $s: X \times A \rightarrow X$	A
3	3 $n \in \mathbb{N}$	A
4	4 $w \in W_{\text{succ}(n)}$	A
1,2,3,4	5 $w \in W \wedge w = \text{succ}(\text{succ}(n))$	
		Th. 22.2.4.3(i)(1, 2, 3, 4)
1,2,3,4	6 $\exists u \in A^* \exists a \in A \ w = u \smallfrown \langle a \rangle$	
		Th. 22.2.4.1(i)($\wedge E1(5)$)
7	7 $u \in A^* \wedge a \in A \wedge w = u \smallfrown \langle a \rangle$	A
3,7	8 $ u = \text{succ}(n)$	
		Th. 22.2.3.3(7), Th. 22.2.2.4(7), Th. 10.4.4.12, Ax. 10.3.7(3, $\wedge E2(5)$)
3,7	9 $ u \neq 0$	Ax. 10.3.5(3, 8)
3,7	10 $u \in W$	Th. 22.2.2.2(7, 9)
1,2,3,7	11 $u \in W_n$	
		Th. 22.2.4.3(i)(1, 2, 3, 8, 10)
1,2,3,7	12 $\exists u \in W_n \exists a \in A \ w = u \smallfrown \langle a \rangle$	
		$\exists I(\wedge I(11, \exists I(\wedge I(\wedge E1(\wedge E2(7)), \wedge E2(\wedge E2(7))))))$
1,2,3,4	13 $\exists u \in W_n \exists a \in A \ w = u \smallfrown \langle a \rangle$	$\exists E(6, 7, 12)$
1,2,3,4	14 $\forall u, v \in W_n \forall a, b \in A \left(\right.$	
	$(w = u \smallfrown \langle a \rangle \wedge w = v \smallfrown \langle b \rangle) \rightarrow (u = v \wedge a = b)$	
		Th. 22.2.4.1(ii)($\wedge E1(5)$), Th. 22.2.4.3(i)(1, 2, 3)
1,2,3,4	15 $\exists! u \in W_n \exists! a \in A \ w = u \smallfrown \langle a \rangle$	Th. 2.10.9.6(13, 14)
1,2,3	16 $\forall w \in W_{\text{succ}(n)} \exists! u \in W_n \exists! a \in A \ w = u \smallfrown \langle a \rangle \forall I(\rightarrow I(4, 15))$	

Stufeninduktion: Anfang

Th. 22.2.4.3(iii)

$$f: A \rightarrow X, \ s: X \times A \rightarrow X \vdash H(0): W_0 \rightarrow X$$

1	1 $f: A \rightarrow X$	A
2	2 $s: X \times A \rightarrow X$	A
1,2	3 $H(0) = G_0$	
		Th. 22.2.4.3(i)(1, 2)
1,2	4 $G_0 \subseteq W_0 \times X$	
		Th. 22.2.4.3(i)(1, 2), Def. 10.4.4.2, Th. 5.2.3.8(1)
1,2	5 $\forall w \in W_0 \exists! x \in X ((w, x) \in G_0)$	

Th. 22.2.4.3(i)(1, 2), Th. 22.2.2.3, Th. 10.5.1.12(1)

1,2 6 $G_0: W_0 \rightarrow X$ *Th. 5.2.6.1(4, 5)*

1,2 7 $H(0): W_0 \rightarrow X = E(3, 6)$

Stufeninduktion: Schritt

Th. 22.2.4.3(iv)

$f: A \rightarrow X, s: X \times A \rightarrow X, n \in \mathbb{N}, H(n): W_n \rightarrow X \vdash$
 $H(\text{succ}(n)): W_{\text{succ}(n)} \rightarrow X$

1 1 $f: A \rightarrow X$ A

2 2 $s: X \times A \rightarrow X$ A

3 3 $n \in \mathbb{N}$ A

4 4 $H(n): W_n \rightarrow X$ A

1,2,3 5 $H(n) \in M$

Th. 5.2.3.8(Th. 22.2.4.3(i)(1, 2), 3)

1,2,3 6 $\mathcal{E}(H(n)) = T(H(n))$

Th. 22.2.4.3(i)(1, 2, 5)

1,2,3 7 $H(\text{succ}(n)) = \mathcal{E}(H(n))$

Th. 22.2.4.3(i)(1, 2, 3)

1,2,3,4 8 $T(H(n)) \subseteq W_{\text{succ}(n)} \times X$

Th. 22.2.4.3(i)(1, 2, 5), Th. 5.2.2.3(4), Th. 22.2.3.7, Th. 22.2.3.3(4), Th. 22.2.2.4, Th. 5.2.3.8(2)

1,2,3,4 9 $\forall w \in W_{\text{succ}(n)} \exists y \in X ((w, y) \in T(H(n)))$

Th. 22.2.4.3(ii)(1, 2, 3), Th. 5.2.3.8(4), Th. 5.2.3.2(4), Th. 5.2.3.8(2), Th. 22.2.4.3(i)(1, 2, 5)

10 10 $w \in W_{\text{succ}(n)}$ A

11 11 $y \in X$ A

12 12 $z \in X$ A

13 13 $(w, y) \in T(H(n)) \wedge (w, z) \in T(H(n))$ A

1,2,3,4,10,11,13 14 $\exists u \in W \exists x \in X \exists a \in A ((u, x) \in H(n) \wedge w = u \smallfrown \langle a \rangle \wedge y = s(x, a))$

Th. 22.2.4.3(i)(1, 2, 5, $\wedge E1(13)$)

1,2,3,4,10,12,13 15 $\exists v \in W \exists t \in X \exists b \in A ((v, t) \in H(n) \wedge w = v \smallfrown \langle b \rangle \wedge z = s(t, b))$

Th. 22.2.4.3(i)(1, 2, 5, $\wedge E2(13)$)

16 16 $u \in W \wedge x \in X \wedge a \in A \wedge (u, x) \in H(n) \wedge$ A

$w = u \smallfrown \langle a \rangle \wedge y = s(x, a)$

17 17 $v \in W \wedge t \in X \wedge b \in A \wedge (v, t) \in H(n) \wedge$ A

$w = v \smallfrown \langle b \rangle \wedge z = s(t, b)$

4,16,17 18 $u \in W_n \wedge v \in W_n$

Th. 5.2.2.3(4, 16, 17)

1,2,3,10,16,17 19 $u = v \wedge a = b$

Th. 22.2.4.3(i)(1, 2, 3, 10), Th. 22.2.4.1(ii)(16, 17)

4,16,17 20 $x = H(n)(u) \wedge t = H(n)(v)$

Th. 5.2.3.9(4, 16, 17)

4,16,17,19,20 21 $x = t$

Th. 2.9.1.7(19, 20)

4,16,17,19,20 22 $s(x, a) = s(t, b)$

Th. 2.9.1.7(19, 21)

16 23 $y = s(x, a)$

		$\wedge E2(\wedge E2(\wedge E2(\wedge E2(16))))$
17	24	$z = s(t, b)$
		$\wedge E2(\wedge E2(\wedge E2(\wedge E2(17))))$
17	25	$s(t, b) = z$
		<i>Th.</i> 2.9.1.1(24)
4,16,17,19,20	26	$y = z$
		<i>Tr.</i> (23, 22, 25)
1,2,3,4,10,11,12,13,16	27	$y = z$
		$\exists E(15, 17, 26)$
1,2,3,4,10,11,12,13	28	$y = z$
		$\exists E(14, 16, 27)$
1,2,3,4,10,11,12	29	$((w, y) \in T(H(n)) \wedge (w, z) \in T(H(n))) \rightarrow y = z \rightarrow I(13, 28)$
1,2,3,4,10	30	$\forall y, z \in X \left(((w, y) \in T(H(n)) \wedge (w, z) \in T(H(n))) \rightarrow y = z \right)$
		$\forall I(\rightarrow I(11, \forall I(\rightarrow I(12, 29))))$
1,2,3,4,10	31	$\exists^{\leq 1} y (y \in X \wedge (w, y) \in T(H(n)))$
		<i>Th.</i> 2.10.8.1(30)
1,2,3,4	32	$\forall w \in W_{\text{succ}(n)} \exists^{\leq 1} y (y \in X \wedge (w, y) \in T(H(n))) \forall I(\rightarrow I(10, 31))$
1,2,3,4	33	$\forall w \in W_{\text{succ}(n)} \exists! y \in X ((w, y) \in T(H(n)))$
		<i>Th.</i> 2.10.9.4(9, 32)
1,2,3,4	34	$T(H(n)): W_{\text{succ}(n)} \rightarrow X$
		<i>Th.</i> 5.2.6.1(8, 33)
1,2,3	35	$H(\text{succ}(n)) = T(H(n))$
		<i>Tr.</i> (7, 6)
1,2,3,4	36	$H(\text{succ}(n)): W_{\text{succ}(n)} \rightarrow X$
		$= E(35, 34)$

Stufeninduktion: Schluss

Th. 22.2.4.3(v)

$$f: A \rightarrow X, \quad s: X \times A \rightarrow X \vdash \forall n \in \mathbb{N} H(n): W_n \rightarrow X$$

1	1	$f: A \rightarrow X$	A
2	2	$s: X \times A \rightarrow X$	A
3	3	$n \in \mathbb{N}$	A
1,2,3	4	$H(n): W_n \rightarrow X$	
			<i>Ax.</i> 10.3.9(<i>Th.</i> 22.2.4.3(iii)(1, 2), <i>Th.</i> 22.2.4.3(iv)(1, 2), 3)
1,2	5	$\forall n \in \mathbb{N} H(n): W_n \rightarrow X \forall I(\rightarrow I(3, 4))$	

Überdeckung durch die Längenstufen

Th. 22.2.4.3(vi)

$$f: A \rightarrow X, \quad s: X \times A \rightarrow X \vdash \forall w \in W \exists! n \in \mathbb{N} (w \in D(n)),$$

$$\bigcup D[\mathbb{N}] = W$$

1	1	$f: A \rightarrow X$	A
2	2	$s: X \times A \rightarrow X$	A
3	3	$w \in W$	A
3	4	$ w \in \mathbb{N} \wedge w \neq 0$	
			<i>Th.</i> 22.2.2.3(3), <i>Th.</i> 22.2.2.2(3)
3	5	$\exists n \in \mathbb{N} w = \text{succ}(n)$	<i>Ax.</i> 10.3.6(4)
6	6	$n \in \mathbb{N} \wedge w = \text{succ}(n)$	A
1,2,3,6	7	$w \in W_n$	
			<i>Th.</i> 22.2.4.3(i)(1, 2, 3, 6)

1,2,3,6	8	$w \in D(n)$			<i>Th.</i> 22.2.4.3(i)(1, 2, 6, 7)
1,2,3,6	9	$\exists n \in \mathbb{N} (w \in D(n))$	$\exists I(\wedge I(\wedge E1(6), 8))$		
1,2,3	10	$\exists n \in \mathbb{N} (w \in D(n))$	$\exists E(5, 6, 9)$		
1,2,3	11	$\forall m, n \in \mathbb{N} ((w \in D(m) \wedge w \in D(n)) \rightarrow m = n)$			<i>Th.</i> 22.2.4.3(i)(1, 2), <i>Ax.</i> 10.3.7
1,2,3	12	$\exists^{\leq 1} n (n \in \mathbb{N} \wedge w \in D(n))$	<i>Th.</i> 2.10.8.1(11)		
1,2,3	13	$\exists! n \in \mathbb{N} (w \in D(n))$	<i>Th.</i> 2.10.9.4(10, 12)		
1,2	14	$\forall w \in W \exists! n \in \mathbb{N} (w \in D(n)) \forall I(\rightarrow I(3, 13))$			
15	15	$x \in \bigcup D[\mathbb{N}]$	<i>A</i>		
15	16	$\exists E \in D[\mathbb{N}] (x \in E)$	<i>Th.</i> 3.13.2.1(15)		
17	17	$E \in D[\mathbb{N}] \wedge x \in E$	<i>A</i>		
1,2,17	18	$\exists n \in \mathbb{N} E = D(n)$			<i>Th.</i> 5.2.4.7(<i>Th.</i> 22.2.4.3(i)(1, 2), $\wedge E1(17)$)
	19	$n \in \mathbb{N} \wedge E = D(n)$	<i>A</i>		
17,19	20	$x \in D(n)$			$= E(\wedge E2(19), \wedge E2(17))$
1,2,19	21	$D(n) = W_n$			<i>Th.</i> 22.2.4.3(i)(1, 2, $\wedge E1(19)$)
1,2,17,19	22	$x \in W_n$	$= E(21, 20)$		
1,2,17,19	23	$x \in W$			<i>Th.</i> 22.2.4.3(i)(1, 2, $\wedge E1(19)$, 22)
1,2,17	24	$x \in W$	$\exists E(18, 19, 23)$		
1,2,15	25	$x \in W$	$\exists E(16, 17, 24)$		
1,2	26	$x \in \bigcup D[\mathbb{N}] \rightarrow x \in W$	$\rightarrow I(15, 25)$		
1,2	27	$\forall x (x \in \bigcup D[\mathbb{N}] \rightarrow x \in W)$	$\forall I(26)$		
1,2	28	$\bigcup D[\mathbb{N}] \subseteq W$	<i>Def.</i> 3.5.1.1(27)		
29	29	$x \in W$	<i>A</i>		
1,2,29	30	$\exists n \in \mathbb{N} (x \in D(n))$			<i>Th.</i> 2.10.9.2($\rightarrow E(\forall E(14), 29)$)
	31	$n \in \mathbb{N} \wedge x \in D(n)$	<i>A</i>		
1,2,31	32	$D(n) \in D[\mathbb{N}]$			<i>Th.</i> 5.2.4.5(<i>Th.</i> 3.5.3.1, <i>Th.</i> 22.2.4.3(i)(1, 2), $\wedge E1(31)$)
1,2,31	33	$D(n) \subseteq \bigcup D[\mathbb{N}]$	<i>Th.</i> 3.13.2.4(32)		
1,2,31	34	$x \in \bigcup D[\mathbb{N}]$			<i>Th.</i> 3.5.2.6(33, $\wedge E2(31)$)
1,2,29	35	$x \in \bigcup D[\mathbb{N}]$	$\exists E(30, 31, 34)$		
1,2	36	$x \in W \rightarrow x \in \bigcup D[\mathbb{N}]$	$\rightarrow I(29, 35)$		
1,2	37	$\forall x (x \in W \rightarrow x \in \bigcup D[\mathbb{N}])$	$\forall I(36)$		
1,2	38	$W \subseteq \bigcup D[\mathbb{N}]$	<i>Def.</i> 3.5.1.1(37)		
1,2	39	$\bigcup D[\mathbb{N}] = W$	<i>Th.</i> 3.5.3.5(28, 38)		

Vereinigung der Stufen

Th. 22.2.4.3(vii)

$$f: A \rightarrow X, \quad s: X \times A \rightarrow X \vdash$$

$$\exists F: W \rightarrow X \left(\begin{array}{l} \forall a \in A \, F(\langle a \rangle) = f(a) \wedge \\ \forall u \in W \, \forall a \in A \, F(u \smallfrown \langle a \rangle) = s(F(u), a) \end{array} \right)$$

$$\begin{array}{ll}
1 & 1 \, f: A \rightarrow X \quad A \\
2 & 2 \, s: X \times A \rightarrow X \quad A \\
1,2 & 3 \, F := \bigcup H[\mathbb{N}] \quad = I \\
1,2 & 4 \, H: \mathbb{N} \rightarrow \mathcal{P}(W \times X) \wedge D: \mathbb{N} \rightarrow \mathcal{P}(W) \\
& \text{Th. 22.2.4.3}(i)(1, 2) \\
1,2 & 5 \, \forall n \in \mathbb{N} \, H(n): W_n \rightarrow X \\
& \text{Th. 22.2.4.3}(v)(1, 2) \\
1,2 & 6 \, \forall n \in \mathbb{N} \, H(n): D(n) \rightarrow X \\
& \text{Th. 22.2.4.3}(i)(1, 2, 5) \\
7 & 7 \, m \in \mathbb{N} \quad A \\
8 & 8 \, n \in \mathbb{N} \quad A \\
9 & 9 \, x \in D(m) \cap D(n) \quad A \\
1,2,7,8,9 & 10 \, x \in W_m \wedge x \in W_n \\
& \text{Th. 22.2.4.3}(i)(1, 2, 7, 8, 9) \\
1,2,7,8,9 & 11 \, \text{succ}(m) = \text{succ}(n) \\
& \text{Th. 2.9.1.4}(\wedge E1(\text{Th. 22.2.4.3}(i)(1, 2, 7, 8, 10)), \wedge E2(\text{Th. 22.2.4.3}(i)(1, 2, 7, 8, 10))) \\
1,2,7,8,9 & 12 \, m = n \quad Ax. 10.3.7(7, 8, 11) \\
1,2,7,8,9 & 13 \, H(m)(x) = H(n)(x) \quad = E(12) \\
1,2 & 14 \, \forall m, n \in \mathbb{N} \, \forall x \in D(m) \cap D(n) \, H(m)(x) = H(n)(x) \\
& \forall I(\rightarrow I(7, \forall I(\rightarrow I(8, \forall I(\rightarrow I(9, 13)))))) \\
1,2 & 15 \, F: \bigcup D[\mathbb{N}] \rightarrow X \wedge \forall n \in \mathbb{N} \, \forall w \in D(n) \, F(w) = H(n)(w) \\
& \text{Th. 5.3.2.22}(\wedge E1(4), \wedge E2(4), 6, 14) \\
1,2 & 16 \, \forall w \in W \, \exists! n \in \mathbb{N} \, (w \in D(n)) \\
& \text{Th. 22.2.4.3}(vi)(1, 2) \\
1,2 & 17 \, \bigcup D[\mathbb{N}] \subseteq W \\
& = E(\text{Th. 22.2.4.3}(vi)(1, 2), \text{Th. 3.5.3.1}) \\
1,2 & 18 \, W \subseteq \bigcup D[\mathbb{N}] \\
& = E(\text{Th. 2.9.1.1}(\text{Th. 22.2.4.3}(vi)(1, 2)), \text{Th. 3.5.3.1}) \\
1,2 & 19 \, \bigcup D[\mathbb{N}] = W \\
& \text{Th. 22.2.4.3}(vi)(1, 2) \\
1,2 & 20 \, F: W \rightarrow X \quad = E(19, \wedge E1(15)) \\
1,2 & 21 \, \forall n \in \mathbb{N} \, \forall w \in W_n \, F(w) = H(n)(w) \\
& \text{Th. 22.2.4.3}(i)(1, 2, \wedge E2(15)) \\
22 & 22 \, a \in A \quad A \\
1,2,22 & 23 \, \langle a \rangle \in W_0 \\
& \text{Th. 22.2.2.4}(22), \text{Th. 22.2.4.3}(i)(1, 2), \text{Def. 10.4.4.2} \\
1,2,22 & 24 \, F(\langle a \rangle) = H(0)(\langle a \rangle)
\end{array}$$

	$\rightarrow E(\forall E(\rightarrow E(\forall E(21), Ax. 10.3.1)), 23)$
1,2,22 25 $H(0) = G_0$	
	<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2)
1,2,22 26 $(\langle a \rangle, f(a)) \in G_0$	
	<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2), <i>Th.</i> 22.2.2.4(22), <i>Th.</i> 5.2.3.8(1, 22)
1,2,22 27 $(\langle a \rangle, f(a)) \in H(0)$	$= E(25, 26)$
1,2,22 28 $H(0)(\langle a \rangle) = f(a)$	
	<i>Th.</i> 5.2.3.10(<i>Th.</i> 22.2.4.3(<i>iii</i>)(1, 2), 27)
1,2,22 29 $F(\langle a \rangle) = f(a)$	<i>Tr.</i> (24, 28)
1,2 30 $\forall a \in A F(\langle a \rangle) = f(a)$	$\forall I(\rightarrow I(22, 29))$
31 31 $u \in W$	<i>A</i>
32 32 $a \in A$	<i>A</i>
1,2,31 33 $\exists n \in \mathbb{N} (u \in D(n))$	
	<i>Th.</i> 2.10.9.2($\rightarrow E(\forall E(16), 31)$)
34 34 $n \in \mathbb{N} \wedge u \in D(n)$	<i>A</i>
1,2,34 35 $u \in W_n$	
	<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2, 34)
1,2,32,34 36 $u \frown \langle a \rangle \in W_{\text{succ}(n)}$	
	<i>Th.</i> 22.2.3.7 (35, 32)
	<i>Th.</i> 22.2.3.3, <i>Th.</i> 22.2.2.4 (35, 32)
	<i>Th.</i> 10.4.4.12, <i>Th.</i> 22.2.4.3(<i>i</i>) (1, 2, 34, 35)
1,2,34 37 $F(u) = H(n)(u)$	
	$\rightarrow E(\forall E(\rightarrow E(\forall E(21), \wedge E1(34))), 35)$
1,2,34 38 $H(n): W_n \rightarrow X$	
	$\rightarrow E(\forall E(5), \wedge E1(34))$
1,2,34 39 $(u, H(n)(u)) \in H(n)$	<i>Th.</i> 5.2.3.2(38, 35)
1,2,34 40 $H(n) \in M$	
	<i>Th.</i> 5.2.3.8(<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2), $\wedge E1(34)$)
1,2,34 41 $\mathcal{E}(H(n)) = T(H(n))$	
	<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2, 40)
1,2,34 42 $H(\text{succ}(n)) = \mathcal{E}(H(n))$	
	<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2, $\wedge E1(34)$)
1,2,34 43 $H(\text{succ}(n)) = T(H(n))$	<i>Tr.</i> (42, 41)
1,2,34 44 $H(n)(u) \in X$	<i>Th.</i> 5.2.3.8(38, 35)
1,2,32,34 45 $(H(n)(u), a) \in X \times A$	<i>Th.</i> 3.16.5.5(44, 32)
1,2,32,34 46 $s(H(n)(u), a) \in X$	<i>Th.</i> 5.2.3.8(2, 45)
1,2,32,34 47 $(u \frown \langle a \rangle, s(H(n)(u), a)) \in W \times X$	
	<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2, 34, 36), <i>Th.</i> 3.16.5.5(36, 46)
1,2,31,32,34 48 $\exists v \in W \exists x \in X \exists b \in A \left((v, x) \in H(n) \wedge \right.$	
$\quad u \frown \langle a \rangle = v \frown \langle b \rangle \wedge$	
$\quad \left. s(H(n)(u), a) = s(x, b) \right)$	
	$\exists I(\wedge I(31, \exists I(\wedge I(44, \exists I(\wedge I(32, \wedge I(39, \wedge I(= I, = I))))))$
1,2,31,32,34 49 $(u \frown \langle a \rangle, s(H(n)(u), a)) \in T(H(n))$	
	<i>Th.</i> 22.2.4.3(<i>i</i>)(1, 2, 40, 47, 48)

$$\begin{array}{ll}
1,2,31,32,34 & 50 \ (u \frown \langle a \rangle, s(H(n)(u), a)) \in H(\text{succ}(n)) = E(43, 49) \\
1,2,34 & 51 \ H(\text{succ}(n)) : W_{\text{succ}(n)} \rightarrow X \\
& \rightarrow E(\forall E(5), Ax. \ 10.3.4(\wedge E1(34))) \\
1,2,31,32,34 & 52 \ H(\text{succ}(n))(u \frown \langle a \rangle) = s(H(n)(u), a) \ \text{Th. } 5.2.3.10(51, 50) \\
1,2,32,34 & 53 \ F(u \frown \langle a \rangle) = H(\text{succ}(n))(u \frown \langle a \rangle) \\
& \rightarrow E(\forall E(\rightarrow E(\forall E(21), Ax. \ 10.3.4(\wedge E1(34))))), 36) \\
1,2,34 & 54 \ H(n)(u) = F(u) \quad \text{Th. } 2.9.1.1(37) \\
1,2,32,34 & 55 \ s(H(n)(u), a) = s(F(u), a) \quad \text{Th. } 2.9.1.7(54) \\
1,2,31,32,34 & 56 \ F(u \frown \langle a \rangle) = s(F(u), a) \quad \text{Tr.}(53, 52, 55) \\
1,2,31,32 & 57 \ F(u \frown \langle a \rangle) = s(F(u), a) \quad \exists E(33, 34, 56) \\
1,2 & 58 \ \forall u \in W \forall a \in A \ F(u \frown \langle a \rangle) = s(F(u), a) \\
& \forall I(\rightarrow I(31, \forall I(\rightarrow I(32, 57)))) \\
1,2 & 59 \ \exists F : W \rightarrow X \left(\forall a \in A \ F(\langle a \rangle) = f(a) \wedge \right. \\
& \quad \left. \forall u \in W \forall a \in A \ F(u \frown \langle a \rangle) = s(F(u), a) \right) \\
& \exists I(\wedge I(20, \wedge I(30, 58)))
\end{array}$$

Eindeutigkeit

Th. 22.2.4.3(viii)

$$\begin{array}{l}
F, F' : W \rightarrow X, \ \forall a \in A \ (F(\langle a \rangle) = f(a) \wedge F'(\langle a \rangle) = f(a)), \\
\forall u \in W \forall a \in A \ \left(F(u \frown \langle a \rangle) = s(F(u), a) \wedge F'(u \frown \langle a \rangle) = s(F'(u), a) \right) \vdash F = F'
\end{array}$$

$$\begin{array}{ll}
1 & 1 \ F, F' : W \rightarrow X \quad A \\
2 & 2 \ \forall a \in A \ (F(\langle a \rangle) = f(a) \wedge F'(\langle a \rangle) = f(a)) A \\
3 & 3 \ \forall u \in W \forall a \in A \ \left(F(u \frown \langle a \rangle) = s(F(u), a) \wedge F'(u \frown \langle a \rangle) = s(F'(u), a) \right) \\
& A \\
4 & 4 \ P(w) := F(w) = F'(w) \quad = I \\
2 & 5 \ \forall a \in A \ P(\langle a \rangle) \quad \text{Th. } 2.9.1.3(2) \\
6 & 6 \ u \in W \quad A \\
7 & 7 \ a \in A \quad A \\
8 & 8 \ P(u) \quad A \\
2,3,6,7,8 & 9 \ P(u \frown \langle a \rangle) \\
& \text{Th. } 2.9.1.7(\rightarrow E(\forall E(\rightarrow E(\forall E(3), 6)), 7), 8) \\
2,3,6,7 & 10 \ P(u) \rightarrow P(u \frown \langle a \rangle) \quad \rightarrow I(8, 9) \\
2,3 & 11 \ \forall u \in W \forall a \in A \ (P(u) \rightarrow P(u \frown \langle a \rangle)) \\
& \forall I(\rightarrow I(6, \forall I(\rightarrow I(7, 10)))) \\
2,3 & 12 \ \forall w \in W \ P(w) \quad \text{Th. } 22.2.4.2(5, 11) \\
1,2,3 & 13 \ F = F' \quad \text{Th. } 5.2.3.16(1, 12)
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 & 1 \ f : A \rightarrow X \quad A \\
2 & 2 \ s : X \times A \rightarrow XA
\end{array}$$

$$\begin{array}{ll}
1,2 \quad 3 \exists F: A^+ \rightarrow X \left(\forall a \in AF(\langle a \rangle) = f(a) \wedge \forall u \in A^+ \forall a \in AF(u \smallfrown \langle a \rangle) = s(F(u), a) \right) & Th. \ 22.2.4.3(vii)(1, 2) \\
1,2 \quad 4 \exists^{\leq 1} F \left(F: A^+ \rightarrow X \wedge \forall a \in AF(\langle a \rangle) = f(a) \wedge \forall u \in A^+ \forall a \in AF(u \smallfrown \langle a \rangle) = s(F(u), a) \right) & Th. \ 2.10.8.1(Th. \ 22.2.4.3(viii)) \\
1,2 \quad 5 \exists ! F: A^+ \rightarrow X \left(\forall a \in AF(\langle a \rangle) = f(a) \wedge \forall u \in A^+ \forall a \in AF(u \smallfrown \langle a \rangle) = s(F(u), a) \right) & Th. \ 2.10.9.4(3, 4)
\end{array}$$

□

2.5 Linksfaltung eines Wortes

Definition 22.2.5.1 (Funktionsgraph eines binären Operators).

Mengen: A, p, x, y
Funktionen: $\star, G, \widehat{\star}_A$
Neues Symbol:
Operatorfunktion: $\triangleright \widehat{\star}_A$

$$\begin{array}{l}
\text{BinOp}(A, \star) \vdash \widehat{\star}_A: A \times A \rightarrow A, \\
\forall x, y \in A \left(\widehat{\star}_A(x, y) := x \star y \right)
\end{array}$$

$$\begin{array}{ll}
1 \quad 1 \text{ BinOp}(A, \star) & A \\
2 \quad 2 p \in A \times A & A \\
2 \quad 3 \pi_1(p), \pi_2(p) \in A & \\
& Th. \ 7.3.1.11(2), Th. \ 7.3.1.12(2) \\
1,2 \quad 4 \pi_1(p) \star \pi_2(p) \in A & Ax. \ 3.17.2.1(1, 3) \\
1 \quad 5 \forall p \in A \times A \pi_1(p) \star \pi_2(p) \in A & \forall I(\rightarrow I(2, 4)) \\
1 \quad 6 \exists ! G: A \times A \rightarrow A \forall p \in A \times A G(p) = \pi_1(p) \star \pi_2(p) & Th. \ 5.2.6.9(5) \\
7 \widehat{\star}_A := \iota G: A \times A \rightarrow A \forall p \in A \times A G(p) = \pi_1(p) \star \pi_2(p) & \\
& = I \\
1 \quad 8 \widehat{\star}_A: A \times A \rightarrow A \wedge \forall p \in A \times A \widehat{\star}_A(p) = \pi_1(p) \star \pi_2(p) & \\
& Th. \ 2.10.9.7(6, 7) \\
1 \quad 9 \widehat{\star}_A: A \times A \rightarrow A & \wedge E1(8) \\
10 \quad 10 x \in A & A \\
11 \quad 11 y \in A & A \\
10,11 \quad 12 (x, y) \in A \times A & \\
& Th. \ 3.16.5.5(\wedge I(10, 11)) \\
1,10,11 \quad 13 \widehat{\star}_A(x, y) = x \star y & \\
& \rightarrow E(\forall E(\wedge E2(8)), 12), Th. \ 7.3.1.3(12), Th. \ 7.3.1.7(12) \\
1 \quad 14 \forall x, y \in A \widehat{\star}_A(x, y) = x \star y & \\
& \forall I(\rightarrow I(10, \forall I(\rightarrow I(11, 13))))
\end{array}$$

Theorem 22.2.5.1 (Existenz und Eindeutigkeit der Linksfaltung).

Mengen: A, a, u
Funktionen: $\star, F$

$$\text{BinOp}(A, \star) \vdash \\
\exists! F: A^+ \rightarrow A \left(\forall a \in A F(\langle a \rangle) = a \wedge \right. \\
\left. \forall u \in A^+ \forall a \in A F(u \frown \langle a \rangle) = F(u) \star a \right)$$

1	1	$\text{BinOp}(A, \star)$	A
	2	$\text{id}_A: A \rightarrow A$	<i>Th.</i> 8.3.1.1
1	3	$\widehat{\star}_A: A \times A \rightarrow A$	<i>Def.</i> 22.2.5.1(1)
	4	$Q(F) := F: A^+ \rightarrow A \wedge \forall a \in A F(\langle a \rangle) = \text{id}_A(a) \wedge$ $\forall u \in A^+ \forall a \in A F(u \frown \langle a \rangle) = \widehat{\star}_A(F(u), a)$	$= I$
	5	$R(F) := F: A^+ \rightarrow A \wedge \forall a \in A F(\langle a \rangle) = a \wedge$ $\forall u \in A^+ \forall a \in A F(u \frown \langle a \rangle) = F(u) \star a$	$= I$
1	6	$\exists! F Q(F)$	<i>Th.</i> 22.2.4.3(2, 3, 4)
1	7	$\forall a \in A \text{id}_A(a) = a$	<i>Th.</i> 8.3.1.3
			<i>Th.</i> 5.2.3.8
1	8	$\forall F: A^+ \rightarrow A \forall u \in A^+ \forall a \in A$ $\widehat{\star}_A(F(u), a) = F(u) \star a$	<i>Def.</i> 22.2.5.1 (1)
1	9	$\forall F (Q(F) \leftrightarrow R(F))$	
			<i>Th.</i> 2.9.1.7(4, 5, 7, 8)
1	10	$\exists F Q(F)$	<i>Th.</i> 2.10.9.2(6)
11	11	$Q(F)$	A
1,11	12	$R(F)$	$\rightarrow E(\forall E(9), 11)$
1,11	13	$\exists F R(F)$	$\exists I(12)$
1	14	$\exists F R(F)$	$\exists E(10, 11, 13)$
1	15	$\exists^{\leq 1} F Q(F)$	<i>Th.</i> 2.10.9.3(6)
1	16	$\exists^{\leq 1} F R(F)$	<i>Th.</i> 2.10.8.6(15, 9)
1	17	$\exists! F R(F)$	<i>Th.</i> 2.10.9.4(14, 16)
1	18	$\exists! F: A^+ \rightarrow A \left(\forall a \in A F(\langle a \rangle) = a \wedge \forall u \in A^+ \forall a \in A F(u \frown \langle a \rangle) = F(u) \star a \right)$	$= E(5, 17)$

Definition 22.2.5.2 (Linksfaltung eines nichtleeren Wortes).

Mengen: A, a, u
Funktionen: $\star, \Pi_\star$
Neues Symbol:
Linksfaltung: $\triangleright \Pi_\star$

$$\text{BinOp}(A, \star) \vdash \Pi_\star := \iota F: A^+ \rightarrow A \left(\right. \\
\forall a \in A F(\langle a \rangle) = a \wedge \\
\left. \forall u \in A^+ \forall a \in A F(u \frown \langle a \rangle) = F(u) \star a \right)$$

Theorem 22.2.5.2 (Kennzeichnung der Linksfaltung).

Mengen: A, a, u
Funktionen: $\star, F, \Pi_\star$

$$\begin{aligned} \text{BinOp}(A, \star) \vdash \Pi_\star : A^+ \rightarrow A \quad \wedge \\ \forall a \in A \quad \Pi_\star(\langle a \rangle) = a \quad \wedge \\ \forall u \in A^+ \quad \forall a \in A \quad \Pi_\star(u \frown \langle a \rangle) = \Pi_\star(u) \star a \end{aligned}$$

$$\begin{aligned} 1 \quad & 1 \text{ BinOp}(A, \star) A \\ 1 \quad & 2 \exists! F : A^+ \rightarrow A \left(\forall a \in A \quad F(\langle a \rangle) = a \wedge \forall u \in A^+ \quad \forall a \in A \quad F(u \frown \langle a \rangle) = F(u) \star a \right) \\ & \text{Th. 22.2.5.1(1)} \\ 1 \quad & 3 \Pi_\star := \iota F : A^+ \rightarrow A \left(\forall a \in A \quad F(\langle a \rangle) = a \wedge \forall u \in A^+ \quad \forall a \in A \quad F(u \frown \langle a \rangle) = F(u) \star a \right) \\ & \text{Def. 22.2.5.2(1)} \\ 1 \quad & 4 \Pi_\star : A^+ \rightarrow A \wedge \forall a \in A \quad \Pi_\star(\langle a \rangle) = a \wedge \forall u \in A^+ \quad \forall a \in A \quad \Pi_\star(u \frown \langle a \rangle) = \Pi_\star(u) \star a \\ & \text{Th. 2.10.9.7(2, 3)} \end{aligned}$$

Theorem 22.2.5.3 (Typisierung der Linksfaltung).

Mengen: A
Funktionen: $\star, \Pi_\star$

$$\text{BinOp}(A, \star) \vdash \Pi_\star : A^+ \rightarrow A$$

$$\begin{aligned} 1 \quad & 1 \text{ BinOp}(A, \star) A \\ 1 \quad & 2 \Pi_\star : A^+ \rightarrow A \\ & \wedge E1(\text{Th. 22.2.5.2(1)}) \end{aligned}$$

Theorem 22.2.5.4 (Rekursionsgleichungen der Linksfaltung).

Mengen: A, a, u
Funktionen: $\star, \Pi_\star$

$$\begin{aligned} \text{BinOp}(A, \star), a \in A \vdash \Pi_\star(\langle a \rangle) = a, \\ \text{BinOp}(A, \star), u \in A^+, a \in A \vdash \Pi_\star(u \frown \langle a \rangle) = \Pi_\star(u) \star a \end{aligned}$$

Beweis.

Buchstabenwort:

$$\begin{aligned} 1 \quad & 1 \text{ BinOp}(A, \star) A \\ 2 \quad & 2 a \in A \quad A \\ 1 \quad & 3 \forall a \in A \quad \Pi_\star(\langle a \rangle) = a \\ & \wedge E1(\wedge E2(\text{Th. 22.2.5.2(1)})) \end{aligned}$$

$$1,2 \quad 4 \quad \Pi_{\star}(\langle a \rangle) = a \rightarrow E(\forall E(3), 2)$$

Rekursionsschritt:

$$1 \quad 1 \quad \text{BinOp}(A, \star)A$$

$$2 \quad 2 \quad u \in A^+ \quad A$$

$$3 \quad 3 \quad a \in A \quad A$$

$$1 \quad 4 \quad \forall u \in A^+ \forall a \in A \Pi_{\star}(u \frown \langle a \rangle) = \Pi_{\star}(u) \star a$$

$$\wedge E2(\wedge E2(Th. 22.2.5.2(1)))$$

$$1,2,3 \quad 5 \quad \Pi_{\star}(u \frown \langle a \rangle) = \Pi_{\star}(u) \star a$$

$$\rightarrow E(\forall E(\rightarrow E(\forall E(4), 2)), 3)$$

□

Kapitel 3

Volle planare Klammerungsbäume

3.1 Eine konkrete Wortcodierung

Die Baumcodierung verwendet zwei Sorten von Zeichen. Ein Blattzeichen trägt ein Element von A ; ein Knotenzeichen speichert zusätzlich die Länge des linken Teilbaums. Diese Längenangabe macht die Zerlegung eines Knotencodes ohne syntaktische Metasprache eindeutig.

Definition 22.3.1.1 (Alphabet der Baumcodes).

Mengen: A, Σ_A

Neues Symbol:

Baumalphabet: $\triangleright \Sigma_A$

$$\Sigma_A := (\{0\} \times A) \cup (\{1\} \times \mathbb{N})$$

Definition 22.3.1.2 (Blattcode).

Mengen: A, a

Neues Symbol:

Blattcode: $\triangleright \text{Blatt}_A(a)$

$$a \in A \vdash \text{Blatt}_A(a) := \langle (0, a) \rangle$$

Theorem 22.3.1.1 (Wohldefiniertheit der Knotencodierten).

Mengen: A, S, T

$$\begin{aligned} S, T \in \Sigma_A^+ \vdash S, T \in \Sigma_A^*, \quad |S|, |T| \in \mathbb{N}, \\ (1, |S|) \in \Sigma_A, \\ \langle (1, |S|) \rangle, S \frown T \in \Sigma_A^+, \\ \langle (1, |S|) \rangle, S \frown T \in \Sigma_A^* \end{aligned}$$

1	1	$S, T \in \Sigma_A^+$	A
1	2	$S, T \in \Sigma_A^*$	<i>Th.</i> 22.2.2.3(1)
1	3	$ S , T \in \mathbb{N}$	<i>Th.</i> 22.2.2.3(1)
1	4	$(1, S) \in \Sigma_A$	<i>Def.</i> 22.3.1.1(3)
1	5	$\langle (1, S) \rangle \in \Sigma_A^+$	<i>Th.</i> 22.2.2.4(4)
1	6	$S \frown T \in \Sigma_A^+$	<i>Th.</i> 22.2.3.7(1)
1	7	$\langle (1, S) \rangle \in \Sigma_A^*$	<i>Th.</i> 22.2.2.3(5)
1	8	$S \frown T \in \Sigma_A^*$	<i>Th.</i> 22.2.2.3(6)

Nach Th. 22.3.1.1 sind die beiden Argumente der äußeren Wortkatenation endliche Wörter. Damit ist die folgende Definition unter ihren angegebenen Voraussetzungen wohldefiniert.

Definition 22.3.1.3 (Knotencode).

Mengen: A, S, T
Neues Symbol:
Knotencode: $\triangleright \text{Knoten}_A(S, T)$

$$S, T \in \Sigma_A^+ \vdash \text{Knoten}_A(S, T) := \langle (1, |S|) \rangle \frown S \frown T$$

Theorem 22.3.1.2 (Konstruktortrennung und -injektivität).

Mengen: A, a, b, S, T, S', T'

$$\begin{aligned} & a, b \in A \vdash (\text{Blatt}_A(a) = \text{Blatt}_A(b) \leftrightarrow a = b), \\ & a \in A, S, T \in \Sigma_A^+ \vdash \text{Blatt}_A(a) \neq \text{Knoten}_A(S, T), \\ & S, T, S', T' \in \Sigma_A^+, \text{Knoten}_A(S, T) = \text{Knoten}_A(S', T') \\ & \vdash S = S' \wedge T = T' \end{aligned}$$

Beweis.

Blattcodes Th. 22.3.1.2(i)

$$a, b \in A \vdash (\text{Blatt}_A(a) = \text{Blatt}_A(b) \leftrightarrow a = b)$$

1	1	$a \in A$	A
2	2	$b \in A$	A
3	3	$\text{Blatt}_A(a) = \text{Blatt}_A(b)$	A
1	4	$(0, a) \in \Sigma_A$	<i>Def.</i> 22.3.1.1(1)
2	5	$(0, b) \in \Sigma_A$	<i>Def.</i> 22.3.1.1(2)
1	6	$\text{Blatt}_A(a)(0) = (0, a)$	
			<i>Def.</i> 22.3.1.2(Th. 22.2.2.4(4))
2	7	$\text{Blatt}_A(b)(0) = (0, b)$	

Def. 22.3.1.2(Th. 22.2.2.4(5))

$$\begin{array}{ll} 1,2,3 & 8 \ (0, a) = (0, b) \\ 1,2,3 & 9 \ a = b \end{array} \quad = E(3, 6, 7)$$

$\wedge E2(Th. 3.11.4.1(8))$

$$\begin{array}{ll} 1,2 & 10 \ \text{Blatt}_A(a) = \text{Blatt}_A(b) \rightarrow a = b \rightarrow I(3, 9) \\ 11 & 11 \ a = b \quad A \\ 1,2,11 & 12 \ \text{Blatt}_A(a) = \text{Blatt}_A(b) \quad = E(11) \\ 1,2 & 13 \ a = b \rightarrow \text{Blatt}_A(a) = \text{Blatt}_A(b) \rightarrow I(11, 12) \\ 1,2 & 14 \ \text{Blatt}_A(a) = \text{Blatt}_A(b) \leftrightarrow a = b \leftrightarrow I(10, 13) \end{array}$$

Trennung der Konstruktoren

Th. 22.3.1.2(ii)

$$a \in A, \ S, T \in \Sigma_A^+ \vdash \text{Blatt}_A(a) \neq \text{Knoten}_A(S, T)$$

$$\begin{array}{ll} 1 & 1 \ a \in A \quad A \\ 2 & 2 \ S, T \in \Sigma_A^+ \quad A \\ 1 & 3 \ (0, a) \in \Sigma_A \quad \text{Def. 22.3.1.1(1)} \\ 2 & 4 \ (1, |S|) \in \Sigma_A \quad \text{Th. 22.3.1.1(2)} \\ 1 & 5 \ \text{Blatt}_A(a) (0) = (0, a) \end{array}$$

Def. 22.3.1.2(Th. 22.2.2.4(3))

$$\begin{array}{ll} 2 & 6 \ \langle (1, |S|) \rangle, S \frown T \in \Sigma_A^* \quad \text{Th. 22.3.1.1(2)} \\ 2 & 7 \ |\langle (1, |S|) \rangle| = 1 \quad \text{Th. 22.2.2.4(4)} \\ & 8 \ 0 \in \mathbb{N}_{<1} \quad \text{Th. 10.5.1.12} \\ 2 & 9 \ 0 \in \mathbb{N}_{<|\langle (1, |S|) \rangle|} \quad = E(6, 7) \\ 2 & 10 \ \langle (1, |S|) \rangle \frown S \frown T(0) \quad \text{Th. 22.2.3.2(5, 8)} \\ & \quad = \langle (1, |S|) \rangle(0) \\ 2 & 11 \ \langle (1, |S|) \rangle(0) = (1, |S|) \quad \text{Th. 22.2.2.4(4)} \\ 2 & 12 \ \text{Knoten}_A(S, T) = \langle (1, |S|) \rangle \frown S \frown T \quad \text{Def. 22.3.1.3(2)} \\ 2 & 13 \ \text{Knoten}_A(S, T) (0) = (1, |S|) \quad = E(11, 9, 10) \\ 13 & 14 \ \text{Blatt}_A(a) = \text{Knoten}_A(S, T) \quad A \\ 1,2,13 & 15 \ (0, a) = (1, |S|) \quad = E(13, 5, 12) \\ 1,2,13 & 16 \ 0 = 1 \end{array}$$

$\wedge E1(Th. 3.11.4.1(14))$

$$\begin{array}{ll} & 17 \ 0 \neq 1 \quad \text{Th. 10.5.1.5} \\ 1,2,13 & 18 \ \perp \quad \perp I(16, 15) \\ 1,2 & 19 \ \text{Blatt}_A(a) \neq \text{Knoten}_A(S, T) \quad \neg I(13, 17) \end{array}$$

Injektivität des Knotenkonstruktors

Th. 22.3.1.2(iii)

$$\begin{array}{l} S, T, S', T' \in \Sigma_A^+, \\ \text{Knoten}_A(S, T) = \text{Knoten}_A(S', T') \vdash S = S' \wedge T = T' \end{array}$$

$$1 \quad 1 \ S, T, S', T' \in \Sigma_A^+ \quad A$$

2	2	$\text{Knoten}_A(S, T) = \text{Knoten}_A(S', T')A$	
1	3	$(1, S), (1, S') \in \Sigma_A$	<i>Th.</i> 22.3.1.1(1)
1	4	$\langle (1, S) \rangle, S \frown T,$ $\langle (1, S') \rangle, S' \frown T' \in \Sigma_A^*$	<i>Th.</i> 22.3.1.1(1)
1	5	$ \langle (1, S) \rangle = 1$	<i>Th.</i> 22.2.2.4(3)
1	6	$ \langle (1, S') \rangle = 1$	<i>Th.</i> 22.2.2.4(3)
1	7	$ \langle (1, S) \rangle = \langle (1, S') \rangle $	<i>Th.</i> 2.9.1.3(5, 6)
1,2	8	$\langle (1, S) \rangle \frown S \frown T$ $= \langle (1, S') \rangle \frown S' \frown T'$	<i>Def.</i> 22.3.1.3(2)
1,2	9	$\langle (1, S) \rangle = \langle (1, S') \rangle$	<i>Th.</i> 22.2.3.4(4, 7, 8)
1,2	10	$S \frown T = S' \frown T'$	<i>Th.</i> 22.2.3.4(4, 7, 8)
1,2	11	$(1, S) = (1, S')$	$= E(9, \text{Th. } 22.2.2.4(3), \text{Th. } 22.2.2.4(3))$
1,2	12	$ S = S' $	$\wedge E2(\text{Th. } 3.11.4.1(11))$
1	13	$S, T, S', T' \in \Sigma_A^*$	<i>Th.</i> 22.2.2.3(1)
1,2	14	$S = S'$	<i>Th.</i> 22.2.3.4(13, 12, 10)
1,2	15	$T = T'$	<i>Th.</i> 22.2.3.4(13, 12, 10)
1,2	16	$S = S' \wedge T = T'$	$\wedge I(14, 15)$

□

3.2 Baumstufen und Baumträger

Definition 22.3.2.1 (Erzeugungsschritt für Baumcodes).

Mengen: A, X, a, S, T

Neues Symbol:

Erzeugungsschritt: $\triangleright \mathcal{C}_A$

$$X \in \mathcal{P}(\Sigma_A^+) \vdash \mathcal{C}_A(X) := \{\text{Blatt}_A(a) \mid a \in A\} \\ \cup \{\text{Knoten}_A(S, T) \mid S, T \in X\}$$

Theorem 22.3.2.1 (Der Erzeugungsschritt ist eine Abbildung).

Mengen: A, X

Funktionen: $\mathcal{C}_A$

$$\mathcal{C}_A: \mathcal{P}(\Sigma_A^+) \rightarrow \mathcal{P}(\Sigma_A^+)$$

1	1	$X \in \mathcal{P}(\Sigma_A^+)$	A
1	2	$X \subseteq \Sigma_A^+$	<i>Th.</i> 3.16.2.1(1)

3	3	$a \in A$	A	
3	4	$(0, a) \in \Sigma_A$	$Def. 22.3.1.1(3)$	
3	5	$Blatt_A(a) \in \Sigma_A^+$		$Def. 22.3.1.2(Th. 22.2.2.4(4))$
	6	$\forall a \in A Blatt_A(a) \in \Sigma_A^+$	$\forall I(\rightarrow I(3, 5))$	
7	7	$S, T \in X$	A	
1,7	8	$S, T \in \Sigma_A^+$	$Th. 3.5.2.6(2, 7)$	
1,7	9	$Knoten_A(S, T) \in \Sigma_A^+$		$Def. 22.3.1.3(Th. 22.3.1.1(8), Th. 22.2.3.7)$
1	10	$\forall S, T \in X Knoten_A(S, T) \in \Sigma_A^+ \forall I(\rightarrow I(7, 9))$		
1	11	$\mathcal{C}_A(X) \subseteq \Sigma_A^+$		$Def. 22.3.2.1(6, 10)$
1	12	$\mathcal{C}_A(X) \in \mathcal{P}(\Sigma_A^+)$	$Th. 3.16.2.1(11)$	
	13	$\forall X \in \mathcal{P}(\Sigma_A^+) \mathcal{C}_A(X) \in \mathcal{P}(\Sigma_A^+) \forall I(\rightarrow I(1, 12))$		
	14	$\mathcal{C}_A: \mathcal{P}(\Sigma_A^+) \rightarrow \mathcal{P}(\Sigma_A^+)$	$Th. 5.2.6.9(13)$	

Theorem 22.3.2.2 (Monotonie des Erzeugungsschritts).

Mengen: A, X, Y, R, a, S, T

$$X, Y \in \mathcal{P}(\Sigma_A^+), X \subseteq Y \vdash \\ \mathcal{C}_A(X) \subseteq \mathcal{C}_A(Y)$$

1	1	$X, Y \in \mathcal{P}(\Sigma_A^+)$	A	
2	2	$X \subseteq Y$	A	
3	3	$R \in \mathcal{C}_A(X)$	A	
1,2,3	4	$\exists a \in A (R = Blatt_A(a)) \vee \exists S, T \in X (R = Knoten_A(S, T))$	$Def. 22.3.2.1(1, 3)$	
5	5	$\exists a \in A (R = Blatt_A(a))$	A	
6	6	$a \in A \wedge R = Blatt_A(a)$	A	
1,6	7	$R \in \mathcal{C}_A(Y)$		$Def. 22.3.2.1(1, \wedge E1(6), \wedge E2(6))$
5	8	$R \in \mathcal{C}_A(Y)$	$\exists E(5, 6, 7)$	
9	9	$\exists S, T \in X (R = Knoten_A(S, T)) A$		
10	10	$S, T \in X \wedge R = Knoten_A(S, T) A$		
1,2,10	11	$S, T \in Y$		$Th. 3.5.2.6(2, \wedge E1(10))$
1,2,10	12	$R \in \mathcal{C}_A(Y)$		$Def. 22.3.2.1(1, 11, \wedge E2(10))$
1,2,9	13	$R \in \mathcal{C}_A(Y)$	$\exists E(9, 10, 12)$	
1,2,3	14	$R \in \mathcal{C}_A(Y)$	$\vee E(4, 5, 8, 9, 13)$	
1,2	15	$\forall R (R \in \mathcal{C}_A(X) \rightarrow R \in \mathcal{C}_A(Y)) \forall I(\rightarrow I(3, 14))$		

$$1,2 \quad 16 \quad \mathcal{C}_A(X) \subseteq \mathcal{C}_A(Y) \quad \text{Def. 3.5.1.1(15)}$$

Definition 22.3.2.2 (Stufen der Klammerungsbäume).

Mengen: $A, n, \mathcal{T}_{A,n}$
Funktionen: $\mathcal{C}_A$
Neues Symbol:
Baumstufe: $\triangleright \mathcal{T}_{A,n}$

$$n \in \mathbb{N} \vdash \mathcal{T}_{A,n} := \text{Rec}_{\emptyset, \mathcal{C}_A}(n)$$

Theorem 22.3.2.3 (Rekursionsgleichungen der Baumstufen).

Mengen: A, n

$$\mathcal{T}_{A,0} = \emptyset, \quad n \in \mathbb{N} \vdash \mathcal{T}_{A,\text{succ}(n)} = \mathcal{C}_A(\mathcal{T}_{A,n})$$

$$1 \quad \mathcal{C}_A: \mathcal{P}(\Sigma_A^+) \rightarrow \mathcal{P}(\Sigma_A^+) \text{ Th. 22.3.2.1}$$

$$2 \quad \emptyset \in \mathcal{P}(\Sigma_A^+)$$

$$\text{Th. 3.16.2.1(Th. 3.6.3.2)}$$

$$3 \quad 0 \in \mathbb{N} \quad \text{Ax. 10.3.1}$$

$$4 \quad \text{Rec}_{\emptyset, \mathcal{C}_A}(0) = \emptyset \quad \text{Th. 10.4.3.23(2, 1)}$$

$$5 \quad \mathcal{T}_{A,0} = \emptyset \quad \text{Def. 22.3.2.2(3, 4)}$$

$$6 \quad 6 \quad n \in \mathbb{N} \quad A$$

$$6 \quad 7 \quad \text{succ}(n) \in \mathbb{N} \quad \text{Ax. 10.3.4(6)}$$

$$6 \quad 8 \quad \text{Rec}_{\emptyset, \mathcal{C}_A}(\text{succ}(n)) = \mathcal{C}_A(\text{Rec}_{\emptyset, \mathcal{C}_A}(n))$$

$$\text{Th. 10.4.3.24(2, 1, 6)}$$

$$6 \quad 9 \quad \mathcal{T}_{A,\text{succ}(n)} = \mathcal{C}_A(\mathcal{T}_{A,n})$$

$$\text{Def. 22.3.2.2(6, 7, 8)}$$

Theorem 22.3.2.4 (Typisierung der Baumstufen).

Mengen: A, n

$$n \in \mathbb{N} \vdash \mathcal{T}_{A,n} \in \mathcal{P}(\Sigma_A^+), \quad \mathcal{T}_{A,n} \subseteq \Sigma_A^+$$

$$1 \quad 1 \quad n \in \mathbb{N} \quad A$$

$$2 \quad \mathcal{C}_A: \mathcal{P}(\Sigma_A^+) \rightarrow \mathcal{P}(\Sigma_A^+) \text{ Th. 22.3.2.1}$$

$$3 \quad \emptyset \in \mathcal{P}(\Sigma_A^+)$$

$$\text{Th. 3.16.2.1(Th. 3.6.3.2)}$$

$$4 \quad \text{Rec}_{\emptyset, \mathcal{C}_A}: \mathbb{N} \rightarrow \mathcal{P}(\Sigma_A^+) \text{ Th. 10.4.3.21(3, 2)}$$

$$1 \quad 5 \quad \text{Rec}_{\emptyset, \mathcal{C}_A}(n) \in \mathcal{P}(\Sigma_A^+) \text{ Th. 5.2.3.8(4, 1)}$$

- $1 \quad 6 \quad \mathcal{T}_{A,n} \in \mathcal{P}(\Sigma_A^+)$ *Def.* 22.3.2.2(1, 5)
 $1 \quad 7 \quad \mathcal{T}_{A,n} \subseteq \Sigma_A^+$ *Th.* 3.16.2.1(6)

Theorem 22.3.2.5 (Monotonie der Baumstufen).

Mengen: A, n

$$n \in \mathbb{N} \vdash \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,\text{succ}(n)}$$

Beweis.

Induktionsanfang

Th. 22.3.2.5(i)

$$\mathcal{T}_{A,0} \subseteq \mathcal{T}_{A,\text{succ}(0)}$$

- $1 \quad \mathcal{T}_{A,0} = \emptyset$ *Th.* 22.3.2.3
 $2 \quad \emptyset \subseteq \mathcal{T}_{A,\text{succ}(0)}$ *Th.* 3.6.3.2
 $3 \quad \mathcal{T}_{A,0} \subseteq \mathcal{T}_{A,\text{succ}(0)} = E(1, 2)$

Induktionsschritt

Th. 22.3.2.5(ii)

$$n \in \mathbb{N}, \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,\text{succ}(n)} \vdash \mathcal{T}_{A,\text{succ}(n)} \subseteq \mathcal{T}_{A,\text{succ}(\text{succ}(n))}$$

- $1 \quad 1 \quad n \in \mathbb{N}$ A
 $2 \quad 2 \quad \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,\text{succ}(n)}$ A
 $1 \quad 3 \quad \text{succ}(n) \in \mathbb{N}$ *Ax.* 10.3.4(1)
 $1 \quad 4 \quad \mathcal{T}_{A,n}, \mathcal{T}_{A,\text{succ}(n)} \in \mathcal{P}(\Sigma_A^+)$ *Th.* 22.3.2.4(1, 3)
 $1,2 \quad 5 \quad \mathcal{C}_A(\mathcal{T}_{A,n}) \subseteq \mathcal{C}_A(\mathcal{T}_{A,\text{succ}(n)})$ *Th.* 22.3.2.2(4, 2)
 $1,2 \quad 6 \quad \mathcal{T}_{A,\text{succ}(n)} \subseteq \mathcal{T}_{A,\text{succ}(\text{succ}(n))}$ *Th.* 22.3.2.3(1, 3, 5)

Induktionsschluss:

- $1 \quad 1 \quad n \in \mathbb{N}A$
 $1 \quad 2 \quad \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,\text{succ}(n)}$

Ax. 10.3.9(*Th.* 22.3.2.5(i), *Th.* 22.3.2.5(ii), 1)

□

Theorem 22.3.2.6 (Additive Verschachtelung der Baumstufen).

Mengen: A, n, k

$$n, k \in \mathbb{N} \vdash \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+k}$$

Beweis.

Induktionsanfang

Th. 22.3.2.6(i)

$$n \in \mathbb{N} \vdash \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+0}$$

$$\begin{array}{ll} 1 & 1 \ n \in \mathbb{N} \quad A \\ 1 & 2 \ n + 0 = n \quad Th. 10.4.4.2(1) \\ & 3 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n} \quad Th. 3.5.3.1 \\ 1 & 4 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+0} = E(2,3) \end{array}$$

Induktionsschritt

Th. 22.3.2.6(ii)

$$n, k \in \mathbb{N}, \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+k} \vdash \\ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+\text{succ}(k)}$$

$$\begin{array}{ll} 1 & 1 \ n, k \in \mathbb{N} \quad A \\ 2 & 2 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+k} \quad A \\ 1 & 3 \ n + k \in \mathbb{N} \quad Th. 10.4.4.1(1) \\ 1 & 4 \ \mathcal{T}_{A,n+k} \subseteq \mathcal{T}_{A,\text{succ}(n+k)} \quad Th. 22.3.2.5(3) \\ 1,2 & 5 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,\text{succ}(n+k)} \quad Th. 3.5.3.6(2,4) \\ 1,2 & 6 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+\text{succ}(k)} \end{array}$$

$$= E(Th. 10.4.4.3(1), 5)$$

Induktionsschluss:

$$\begin{array}{ll} 1 & 1 \ n, k \in \mathbb{N} \quad A \\ 1 & 2 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,n+k} \end{array}$$

$$Ax. 10.3.9(Th. 22.3.2.6(i)(1), Th. 22.3.2.6(ii)(1), 1)$$

□

Theorem 22.3.2.7 (Verschachtelung der Baumstufen).

Mengen: A, m, n, k

Relationsoperatoren: $\leq_{\mathbb{N}}$

$$m, n \in \mathbb{N}, m \leq n \vdash \mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,n}$$

$$\begin{array}{ll} 1 & 1 \ m, n \in \mathbb{N} \quad A \\ 2 & 2 \ m \leq n \quad A \\ 1,2 & 3 \ \exists k \in \mathbb{N} (m + k = n) \quad Th. 10.4.5.6(1,2) \\ 4 & 4 \ k \in \mathbb{N} \wedge m + k = n \quad A \end{array}$$

4	5	$k \in \mathbb{N}$	$\wedge E1(4)$
4	6	$m + k = n$	$\wedge E2(4)$
1,4	7	$\mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,m+k}$	<i>Th.</i> 22.3.2.6(1, 5)
1,4	8	$\mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,n}$	$= E(6, 7)$
1,2	9	$\mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,n}$	$\exists E(3, 4, 8)$

Definition 22.3.2.3 (Menge der vollen planaren Klammerungsbäume).

Mengen: $A, R, n, \mathcal{T}(A)$

Neues Symbol:

Klammerungsbäume: $\triangleright \mathcal{T}(A)$

$$\mathcal{T}(A) := \{R \in \Sigma_A^+ \mid \exists n \in \mathbb{N} (R \in \mathcal{T}_{A,n})\}$$

Theorem 22.3.2.8 (Jede Baumstufe liegt im Baumträger).

Mengen: A, n, R

$$n \in \mathbb{N} \vdash \mathcal{T}_{A,n} \subseteq \mathcal{T}(A)$$

1	1	$n \in \mathbb{N}$	A
2	2	$R \in \mathcal{T}_{A,n}$	A
1	3	$\mathcal{T}_{A,n} \subseteq \Sigma_A^+$	<i>Th.</i> 22.3.2.4(1)
1,2	4	$R \in \Sigma_A^+$	<i>Th.</i> 3.5.2.6(3, 2)
1,2	5	$\exists k \in \mathbb{N} (R \in \mathcal{T}_{A,k})$	$\exists I(\wedge I(1, 2))$
1,2	6	$R \in \mathcal{T}(A)$	<i>Def.</i> 22.3.2.3(4, 5)
1	7	$\forall R (R \in \mathcal{T}_{A,n} \rightarrow R \in \mathcal{T}(A)) \forall I(\rightarrow I(2, 6))$	
1	8	$\mathcal{T}_{A,n} \subseteq \mathcal{T}(A)$	<i>Def.</i> 3.5.1.1(7)

Theorem 22.3.2.9 (Gemeinsame Baumstufe).

Mengen: A, S, T, m, n, q

$$S, T \in \mathcal{T}(A) \vdash \exists q \in \mathbb{N} (S, T \in \mathcal{T}_{A,q})$$

1	1	$S, T \in \mathcal{T}(A)$	A
1	2	$\exists m \in \mathbb{N} (S \in \mathcal{T}_{A,m})$	<i>Def.</i> 22.3.2.3(1)
1	3	$\exists n \in \mathbb{N} (T \in \mathcal{T}_{A,n})$	<i>Def.</i> 22.3.2.3(1)
4	4	$m \in \mathbb{N} \wedge S \in \mathcal{T}_{A,m}$	A
5	5	$n \in \mathbb{N} \wedge T \in \mathcal{T}_{A,n}$	A

4,5	6	$m, n \in \mathbb{N}$		$\wedge I(\wedge E1(4), \wedge E1(5))$
4,5	7	$S \in \mathcal{T}_{A,m} \wedge T \in \mathcal{T}_{A,n}$		$\wedge I(\wedge E2(4), \wedge E2(5))$
4,5	8	$m \leq n \vee n \leq m$	Th. 10.4.5.22(6)	
9	9	$m \leq n$	A	
4,5,9	10	$\mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,n}$	Th. 22.3.2.7(6, 9)	
4,5,9	11	$S \in \mathcal{T}_{A,n}$		Th. 3.5.2.6(10, $\wedge E1(7)$)
4,5,9	12	$\exists q \in \mathbb{N} (S, T \in \mathcal{T}_{A,q})$		$\exists I(\wedge I(\wedge E1(5), \wedge I(11, \wedge E2(7))))$
13	13	$n \leq m$	A	
4,5,13	14	$\mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,m}$	Th. 22.3.2.7(6, 13)	
4,5,13	15	$T \in \mathcal{T}_{A,m}$		Th. 3.5.2.6(14, $\wedge E2(7)$)
4,5,13	16	$\exists q \in \mathbb{N} (S, T \in \mathcal{T}_{A,q})$		$\exists I(\wedge I(\wedge E1(4), \wedge I(\wedge E1(7), 15)))$
4,5	17	$\exists q \in \mathbb{N} (S, T \in \mathcal{T}_{A,q}) \vee E(8, 9, 12, 13, 16)$		
4	18	$\exists q \in \mathbb{N} (S, T \in \mathcal{T}_{A,q}) \exists E(3, 5, 17)$		
1	19	$\exists q \in \mathbb{N} (S, T \in \mathcal{T}_{A,q}) \exists E(2, 4, 18)$		

Theorem 22.3.2.10 (Konstruktorabschluss der Klammerungsbäume).

Mengen: A, a, S, T, q

$$a \in A \vdash \text{Blatt}_A(a) \in \mathcal{T}(A),$$

$$S, T \in \mathcal{T}(A) \vdash \text{Knoten}_A(S, T) \in \mathcal{T}(A)$$

Beweis.

Blattabschluss

Th. 22.3.2.10(i)

$$a \in A \vdash \text{Blatt}_A(a) \in \mathcal{T}(A)$$

1	1	$a \in A$	A
	2	$0 \in \mathbb{N}$	Ax. 10.3.1
	3	$\mathcal{T}_{A,0} \in \mathcal{P}(\Sigma_A^+)$	Th. 22.3.2.4(2)
1	4	$\text{Blatt}_A(a) \in \mathcal{C}_A(\mathcal{T}_{A,0})$	Def. 22.3.2.1(3, 1)
1	5	$\text{Blatt}_A(a) \in \mathcal{T}_{A,\text{succ}(0)}$	Th. 22.3.2.3(2, 4)
	6	$\text{succ}(0) \in \mathbb{N}$	Ax. 10.3.4(2)
	7	$\mathcal{T}_{A,\text{succ}(0)} \subseteq \mathcal{T}(A)$	Th. 22.3.2.8(6)
1	8	$\text{Blatt}_A(a) \in \mathcal{T}(A)$	Th. 3.5.2.6(7, 5)

Knotenabschluss

Th. 22.3.2.10(ii)

$$S, T \in \mathcal{T}(A) \vdash \text{Knoten}_A(S, T) \in \mathcal{T}(A)$$

1	1	$S, T \in \mathcal{T}(A)$	A
1	2	$\exists q \in \mathbb{N} (S, T \in \mathcal{T}_{A,q})$	Th. 22.3.2.9(1)
3	3	$q \in \mathbb{N} \wedge S, T \in \mathcal{T}_{A,q}$	A
3	4	$q \in \mathbb{N}$	$\wedge E1(3)$
3	5	$S, T \in \mathcal{T}_{A,q}$	$\wedge E2(3)$
3	6	$\mathcal{T}_{A,q} \in \mathcal{P}(\Sigma_A^+)$	Th. 22.3.2.4(4)
3	7	$\text{Knoten}_A(S, T) \in \mathcal{C}_A(\mathcal{T}_{A,q})$	Def. 22.3.2.1(6, 5)
3	8	$\text{Knoten}_A(S, T) \in \mathcal{T}_{A, \text{succ}(q)}$	Th. 22.3.2.3(4, 7)
3	9	$\text{succ}(q) \in \mathbb{N}$	Ax. 10.3.4(4)
3	10	$\mathcal{T}_{A, \text{succ}(q)} \subseteq \mathcal{T}(A)$	Th. 22.3.2.8(9)
3	11	$\text{Knoten}_A(S, T) \in \mathcal{T}(A)$	Th. 3.5.2.6(10, 8)
1	12	$\text{Knoten}_A(S, T) \in \mathcal{T}(A)$	$\exists E(2, 3, 11)$

Schluss:

1	1	$a \in A$	A
1	2	$\text{Blatt}_A(a) \in \mathcal{T}(A)$	Th. 22.3.2.10(i)(1)
3	3	$S, T \in \mathcal{T}(A)$	A
3	4	$\text{Knoten}_A(S, T) \in \mathcal{T}(A)$	

Th. 22.3.2.10(ii)(3)

□

Theorem 22.3.2.11 (Konstruktorzerlegung).**Mengen:** A, R, a, S, T, n, k

$$R \in \mathcal{T}(A) \dashv\vdash \exists a \in A (R = \text{Blatt}_A(a)) \vee \exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$$

*Beweis.***Hinrichtung**

Th. 22.3.2.11(i)

$$R \in \mathcal{T}(A) \vdash \exists a \in A (R = \text{Blatt}_A(a)) \vee \exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$$

1	1	$R \in \mathcal{T}(A)$	A
1	2	$\exists n \in \mathbb{N} (R \in \mathcal{T}_{A,n})$	Def. 22.3.2.3(1)

3	3	$n \in \mathbb{N} \wedge R \in \mathcal{T}_{A,n}$	A	
3	4	$n \in \mathbb{N}$	$\wedge E1(3)$	
3	5	$R \in \mathcal{T}_{A,n}$	$\wedge E2(3)$	
6	6	$n = 0$	A	
3,6	7	$R \in \mathcal{T}_{A,0}$	$= E(6, 5)$	
	8	$\mathcal{T}_{A,0} = \emptyset$	$Th. 22.3.2.3$	
3,6	9	$R \in \emptyset$	$= E(8, 7)$	
	10	$R \notin \emptyset$	$Th. 3.6.2.2$	
3,6	11	$\perp$	$\perp I(10, 9)$	
3	12	$n \neq 0$	$\neg I(6, 11)$	
3	13	$\exists k \in \mathbb{N} (n = \text{succ}(k))$	$Ax. 10.3.6(4, 12)$	
14	14	$k \in \mathbb{N} \wedge n = \text{succ}(k)$	A	
14	15	$k \in \mathbb{N}$	$\wedge E1(14)$	
14	16	$n = \text{succ}(k)$	$\wedge E2(14)$	
3,14	17	$R \in \mathcal{T}_{A,\text{succ}(k)}$	$= E(16, 5)$	
3,14	18	$R \in \mathcal{C}_A(\mathcal{T}_{A,k})$	$Th. 22.3.2.3(15, 17)$	
14	19	$\mathcal{T}_{A,k} \in \mathcal{P}(\Sigma_A^+)$	$Th. 22.3.2.4(15)$	
3,14	20	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}_{A,k} (R = \text{Knoten}_A(S, T))$		$Def. 22.3.2.1(19, 18)$
21	21	$\exists a \in A (R = \text{Blatt}_A(a))$	A	
21	22	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$	$\vee I1(21)$	
23	23	$\exists S, T \in \mathcal{T}_{A,k} (R = \text{Knoten}_A(S, T))$	A	
24	24	$S, T \in \mathcal{T}_{A,k} \wedge R = \text{Knoten}_A(S, T)$	A	
14	25	$\mathcal{T}_{A,k} \subseteq \mathcal{T}(A)$	$Th. 22.3.2.8(15)$	
14,24	26	$S, T \in \mathcal{T}(A)$		$Th. 3.5.2.6(25, \wedge E1(24))$
14,24	27	$\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$		$\exists I(\wedge I(26, \wedge E2(24)))$
14,24	28	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$	$\vee I2(27)$	
14,23	29	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$	$\exists E(23, 24, 28)$	
3,14	30	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$		$\vee E(20, 21, 22, 23, 29)$
3	31	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$	$\exists E(13, 14, 30)$	
1	32	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T))$	$\exists E(2, 3, 31)$	

Rückrichtung

Th. 22.3.2.11(ii)

$$\begin{aligned} & \exists a \in A (R = \text{Blatt}_A(a)) \vee \\ & \exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T)) \vdash R \in \mathcal{T}(A) \end{aligned}$$

$$\begin{array}{ll}
1 & 1 \exists a \in A (R = \text{Blatt}_A(a)) \vee A \\
& \exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T)) \\
2 & 2 \exists a \in A (R = \text{Blatt}_A(a)) A \\
3 & 3 a \in A \wedge R = \text{Blatt}_A(a) A \\
3 & 4 a \in A \wedge E1(3) \\
3 & 5 R = \text{Blatt}_A(a) \wedge E2(3) \\
3 & 6 R \in \mathcal{T}(A) \\
& = E(5, Th. 22.3.2.10(4)) \\
2 & 7 R \in \mathcal{T}(A) \exists E(2, 3, 6) \\
8 & 8 \exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T)) A \\
9 & 9 S, T \in \mathcal{T}(A) \wedge R = \text{Knoten}_A(S, T) A \\
9 & 10 S, T \in \mathcal{T}(A) \wedge E1(9) \\
9 & 11 R = \text{Knoten}_A(S, T) \wedge E2(9) \\
9 & 12 R \in \mathcal{T}(A) \\
& = E(11, Th. 22.3.2.10(10)) \\
8 & 13 R \in \mathcal{T}(A) \exists E(8, 9, 12) \\
1 & 14 R \in \mathcal{T}(A) \vee E(1, 2, 7, 8, 13)
\end{array}$$

Schluss:

$$\begin{array}{l}
1 R \in \mathcal{T}(A) \leftrightarrow \exists a \in A (R = \text{Blatt}_A(a)) \vee \\
\exists S, T \in \mathcal{T}(A) (R = \text{Knoten}_A(S, T)) \\
\leftrightarrow I(Th. 22.3.2.11(i), Th. 22.3.2.11(ii))
\end{array}$$

□

3.3 Strukturinduktion und Strukturrekursion

Theorem 22.3.3.1 (Strukturinduktion auf Klammerungsbäumen).

Mengen: A, a, R, S, T, n

Prädikate: P

$$\begin{array}{l}
\forall a \in A P(\text{Blatt}_A(a)), \\
\forall S, T \in \mathcal{T}(A) (P(S) \wedge P(T) \rightarrow P(\text{Knoten}_A(S, T))) \\
\vdash \forall R \in \mathcal{T}(A) P(R)
\end{array}$$

Beweis.

Induktionsanfang

Th. 22.3.3.1(i)

$$\forall R \in \mathcal{T}_{A,0} P(R)$$

$$1 \quad 1 R \in \mathcal{T}_{A,0} \quad A$$

2	$\mathcal{T}_{A,0} = \emptyset$	<i>Th.</i> 22.3.2.3
1 3	$R \in \emptyset$	$= E(2, 1)$
4	$R \notin \emptyset$	<i>Th.</i> 3.6.2.2
1 5	$\perp$	$\perp I(4, 3)$
1 6	$P(R)$	<i>Th.</i> 2.1.7.4(5)
7	$\forall R \in \mathcal{T}_{A,0} P(R) \forall I(\rightarrow I(1, 6))$	

Induktionsschritt

Th. 22.3.3.1(ii)

$$\begin{aligned}
& n \in \mathbb{N}, \forall a \in A P(\text{Blatt}_A(a)), \\
& \forall S, T \in \mathcal{T}(A) (P(S) \wedge P(T) \rightarrow P(\text{Knoten}_A(S, T))), \\
& \forall R \in \mathcal{T}_{A,n} P(R) \vdash \forall R \in \mathcal{T}_{A,\text{succ}(n)} P(R)
\end{aligned}$$

1	1	$n \in \mathbb{N}$	A
2	2	$\forall a \in A P(\text{Blatt}_A(a))$	A
3	3	$\forall S, T \in \mathcal{T}(A) (P(S) \wedge P(T) \rightarrow P(\text{Knoten}_A(S, T)))$	A
4	4	$\forall R \in \mathcal{T}_{A,n} P(R)$	A
5	5	$R \in \mathcal{T}_{A,\text{succ}(n)}$	A
1,5	6	$R \in \mathcal{C}_A(\mathcal{T}_{A,n})$	<i>Th.</i> 22.3.2.3(1, 5)
1,5	7	$\exists a \in A (R = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}_{A,n} (R = \text{Knoten}_A(S, T))$	<i>Def.</i> 22.3.2.1(<i>Th.</i> 22.3.2.4(1), 6)
8	8	$\exists a \in A (R = \text{Blatt}_A(a))$	A
9	9	$a \in A \wedge R = \text{Blatt}_A(a)$	A
2,9	10	$P(\text{Blatt}_A(a))$	$\rightarrow E(\wedge E1(9), \forall E(2))$
2,9	11	$P(R)$	$= E(\wedge E2(9), 10)$
2,8	12	$P(R)$	$\exists E(8, 9, 11)$
13	13	$\exists S, T \in \mathcal{T}_{A,n} (R = \text{Knoten}_A(S, T))$	A
14	14	$S \in \mathcal{T}_{A,n} \wedge T \in \mathcal{T}_{A,n} \wedge R = \text{Knoten}_A(S, T)$	A
14	15	$S \in \mathcal{T}_{A,n}$	$\wedge E1(14)$
14	16	$T \in \mathcal{T}_{A,n}$	$\wedge E1(\wedge E2(14))$
14	17	$R = \text{Knoten}_A(S, T)$	$\wedge E2(\wedge E2(14))$
1,14	18	$S \in \mathcal{T}(A)$	<i>Th.</i> 3.5.2.6(<i>Th.</i> 22.3.2.8(1), 15)
1,14	19	$T \in \mathcal{T}(A)$	<i>Th.</i> 3.5.2.6(<i>Th.</i> 22.3.2.8(1), 16)
4,14	20	$P(S) \wedge P(T)$	$\wedge I(\rightarrow E(15, \forall E(4)), \rightarrow E(16, \forall E(4)))$
3,4,14	21	$P(\text{Knoten}_A(S, T))$	$\rightarrow E(20, \rightarrow E(19, \forall E(\rightarrow E(18, \forall E(3)))))$
3,4,14	22	$P(R)$	$= E(17, 21)$
1,3,4,13	23	$P(R)$	$\exists E(13, 14, 22)$

$$\begin{array}{ll}
1,2,3,4,5 & 24 \ P(R) & \forall E(7, 8, 12, 13, 23) \\
1,2,3,4 & 25 \ \forall R \in \mathcal{T}_{A, \text{succ}(n)} \ P(R) & \forall I(\rightarrow I(5, 24))
\end{array}$$

Induktionsschluss:

$$\begin{array}{ll}
1 & 1 \ \forall a \in A \ P(\text{Blatt}_A(a)) & A \\
2 & 2 \ \forall S, T \in \mathcal{T}(A) \ (P(S) \wedge P(T) \rightarrow P(\text{Knoten}_A(S, T))) & A \\
1,2 & 3 \ \forall n \in \mathbb{N} \ \forall R \in \mathcal{T}_{A,n} \ P(R) & Ax. \ 10.3.9(Th. \ 22.3.3.1(i), Th. \ 22.3.3.1(ii)(1, 2)) \\
4 & 4 \ R \in \mathcal{T}(A) & A \\
4 & 5 \ \exists n \in \mathbb{N} \ (R \in \mathcal{T}_{A,n}) & Def. \ 22.3.2.3(4) \\
6 & 6 \ n \in \mathbb{N} \wedge R \in \mathcal{T}_{A,n} & A \\
1,2,6 & 7 \ \forall R \in \mathcal{T}_{A,n} \ P(R) & \\
& & \rightarrow E(\wedge E1(6), \forall E(3)) \\
1,2,6 & 8 \ P(R) & \\
& & \rightarrow E(\wedge E2(6), \forall E(7)) \\
1,2,4 & 9 \ P(R) & \exists E(5, 6, 8) \\
1,2 & 10 \ \forall R \in \mathcal{T}(A) \ P(R) & \forall I(\rightarrow I(4, 9))
\end{array}$$

□

Definition 22.3.3.1 (Baum-Rekursionsoperator).

Mengen: A, X, a, S, T, x, y, G
Funktionen: $f, g, \mathcal{R}_{f,g}$
Neues Symbol:
Baum-Rekursionsoperator: $\triangleright \mathcal{R}_{f,g}$

$$\begin{aligned}
f: A \rightarrow X, \ g: X \times X \rightarrow X, \ G \in \mathcal{P}(\mathcal{T}(A) \times X) \vdash \\
\mathcal{R}_{f,g}(G) := \{(\text{Blatt}_A(a), f(a)) \mid a \in A\} \\
\cup \{(\text{Knoten}_A(S, T), g(x, y)) \mid (S, x) \in G \wedge (T, y) \in G\}
\end{aligned}$$

Theorem 22.3.3.2 (Transformation eines typisierten Baumgraphen).

Mengen: $A, X, D, G, Q, p, a, b, S, T, S', T', x, y, z, w$
Funktionen: $f, g, \mathcal{R}_{f,g}$

$$\begin{aligned}
f: A \rightarrow X, \ g: X \times X \rightarrow X, \ D \in \mathcal{P}(\mathcal{T}(A)), \ G: D \rightarrow X \vdash \\
\mathcal{R}_{f,g}(G): \mathcal{C}_A(D) \rightarrow X, \\
\forall a \in A \ \mathcal{R}_{f,g}(G)(\text{Blatt}_A(a)) = f(a), \\
\forall S, T \in D \ \mathcal{R}_{f,g}(G)(\text{Knoten}_A(S, T)) = g(G(S), G(T))
\end{aligned}$$

Beweis.

Vorbereitung

Th. 22.3.3.2(i)

$$f: A \rightarrow X, g: X \times X \rightarrow X, D \in \mathcal{P}(\mathcal{T}(A)), G: D \rightarrow X \vdash \\ D \in \mathcal{P}(\Sigma_A^+), G \in \mathcal{P}(\mathcal{T}(A) \times X)$$

$$\begin{array}{ll} 1 & 1 \ f: A \rightarrow X \quad A \\ 2 & 2 \ g: X \times X \rightarrow X \quad A \\ 3 & 3 \ D \in \mathcal{P}(\mathcal{T}(A)) \quad A \\ 4 & 4 \ G: D \rightarrow X \quad A \\ 3 & 5 \ D \subseteq \mathcal{T}(A) \quad Th. \ 3.16.2.1(3) \\ & 6 \ \mathcal{T}(A) \subseteq \Sigma_A^+ \quad Def. \ 22.3.2.3 \\ 3 & 7 \ D \subseteq \Sigma_A^+ \quad Th. \ 3.5.3.6(5, 6) \\ 3 & 8 \ D \in \mathcal{P}(\Sigma_A^+) \quad Th. \ 3.16.2.1(7) \\ 3,4 & 9 \ G \subseteq \mathcal{T}(A) \times X \\ & Def. \ 5.2.1.1(4, 5), Th. \ 3.16.5.5, Th. \ 3.5.2.6 \\ 3,4 & 10 \ G \in \mathcal{P}(\mathcal{T}(A) \times X) Th. \ 3.16.2.1(9) \end{array}$$

Trägertypisierung

Th. 22.3.3.2(ii)

$$f: A \rightarrow X, g: X \times X \rightarrow X, D \in \mathcal{P}(\mathcal{T}(A)), G: D \rightarrow X \vdash \\ \mathcal{R}_{f,g}(G) \subseteq \mathcal{C}_A(D) \times X$$

$$\begin{array}{ll} 1 & 1 \ f: A \rightarrow X \quad A \\ 2 & 2 \ g: X \times X \rightarrow X \quad A \\ 3 & 3 \ D \in \mathcal{P}(\mathcal{T}(A)) \quad A \\ 4 & 4 \ G: D \rightarrow X \quad A \\ 5 & 5 \ p \in \mathcal{R}_{f,g}(G) \quad A \\ 1,2,3,4,5 & 6 \ \exists a \in A \ p = (\text{Blatt}_A(a), f(a)) \vee \\ & \exists S, T \in D \ \exists x, y \in X \ ((S, x) \in G \wedge (T, y) \in G \wedge p = (\text{Knoten}_A(S, T), g(x, y))) \\ & Def. \ 22.3.3.1(Th. \ 22.3.3.2(i)(1, 2, 3, 4), 5), Def. \ 5.2.1.1(4) \\ 7 & 7 \ \exists a \in A \ p = (\text{Blatt}_A(a), f(a)) \quad A \\ 8 & 8 \ a \in A \wedge p = (\text{Blatt}_A(a), f(a)) \quad A \\ 1,3,8 & 9 \ p \in \mathcal{C}_A(D) \times X \\ & Th. \ 3.16.5.9(Def. \ 22.3.2.1(Th. \ 22.3.3.2(i)(1, 2, 3, 4), \wedge E1(8)), Th. \ 5.2.3.8(1, \wedge E1(8))), \wedge E2(8) \\ 1,3,7 & 10 \ p \in \mathcal{C}_A(D) \times X \quad \exists E(7, 8, 9) \\ 11 & 11 \ \exists S, T \in D \ \exists x, y \in X \ ((S, x) \in G \wedge (T, y) \in G \wedge p = (\text{Knoten}_A(S, T), g(x, y))) \\ & A \\ 12 & 12 \ S, T \in D \wedge x, y \in X \wedge \\ & (S, x) \in G \wedge (T, y) \in G \wedge p = (\text{Knoten}_A(S, T), g(x, y)) \\ & A \\ 3,12 & 13 \ \text{Knoten}_A(S, T) \in \mathcal{C}_A(D) \\ & Def. \ 22.3.2.1(Th. \ 22.3.3.2(i)(1, 2, 3, 4), \wedge E1(12)) \\ 12 & 14 \ (x, y) \in X \times X \\ & Th. \ 3.16.5.9(\wedge E1(\wedge E2(12)), \wedge E2(\wedge E2(12))) \end{array}$$

2,12	15	$g(x, y) \in X$	<i>Th.</i> 5.2.3.8(2, 14)
2,3,12	16	$(\text{Knoten}_A(S, T), g(x, y)) \in \mathcal{C}_A(D) \times X$	<i>Th.</i> 3.16.5.9(13, 15)
2,3,12	17	$p \in \mathcal{C}_A(D) \times X$	$= E(\wedge E2(\wedge E2(\wedge E2(\wedge E2(12))))), 16)$
2,3,11	18	$p \in \mathcal{C}_A(D) \times X$	$\exists E(11, 12, 17)$
1,2,3,4,5	19	$p \in \mathcal{C}_A(D) \times X$	$\forall E(6, 7, 10, 11, 18)$
1,2,3,4	20	$\mathcal{R}_{f,g}(G) \subseteq \mathcal{C}_A(D) \times X$	<i>Def.</i> 3.5.1.1($\forall I(\rightarrow I(5, 19))$)

Blattfasern

Th. 22.3.3.2(iii)

$f: A \rightarrow X, g: X \times X \rightarrow X, D \in \mathcal{P}(\mathcal{T}(A)), G: D \rightarrow X \vdash$
 $\forall a \in A \forall z ((\text{Blatt}_A(a), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = f(a))$

1	1	$f: A \rightarrow X$	A
2	2	$g: X \times X \rightarrow X$	A
3	3	$D \in \mathcal{P}(\mathcal{T}(A))$	A
4	4	$G: D \rightarrow X$	A
5	5	$a \in A$	A
6	6	$(\text{Blatt}_A(a), z) \in \mathcal{R}_{f,g}(G)$	A
1,2,3,4,5,6	7	$\exists b \in A (\text{Blatt}_A(a), z) = (\text{Blatt}_A(b), f(b)) \vee$ $\exists S, T \in D \exists x, y \in X ((S, x) \in G \wedge (T, y) \in G \wedge$ $(\text{Blatt}_A(a), z) = (\text{Knoten}_A(S, T), g(x, y)))$	<i>Def.</i> 22.3.3.1(<i>Th.</i> 22.3.3.2(i)(1, 2, 3, 4), 6), <i>Def.</i> 5.2.1.1(4)
8	8	$\exists b \in A (\text{Blatt}_A(a), z) = (\text{Blatt}_A(b), f(b))$	A
9	9	$b \in A \wedge (\text{Blatt}_A(a), z) = (\text{Blatt}_A(b), f(b))$	A
5,9	10	$\text{Blatt}_A(a) = \text{Blatt}_A(b) \wedge z = f(b)$	<i>Th.</i> 3.11.4.1($\wedge E2(9)$)
5,9	11	$a = b$	<i>Th.</i> 22.3.1.2(5, $\wedge E1(9)$, $\wedge E1(10)$) $= E(11, \wedge E2(10))$
5,9	12	$z = f(a)$	$\exists E(8, 9, 12)$
5,8	13	$z = f(a)$	$\exists E(8, 9, 12)$
14	14	$\exists S, T \in D \exists x, y \in X ((S, x) \in G \wedge (T, y) \in G \wedge$ $(\text{Blatt}_A(a), z) = (\text{Knoten}_A(S, T), g(x, y)))$	A
15	15	$S, T \in D \wedge x, y \in X \wedge (S, x) \in G \wedge$ $(T, y) \in G \wedge (\text{Blatt}_A(a), z) = (\text{Knoten}_A(S, T), g(x, y))$	A
3	16	$D \subseteq \Sigma_A^+$	<i>Th.</i> 3.16.2.1(<i>Th.</i> 22.3.3.2(i)(1, 2, 3, 4))
3,15	17	$S, T \in \Sigma_A^+$	<i>Th.</i> 3.5.2.6(16, $\wedge E1(15)$)
15	18	$(\text{Blatt}_A(a), z) = (\text{Knoten}_A(S, T), g(x, y))$	$\wedge E2(\wedge E2(\wedge E2(\wedge E2(15))))$
15	19	$\text{Blatt}_A(a) = \text{Knoten}_A(S, T)$	

		$\wedge E1(Th. 3.11.4.1(18))$
3,5,15	20	$Blatt_A(a) \neq Knoten_A(S, T)$
		$Th. 22.3.1.2(5, 17)$
3,5,15	21	$\perp$
		$\perp I(20, 19)$
1,2,3,4,5,15	22	$z = f(a)$
		$Th. 2.1.7.4(21)$
1,2,3,4,5,14	23	$z = f(a)$
		$\exists E(14, 15, 22)$
1,2,3,4,5,6	24	$z = f(a)$
		$\vee E(7, 8, 13, 14, 23)$
1,2,3,4,5	25	$(Blatt_A(a), z) \in \mathcal{R}_{f,g}(G) \rightarrow z = f(a)$
		$\rightarrow I(6, 24)$
26	26	$z = f(a)$
		A
1,2,3,4,5	27	$(Blatt_A(a), f(a)) \in \mathcal{R}_{f,g}(G)$
		$Def. 22.3.3.1(Th. 22.3.3.2(i)(1, 2, 3, 4), 5)$
1,2,3,4,5,26	28	$(Blatt_A(a), z) \in \mathcal{R}_{f,g}(G)$
		$= E(26, 27)$
1,2,3,4,5	29	$z = f(a) \rightarrow (Blatt_A(a), z) \in \mathcal{R}_{f,g}(G)$
		$\rightarrow I(26, 28)$
1,2,3,4,5	30	$(Blatt_A(a), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = f(a)$
		$\leftrightarrow I(25, 29)$
1,2,3,4,5	31	$\forall z ((Blatt_A(a), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = f(a))$
		$\forall I(30)$
1,2,3,4	32	$\forall a \in A \forall z ((Blatt_A(a), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = f(a)) \forall I(\rightarrow I(5, 31))$

Knotenfasern

Th. 22.3.3.2(iv)

$f: A \rightarrow X, g: X \times X \rightarrow X, D \in \mathcal{P}(\mathcal{T}(A)), G: D \rightarrow X \vdash$
 $\forall S, T \in D \forall z ((Knoten_A(S, T), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = g(G(S), G(T)))$

1	1	$f: A \rightarrow X$	A
2	2	$g: X \times X \rightarrow X$	A
3	3	$D \in \mathcal{P}(\mathcal{T}(A))$	A
4	4	$G: D \rightarrow X$	A
5	5	$S \in D$	A
6	6	$T \in D$	A
7	7	$(Knoten_A(S, T), z) \in \mathcal{R}_{f,g}(G)$	A
1,2,3,4,5,6,7	8	$\exists a \in A (Knoten_A(S, T), z) = (Blatt_A(a), f(a)) \vee$ $\exists S', T' \in D \exists x, y \in X ((S', x) \in G \wedge (T', y) \in G \wedge$ $(Knoten_A(S, T), z) = (Knoten_A(S', T'), g(x, y)))$	$Def. 22.3.3.1(Th. 22.3.3.2(i)(1, 2, 3, 4), 7), Def. 5.2.1.1(4)$
9	9	$\exists a \in A (Knoten_A(S, T), z) = (Blatt_A(a), f(a))$	A
10	10	$a \in A \wedge (Knoten_A(S, T), z) = (Blatt_A(a), f(a))$	A
10	11	$Knoten_A(S, T) = Blatt_A(a)$	$\wedge E1(Th. 3.11.4.1(\wedge E2(10)))$
10	12	$Blatt_A(a) = Knoten_A(S, T)$	$Th. 2.9.1.1(11)$
3	13	$D \subseteq \Sigma_A^+$	$Th. 3.16.2.1(Th. 22.3.3.2(i)(1, 2, 3, 4))$
3,5,6	14	$S, T \in \Sigma_A^+$	$Th. 3.5.2.6(13, \wedge I(5, 6))$
3,5,6,10	15	$Blatt_A(a) \neq Knoten_A(S, T)$	$Th. 22.3.1.2(\wedge E1(10), 14)$

3,5,6,10	16	$\perp$	$\perp I(15, 12)$
1,2,3,4,5,6,10	17	$z = g(G(S), G(T))$	<i>Th.</i> 2.1.7.4(16)
1,2,3,4,5,6,9	18	$z = g(G(S), G(T))$	$\exists E(9, 10, 17)$
19	19	$\exists S', T' \in D \exists x, y \in X ((S', x) \in G \wedge (T', y) \in G \wedge$ $(\text{Knoten}_A(S, T), z) = (\text{Knoten}_A(S', T'), g(x, y)))$	A
20	20	$S', T' \in D \wedge x, y \in X \wedge (S', x) \in G \wedge$ $(T', y) \in G \wedge (\text{Knoten}_A(S, T), z) = (\text{Knoten}_A(S', T'), g(x, y))$	A
3,20	21	$S', T' \in \Sigma_A^+$	<i>Th.</i> 3.5.2.6(13, $\wedge E1(20)$)
20	22	$(\text{Knoten}_A(S, T), z) = (\text{Knoten}_A(S', T'), g(x, y))$	$\wedge E2(\wedge E2(\wedge E2(\wedge E2(20))))$
20	23	$\text{Knoten}_A(S, T) = \text{Knoten}_A(S', T') \wedge z = g(x, y)$	<i>Th.</i> 3.11.4.1(22)
3,5,6,20	24	$S = S' \wedge T = T'$	<i>Th.</i> 22.3.1.2(14, 21, $\wedge E1(23)$)
20	25	$(S', x) \in G$	$\wedge E1(\wedge E2(\wedge E2(20)))$
20	26	$(T', y) \in G$	$\wedge E1(\wedge E2(\wedge E2(\wedge E2(20))))$
4,20	27	$G(S') = x$	<i>Th.</i> 5.2.3.10(4, 25)
4,20	28	$G(T') = y$	<i>Th.</i> 5.2.3.10(4, 26)
3,4,5,6,20	29	$z = g(G(S), G(T))$	$= E(\wedge E2(23), 24, 27, 28)$
1,2,3,4,5,6,19	30	$z = g(G(S), G(T))$	$\exists E(19, 20, 29)$
1,2,3,4,5,6,7	31	$z = g(G(S), G(T))$	$\vee E(8, 9, 18, 19, 30)$
1,2,3,4,5,6	32	$(\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G) \rightarrow z = g(G(S), G(T)) \rightarrow I(7, 31)$	
33	33	$z = g(G(S), G(T))$	A
4,5	34	$(S, G(S)) \in G$	<i>Th.</i> 5.2.3.2(4, 5)
4,6	35	$(T, G(T)) \in G$	<i>Th.</i> 5.2.3.2(4, 6)
1,2,3,4,5,6	36	$(\text{Knoten}_A(S, T), g(G(S), G(T))) \in \mathcal{R}_{f,g}(G)$	<i>Def.</i> 22.3.3.1(<i>Th.</i> 22.3.3.2(i)(1, 2, 3, 4), 34, 35)
1,2,3,4,5,6,33	37	$(\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G)$	$= E(33, 36)$
1,2,3,4,5,6	38	$z = g(G(S), G(T)) \rightarrow (\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G) \rightarrow I(33, 37)$	
1,2,3,4,5,6	39	$(\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = g(G(S), G(T)) \leftrightarrow I(32, 38)$	
1,2,3,4,5,6	40	$\forall z ((\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = g(G(S), G(T)))$	$\forall I(39)$
1,2,3,4,5	41	$\forall T \in D \forall z ((\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = g(G(S), G(T)))$	$\forall I(\rightarrow I(6, 40))$
1,2,3,4	42	$\forall S, T \in D \forall z ((\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G) \leftrightarrow z = g(G(S), G(T)))$	$\forall I(\rightarrow I(5, 41))$

Schluss:

$$1 \quad 1 \quad f: A \rightarrow X \quad A$$

2	$2\ g: X \times X \rightarrow X$	A
3	$3\ D \in \mathcal{P}(\mathcal{T}(A))$	A
4	$4\ G: D \rightarrow X$	A
1,2,3,4	$5\ \mathcal{R}_{f,g}(G) \subseteq \mathcal{C}_A(D) \times X$	$Th. 22.3.3.2(ii)(1, 2, 3, 4)$
6	$6\ a \in A$	A
1,6	$7\ f(a) \in X$	$Th. 5.2.3.8(1, 6)$
1,2,3,4,6	$8\ (\text{Blatt}_A(a), f(a)) \in \mathcal{R}_{f,g}(G)$	$Th. 22.3.3.2(iii)(1, 2, 3, 4, 6)$
1,2,3,4,6	$9\ \exists z (z \in X \wedge (\text{Blatt}_A(a), z) \in \mathcal{R}_{f,g}(G))$	$\exists I(\wedge I(7, 8))$
10	$10\ z \in X \wedge (\text{Blatt}_A(a), z) \in \mathcal{R}_{f,g}(G)$	A
11	$11\ w \in X \wedge (\text{Blatt}_A(a), w) \in \mathcal{R}_{f,g}(G)$	A
1,2,3,4,6,10	$12\ z = f(a)$	$Th. 22.3.3.2(iii)(1, 2, 3, 4, 6, \wedge E2(10))$
1,2,3,4,6,11	$13\ w = f(a)$	$Th. 22.3.3.2(iii)(1, 2, 3, 4, 6, \wedge E2(11))$
1,2,3,4,6,10,11	$14\ z = w$	$Th. 2.9.1.3(12, 13)$
1,2,3,4,6	$15\ \exists! z \in X ((\text{Blatt}_A(a), z) \in \mathcal{R}_{f,g}(G))$	$\exists! I(9, 10, 11, 14)$
1,2,3,4	$16\ \forall a \in A \exists! z \in X ((\text{Blatt}_A(a), z) \in \mathcal{R}_{f,g}(G))$	$\forall I(\rightarrow I(6, 15))$
17	$17\ S \in D$	A
18	$18\ T \in D$	A
4,17	$19\ G(S) \in X$	$Th. 5.2.3.8(4, 17)$
4,18	$20\ G(T) \in X$	$Th. 5.2.3.8(4, 18)$
2,4,17,18	$21\ g(G(S), G(T)) \in X$	$Th. 5.2.3.8(2, Th. 3.16.5.9(19, 20))$
1,2,3,4,17,18	$22\ (\text{Knoten}_A(S, T), g(G(S), G(T))) \in \mathcal{R}_{f,g}(G)$	$Th. 22.3.3.2(iv)(1, 2, 3, 4, 17, 18)$
1,2,3,4,17,18	$23\ \exists z (z \in X \wedge (\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G))$	$\exists I(\wedge I(21, 22))$
24	$24\ z \in X \wedge (\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G)$	A
25	$25\ w \in X \wedge (\text{Knoten}_A(S, T), w) \in \mathcal{R}_{f,g}(G)$	A
1,2,3,4,17,18,24	$26\ z = g(G(S), G(T))$	$Th. 22.3.3.2(iv)(1, 2, 3, 4, 17, 18, \wedge E2(24))$
1,2,3,4,17,18,25	$27\ w = g(G(S), G(T))$	$Th. 22.3.3.2(iv)(1, 2, 3, 4, 17, 18, \wedge E2(25))$
1,2,3,4,17,18,24,25	$28\ z = w$	$Th. 2.9.1.3(26, 27)$
1,2,3,4,17,18	$29\ \exists! z \in X ((\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G))$	$\exists! I(23, 24, 25, 28)$
1,2,3,4,17	$30\ \forall T \in D \exists! z \in X ((\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G))$	$\forall I(\rightarrow I(18, 29))$
1,2,3,4	$31\ \forall S, T \in D \exists! z \in X ((\text{Knoten}_A(S, T), z) \in \mathcal{R}_{f,g}(G)) \forall I(\rightarrow I(17, 30))$	
32	$32\ Q \in \mathcal{C}_A(D)$	A
3,32	$33\ \exists a \in A (Q = \text{Blatt}_A(a)) \vee \exists S, T \in D (Q = \text{Knoten}_A(S, T))$	$Def. 22.3.2.1(Th. 22.3.3.2(i)(1, 2, 3, 4), 32)$
34	$34\ \exists a \in A (Q = \text{Blatt}_A(a))$	A
35	$35\ a \in A \wedge Q = \text{Blatt}_A(a)$	A

1,2,3,4,35	36	$\exists! z \in X ((Q, z) \in \mathcal{R}_{f,g}(G))$	$= E(\wedge E2(35), \rightarrow E(\wedge E1(35), \forall E(16)))$
1,2,3,4,34	37	$\exists! z \in X ((Q, z) \in \mathcal{R}_{f,g}(G))$	$\exists E(34, 35, 36)$
	38	$\exists S, T \in D (Q = \text{Knoten}_A(S, T))$	A
	39	$S, T \in D \wedge Q = \text{Knoten}_A(S, T)$	A
1,2,3,4,39	40	$\exists! z \in X ((Q, z) \in \mathcal{R}_{f,g}(G))$	$= E(\wedge E2(39), \rightarrow E(\wedge E2(\wedge E1(39)), \forall E(\rightarrow E(\wedge E1(\wedge E1(39)), \forall E(31))))$
1,2,3,4,38	41	$\exists! z \in X ((Q, z) \in \mathcal{R}_{f,g}(G))$	$\exists E(38, 39, 40)$
1,2,3,4,32	42	$\exists! z \in X ((Q, z) \in \mathcal{R}_{f,g}(G))$	$\forall E(33, 34, 37, 38, 41)$
1,2,3,4	43	$\forall Q \in \mathcal{C}_A(D) \exists! z \in X ((Q, z) \in \mathcal{R}_{f,g}(G))$	$\forall I(\rightarrow I(32, 42))$
1,2,3,4	44	$\mathcal{R}_{f,g}(G): \mathcal{C}_A(D) \rightarrow X$	$Th. 5.2.6.1(5, 43)$
1,2,3,4	45	$\forall a \in A \mathcal{R}_{f,g}(G)(\text{Blatt}_A(a)) = f(a)$	$Th. 22.3.3.2(iii)(1, 2, 3, 4), Th. 5.2.3.10(44)$
1,2,3,4	46	$\forall S, T \in D \mathcal{R}_{f,g}(G)(\text{Knoten}_A(S, T)) = g(G(S), G(T))$	$Th. 22.3.3.2(iv)(1, 2, 3, 4), Th. 5.2.3.10(44)$

□

Theorem 22.3.3.3 (Strukturrekursion auf Klammerungsbäumen).

Mengen: A, X, B, M, G
Mengen: $a, S, T, Q, U, Y, p, x, y, m, n, k$
Funktionen: $f, g, D, J, H, E, E_1, E_2$
Funktionsoperatoren: $\mathcal{R}_{f,g}$

$$\begin{aligned}
 & f: A \rightarrow X, \quad g: X \times X \rightarrow X \vdash \\
 & \exists! E: \mathcal{T}(A) \rightarrow X \left(\forall a \in A E(\text{Blatt}_A(a)) = f(a) \wedge \right. \\
 & \quad \left. \forall S, T \in \mathcal{T}(A) E(\text{Knoten}_A(S, T)) = g(E(S), E(T)) \right)
 \end{aligned}$$

Beweis.

Abkürzungen:

1	$B := \mathcal{T}(A)$	Abkürzung
2	$M := \mathcal{P}(B \times X)$	Abkürzung
3	$D := \iota J: \mathbb{N} \rightarrow \mathcal{P}(B) \forall n \in \mathbb{N} J(n) = \mathcal{T}_{A,n}$	Abkürzung
4	$H := \text{Rec}_{\emptyset, \mathcal{R}_{f,g}}$	Abkürzung
5	$E := \bigcup H[\mathbb{N}]$	Abkürzung

Typisierung des Stufenoperators

Th. 22.3.3.3(i)

$$f: A \rightarrow X, \quad g: X \times X \rightarrow X \vdash \mathcal{R}_{f,g}: M \rightarrow M, \quad \emptyset \in M$$

1	1	$f: A \rightarrow X$	A
2	2	$g: X \times X \rightarrow X$	A
3	3	$G \in M$	A
3	4	$G \subseteq B \times X$	$Th. 3.16.2.1(3)$
5	5	$p \in \mathcal{R}_{f,g}(G)$	A
1,2,3,5	6	$\exists a \in Ap = (\text{Blatt}_A(a), f(a)) \vee$ $\exists S, T \in B \exists x, y \in X ((S, x) \in G \wedge (T, y) \in G \wedge p = (\text{Knoten}_A(S, T), g(x, y)))$	$Def. 22.3.3.1(1, 2, 3, 5), Th. 3.5.2.6(4), Th. 3.16.5.5$
7	7	$\exists a \in Ap = (\text{Blatt}_A(a), f(a))$	A
8	8	$a \in A \wedge p = (\text{Blatt}_A(a), f(a))$	A
1,8	9	$p \in B \times X$	$Th. 3.16.5.9(Th. 22.3.2.10(\wedge E1(8)), Th. 5.2.3.8(1, \wedge E1(8))), \wedge E2(8)$
1,7	10	$p \in B \times X$	$\exists E(7, 8, 9)$
11	11	$\exists S, T \in B \exists x, y \in X ((S, x) \in G \wedge (T, y) \in G \wedge p = (\text{Knoten}_A(S, T), g(x, y)))$	A
12	12	$S, T \in B \wedge x, y \in X \wedge$ $(S, x) \in G \wedge (T, y) \in G \wedge p = (\text{Knoten}_A(S, T), g(x, y))$	A
12	13	$\text{Knoten}_A(S, T) \in B$	$Th. 22.3.2.10(\wedge E1(12))$
12	14	$(x, y) \in X \times X$	$Th. 3.16.5.9(\wedge E1(\wedge E2(12)), \wedge E2(\wedge E2(12)))$
2,12	15	$g(x, y) \in X$	$Th. 5.2.3.8(2, 14)$
2,12	16	$(\text{Knoten}_A(S, T), g(x, y)) \in B \times X$	$Th. 3.16.5.9(13, 15)$
2,12	17	$p \in B \times X$	$= E(\wedge E2(\wedge E2(\wedge E2(\wedge E2(12))))), 16)$
2,11	18	$p \in B \times X$	$\exists E(11, 12, 17)$
1,2,3,5	19	$p \in B \times X$	$\vee E(6, 7, 10, 11, 18)$
1,2,3	20	$\mathcal{R}_{f,g}(G) \subseteq B \times X$	$Def. 3.5.1.1(\forall I(\rightarrow I(5, 19)))$
1,2,3	21	$\mathcal{R}_{f,g}(G) \in M$	$Th. 3.16.2.1(20)$
1,2	22	$\forall G \in M \mathcal{R}_{f,g}(G) \in M$	$\forall I(\rightarrow I(3, 21))$
1,2	23	$\mathcal{R}_{f,g}: M \rightarrow M$	$Th. 5.2.6.9(22)$
	24	$\emptyset \in M$	$Th. 3.16.2.1(Th. 3.6.3.2)$

Rekursionsgleichungen der Stufenfolge

Th. 22.3.3.3(ii)

$$f: A \rightarrow X, g: X \times X \rightarrow X \vdash$$

$$H: \mathbb{N} \rightarrow M, H(0) = \emptyset, \forall n \in \mathbb{N} H(\text{succ}(n)) = \mathcal{R}_{f,g}(H(n))$$

1	1	$f: A \rightarrow X$	A
2	2	$g: X \times X \rightarrow X$	A
1,2	3	$\mathcal{R}_{f,g}: M \rightarrow M$	

Th. 22.3.3.3(i)(1, 2)

$$4 \ \emptyset \in M$$

Th. 3.16.2.1(Th. 3.6.3.2)

$$1,2 \ 5 \ H: \mathbb{N} \rightarrow M \quad \text{Th. 10.4.3.21(4, 3)}$$

$$1,2 \ 6 \ H(0) = \emptyset \quad \text{Th. 10.4.3.23(4, 3)}$$

$$7 \ 7 \ n \in \mathbb{N} \quad A$$

$$1,2,7 \ 8 \ H(\text{succ}(n)) = \mathcal{R}_{f,g}(H(n))$$

Th. 10.4.3.24(4, 3, 7)

$$1,2 \ 9 \ \forall n \in \mathbb{N} \ H(\text{succ}(n)) = \mathcal{R}_{f,g}(H(n)) \forall I(\rightarrow I(7, 8))$$

Stufentypisierung: Anfang

Th. 22.3.3.3(iii)

$$f: A \rightarrow X, \ g: X \times X \rightarrow X \vdash H(0): \mathcal{T}_{A,0} \rightarrow X$$

$$1 \ 1 \ f: A \rightarrow X \quad A$$

$$2 \ 2 \ g: X \times X \rightarrow X \quad A$$

$$1,2 \ 3 \ H(0) = \emptyset$$

Th. 22.3.3.3(ii)(1, 2)

$$4 \ \mathcal{T}_{A,0} = \emptyset \quad \text{Th. 22.3.2.3}$$

$$5 \ \emptyset: \emptyset \rightarrow X \quad \text{Th. 5.3.4.7}$$

$$1,2 \ 6 \ H(0): \mathcal{T}_{A,0} \rightarrow X = E(3, 4, 5)$$

Stufentypisierung: Schritt

Th. 22.3.3.3(iv)

$$f: A \rightarrow X, \ g: X \times X \rightarrow X, \ n \in \mathbb{N}, \\ H(n): \mathcal{T}_{A,n} \rightarrow X \vdash H(\text{succ}(n)): \mathcal{T}_{A,\text{succ}(n)} \rightarrow X$$

$$1 \ 1 \ f: A \rightarrow X \quad A$$

$$2 \ 2 \ g: X \times X \rightarrow X \quad A$$

$$3 \ 3 \ n \in \mathbb{N} \quad A$$

$$4 \ 4 \ H(n): \mathcal{T}_{A,n} \rightarrow X \quad A$$

$$1,2,3 \ 5 \ H(\text{succ}(n)) = \mathcal{R}_{f,g}(H(n))$$

$\rightarrow E(3, \forall E(\text{Th. 22.3.3.3(ii)(1, 2)))$

$$3 \ 6 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}(A) \quad \text{Th. 22.3.2.8(3)}$$

$$3 \ 7 \ \mathcal{T}_{A,n} \in \mathcal{P}(\mathcal{T}(A)) \quad \text{Th. 3.16.2.1(6)}$$

$$1,2,3,4 \ 8 \ \mathcal{R}_{f,g}(H(n)): \mathcal{C}_A(\mathcal{T}_{A,n}) \rightarrow X$$

Th. 22.3.3.2(1, 2, 7, 4)

$$3 \ 9 \ \mathcal{T}_{A,\text{succ}(n)} = \mathcal{C}_A(\mathcal{T}_{A,n}) \quad \text{Th. 22.3.2.3(3)}$$

$$1,2,3,4 \ 10 \ H(\text{succ}(n)): \mathcal{T}_{A,\text{succ}(n)} \rightarrow X = E(5, 8, 9)$$

Stufentypisierung

Th. 22.3.3.3(v)

$$f: A \rightarrow X, \ g: X \times X \rightarrow X \vdash \forall n \in \mathbb{N} \ H(n): \mathcal{T}_{A,n} \rightarrow X$$

1 1 $f: A \rightarrow X$ A
 2 2 $g: X \times X \rightarrow X$ A
 1,2 3 $\forall n \in \mathbb{N} H(n): \mathcal{T}_{A,n} \rightarrow X$

Ax. 10.3.9(*Th.* 22.3.3.3(*iii*)(1,2), *Th.* 22.3.3.3(*iv*)(1,2))

Kohärenz: Anfang

Th. 22.3.3.3(vi)

$f: A \rightarrow X, g: X \times X \rightarrow X \vdash H(\text{succ}(0)) \upharpoonright_{\mathcal{T}_{A,0}} = H(0)$

1 1 $f: A \rightarrow X$ A
 2 2 $g: X \times X \rightarrow X$ A
 3 0 $\in \mathbb{N}$ *Ax.* 10.3.1
 4 $\text{succ}(0) \in \mathbb{N}$ *Ax.* 10.3.4(3)
 1,2 5 $H(\text{succ}(0)): \mathcal{T}_{A,\text{succ}(0)} \rightarrow X$
 $\rightarrow E(4, \forall E(\text{Th. } 22.3.3.3(v)(1,2)))$
 1,2 6 $H(0): \mathcal{T}_{A,0} \rightarrow X$
 $\rightarrow E(3, \forall E(\text{Th. } 22.3.3.3(v)(1,2)))$
 7 $\mathcal{T}_{A,0} \subseteq \mathcal{T}_{A,\text{succ}(0)}$ *Th.* 22.3.2.5(3)
 8 $\mathcal{T}_{A,0} = \emptyset$ *Th.* 22.3.2.3
 9 9 $Q \in \mathcal{T}_{A,0}$ A
 9 10 $Q \in \emptyset$ $= E(8, 9)$
 11 $Q \notin \emptyset$ *Th.* 3.6.2.2
 9 12 $\perp$ $\perp I(11, 10)$
 1,2,9 13 $H(\text{succ}(0))(Q) = H(0)(Q)$ *Th.* 2.1.7.4(12)
 1,2 14 $\forall Q \in \mathcal{T}_{A,0} H(\text{succ}(0))(Q) = H(0)(Q) \forall I(\rightarrow I(9, 13))$
 1,2 15 $H(\text{succ}(0)) \upharpoonright_{\mathcal{T}_{A,0}} = H(0)$
 $\rightarrow E(14, \leftrightarrow E(2(\text{Th. } 5.3.2.15(5, 6, 7))))$

Kohärenz: Schritt

Th. 22.3.3.3(vii)

$f: A \rightarrow X, g: X \times X \rightarrow X, n \in \mathbb{N},$
 $H(\text{succ}(n)) \upharpoonright_{\mathcal{T}_{A,n}} = H(n) \vdash$
 $H(\text{succ}(\text{succ}(n))) \upharpoonright_{\mathcal{T}_{A,\text{succ}(n)}} = H(\text{succ}(n))$

1 1 $f: A \rightarrow X$ A
 2 2 $g: X \times X \rightarrow X$ A
 3 3 $n \in \mathbb{N}$ A
 4 4 $H(\text{succ}(n)) \upharpoonright_{\mathcal{T}_{A,n}} = H(n)$ A
 3 5 $\text{succ}(n) \in \mathbb{N}$ *Ax.* 10.3.4(3)
 3 6 $\mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,\text{succ}(n)}$ *Th.* 22.3.2.8(3)
 3 7 $\mathcal{T}_{A,n} \in \mathcal{P}(B)$ *Th.* 3.16.2.1(6)
 1,2,3 8 $H(n): \mathcal{T}_{A,n} \rightarrow X$
 $\rightarrow E(3, \forall E(\text{Th. } 22.3.3.3(v)(1,2)))$

3	9	$\mathcal{T}_{A,\text{succ}(n)} \subseteq B$	<i>Th.</i> 22.3.2.8(5)
3	10	$\mathcal{T}_{A,\text{succ}(n)} \in \mathcal{P}(B)$	<i>Th.</i> 3.16.2.1(9)
1,2,3	11	$H(\text{succ}(n)) : \mathcal{T}_{A,\text{succ}(n)} \rightarrow X$	$\rightarrow E(5, \forall E(\text{Th. 22.3.3.3}(v)(1, 2)))$
3	12	$\mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,\text{succ}(n)}$	<i>Th.</i> 22.3.2.5(3)
1,2,3,4	13	$\forall U \in \mathcal{T}_{A,n} H(\text{succ}(n))(U) = H(n)(U)$	$\rightarrow E(4, \leftrightarrow E1(\text{Th. 5.3.2.15}(11, 8, 12)))$
14	14	$Q \in \mathcal{T}_{A,\text{succ}(n)}$	<i>A</i>
3,14	15	$\exists a \in A (Q = \text{Blatt}_A(a)) \vee$ $\exists S, T \in \mathcal{T}_{A,n} (Q = \text{Knoten}_A(S, T))$	<i>Def.</i> 22.3.2.1(<i>Th.</i> 22.3.2.3(3, 14))
16	16	$\exists a \in A (Q = \text{Blatt}_A(a))$	<i>A</i>
17	17	$a \in A \wedge Q = \text{Blatt}_A(a)$	<i>A</i>
1,2,3,17	18	$H(\text{succ}(\text{succ}(n)))(Q) = f(a)$ $= E(\wedge E2(17), \rightarrow E(5, \forall E(\text{Th. 22.3.3.3}(ii)(1, 2))), \text{Th. 22.3.3.2}(1, 2, 10, 11, \wedge E1(17)))$	
1,2,3,17	19	$H(\text{succ}(n))(Q) = f(a)$ $= E(\wedge E2(17), \rightarrow E(3, \forall E(\text{Th. 22.3.3.3}(ii)(1, 2))), \text{Th. 22.3.3.2}(1, 2, 7, 8, \wedge E1(17)))$	
1,2,3,17	20	$H(\text{succ}(\text{succ}(n)))(Q) = H(\text{succ}(n))(Q)$	<i>Th.</i> 2.9.1.3(18, 19)
1,2,3,16	21	$H(\text{succ}(\text{succ}(n)))(Q) = H(\text{succ}(n))(Q) \exists E(16, 17, 20)$	
22	22	$\exists S, T \in \mathcal{T}_{A,n} (Q = \text{Knoten}_A(S, T))$	<i>A</i>
23	23	$S, T \in \mathcal{T}_{A,n} \wedge Q = \text{Knoten}_A(S, T)$	<i>A</i>
3,23	24	$S, T \in \mathcal{T}_{A,\text{succ}(n)}$	<i>Th.</i> 3.5.2.6(12, $\wedge E1(23)$)
1,2,3,23	25	$H(\text{succ}(\text{succ}(n)))(Q) = g(H(\text{succ}(n))(S), H(\text{succ}(n))(T))$ $= E(\wedge E2(23), \rightarrow E(5, \forall E(\text{Th. 22.3.3.3}(ii)(1, 2))), \text{Th. 22.3.3.2}(1, 2, 10, 11, 24))$	
1,2,3,23	26	$H(\text{succ}(n))(Q) = g(H(n)(S), H(n)(T))$ $= E(\wedge E2(23), \rightarrow E(3, \forall E(\text{Th. 22.3.3.3}(ii)(1, 2))), \text{Th. 22.3.3.2}(1, 2, 7, 8, \wedge E1(23)))$	
1,2,3,4,23	27	$g(H(\text{succ}(n))(S), H(\text{succ}(n))(T)) = g(H(n)(S), H(n)(T))$ $= E(\rightarrow E(\wedge E1(\wedge E1(23)), \forall E(13)), \rightarrow E(\wedge E2(\wedge E1(23)), \forall E(13)))$	
1,2,3,4,23	28	$H(\text{succ}(\text{succ}(n)))(Q) = H(\text{succ}(n))(Q) = E(25, 27, 26)$	
1,2,3,4,22	29	$H(\text{succ}(\text{succ}(n)))(Q) = H(\text{succ}(n))(Q) \exists E(22, 23, 28)$	
1,2,3,4,14	30	$H(\text{succ}(\text{succ}(n)))(Q) = H(\text{succ}(n))(Q)$	$\vee E(15, 16, 21, 22, 29)$
1,2,3,4	31	$\forall Q \in \mathcal{T}_{A,\text{succ}(n)} H(\text{succ}(\text{succ}(n)))(Q) = H(\text{succ}(n))(Q)$	$\forall I(\rightarrow I(14, 30))$
3	32	$\text{succ}(\text{succ}(n)) \in \mathbb{N}$	<i>Ax.</i> 10.3.4(5)
1,2,3	33	$H(\text{succ}(\text{succ}(n))) : \mathcal{T}_{A,\text{succ}(\text{succ}(n))} \rightarrow X$	$\rightarrow E(32, \forall E(\text{Th. 22.3.3.3}(v)(1, 2)))$
3	34	$\mathcal{T}_{A,\text{succ}(n)} \subseteq \mathcal{T}_{A,\text{succ}(\text{succ}(n))}$	<i>Th.</i> 22.3.2.5(5)
1,2,3,4	35	$H(\text{succ}(\text{succ}(n))) \upharpoonright_{\mathcal{T}_{A,\text{succ}(n)}} = H(\text{succ}(n))$	$\rightarrow E(31, \leftrightarrow E2(\text{Th. 5.3.2.15}(33, 11, 34)))$

Nachfolgerkohärenz

Th. 22.3.3.3(viii)

$$f: A \rightarrow X, g: X \times X \rightarrow X \vdash \\ \forall n \in \mathbb{N} H(\text{succ}(n)) \upharpoonright_{\mathcal{T}_{A,n}} = H(n)$$

$$\begin{array}{ll} 1 & 1 \ f: A \rightarrow X \quad A \\ 2 & 2 \ g: X \times X \rightarrow X \ A \\ 1,2 & 3 \ \forall n \in \mathbb{N} H(\text{succ}(n)) \upharpoonright_{\mathcal{T}_{A,n}} = H(n) \end{array} \quad \text{Ax. 10.3.9(Th. 22.3.3.3(vi)(1,2), Th. 22.3.3.3(vii)(1,2))}$$

Iterierte Kohärenz: Anfang

Th. 22.3.3.3(ix)

$$f: A \rightarrow X, g: X \times X \rightarrow X, m \in \mathbb{N} \vdash \\ H(m+0) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$$

$$\begin{array}{ll} 1 & 1 \ f: A \rightarrow X \quad A \\ 2 & 2 \ g: X \times X \rightarrow X \quad A \\ 3 & 3 \ m \in \mathbb{N} \quad A \\ 3 & 4 \ m+0 = m \quad \text{Th. 10.4.4.2(3)} \\ 1,2,3 & 5 \ H(m+0): \mathcal{T}_{A,m+0} \rightarrow X \\ & = E(4, \rightarrow E(3, \forall E(\text{Th. 22.3.3.3(v)(1,2)}))) \\ 1,2,3 & 6 \ H(m): \mathcal{T}_{A,m} \rightarrow X \\ & \rightarrow E(3, \forall E(\text{Th. 22.3.3.3(v)(1,2)})) \\ 3 & 7 \ \mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,m+0} \quad = E(4, \text{Th. 3.5.3.1}) \\ 8 & 8 \ Q \in \mathcal{T}_{A,m} \quad A \\ 3,8 & 9 \ H(m+0)(Q) = H(m)(Q) \quad = E(4) \\ 1,2,3 & 10 \ \forall Q \in \mathcal{T}_{A,m} H(m+0)(Q) = H(m)(Q) \forall I(\rightarrow I(8,9)) \\ 1,2,3 & 11 \ H(m+0) \upharpoonright_{\mathcal{T}_{A,m}} = H(m) \\ & \rightarrow E(10, \leftrightarrow E(2(\text{Th. 5.3.2.15(5,6,7)}))) \end{array}$$

Iterierte Kohärenz: Schritt

Th. 22.3.3.3(x)

$$f: A \rightarrow X, g: X \times X \rightarrow X, m, k \in \mathbb{N}, \\ H(m+k) \upharpoonright_{\mathcal{T}_{A,m}} = H(m) \vdash \\ H(m+\text{succ}(k)) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$$

$$\begin{array}{ll} 1 & 1 \ f: A \rightarrow X \quad A \\ 2 & 2 \ g: X \times X \rightarrow X \quad A \\ 3 & 3 \ m, k \in \mathbb{N} \quad A \\ 4 & 4 \ H(m+k) \upharpoonright_{\mathcal{T}_{A,m}} = H(m) \quad A \\ 3 & 5 \ m+k \in \mathbb{N} \quad \text{Th. 10.4.4.1(3)} \\ 1,2,3 & 6 \ H(m+k): \mathcal{T}_{A,m+k} \rightarrow X \\ & \rightarrow E(5, \forall E(\text{Th. 22.3.3.3(v)(1,2)})) \\ 3 & 7 \ \text{succ}(m+k) \in \mathbb{N} \quad \text{Ax. 10.3.4(5)} \end{array}$$

1,2,3	8	$H(\text{succ}(m+k)) : \mathcal{T}_{A,\text{succ}(m+k)} \rightarrow X$	$\rightarrow E(7, \forall E(\text{Th. 22.3.3.3}(v)(1,2)))$
3	9	$m + \text{succ}(k) \in \mathbb{N}$	$\text{Th. 10.4.4.1}(\wedge E1(3), \text{Ax. 10.3.4}(\wedge E2(3)))$
1,2,3	10	$H(m + \text{succ}(k)) : \mathcal{T}_{A,m+\text{succ}(k)} \rightarrow X$	$\rightarrow E(9, \forall E(\text{Th. 22.3.3.3}(v)(1,2)))$
3	11	$\mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,m+k}$	$\text{Th. 22.3.2.6}(3)$
3	12	$m + \text{succ}(k) = \text{succ}(m+k)$	$\text{Th. 10.4.4.3}(3)$
1,2,3	13	$H(\text{succ}(m+k)) \upharpoonright_{\mathcal{T}_{A,m+k}} = H(m+k)$	$\rightarrow E(5, \forall E(\text{Th. 22.3.3.3}(viii)(1,2)))$
3	14	$\mathcal{T}_{A,m+k} \subseteq \mathcal{T}_{A,\text{succ}(m+k)}$	$\text{Th. 22.3.2.5}(5)$
1,2,3	15	$\forall Q \in \mathcal{T}_{A,m+k} H(\text{succ}(m+k))(Q) = H(m+k)(Q)$	$\rightarrow E(13, \leftrightarrow E1(\text{Th. 5.3.2.15}(8,6,14)))$
1,2,3	16	$H(m) : \mathcal{T}_{A,m} \rightarrow X$	$\rightarrow E(\wedge E1(3), \forall E(\text{Th. 22.3.3.3}(v)(1,2)))$
1,2,3,4	17	$\forall Q \in \mathcal{T}_{A,m} H(m+k)(Q) = H(m)(Q)$	$\rightarrow E(4, \leftrightarrow E1(\text{Th. 5.3.2.15}(6,16,11)))$
3	18	$\mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,m+\text{succ}(k)}$	$\text{Th. 22.3.2.6}(3)$
19	19	$Q \in \mathcal{T}_{A,m}$	A
3,19	20	$Q \in \mathcal{T}_{A,m+k}$	$\text{Th. 3.5.2.6}(11,19)$
1,2,3,19	21	$H(\text{succ}(m+k))(Q) = H(m+k)(Q)$	$\rightarrow E(20, \forall E(15))$
1,2,3,4,19	22	$H(m+k)(Q) = H(m)(Q)$	$\rightarrow E(19, \forall E(17))$
1,2,3,4,19	23	$H(m + \text{succ}(k))(Q) = H(m)(Q)$	$= E(12, 21, 22)$
1,2,3,4	24	$\forall Q \in \mathcal{T}_{A,m} H(m + \text{succ}(k))(Q) = H(m)(Q) \forall I(\rightarrow I(19, 23))$	
1,2,3,4	25	$H(m + \text{succ}(k)) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$	$\rightarrow E(24, \leftrightarrow E2(\text{Th. 5.3.2.15}(10,16,18)))$

Iterierte Kohärenz

Th. 22.3.3.3(xi)

$$f : A \rightarrow X, \ g : X \times X \rightarrow X, \ m, k \in \mathbb{N} \vdash \\ H(m+k) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$$

1	1	$f : A \rightarrow X$	A
2	2	$g : X \times X \rightarrow X$	A
3	3	$m, k \in \mathbb{N}$	A
1,2,3	4	$H(m+k) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$	$\text{Ax. 10.3.9}(\text{Th. 22.3.3.3}(ix)(1,2,3), \text{Th. 22.3.3.3}(x)(1,2,3))$

Fernkohärenz

Th. 22.3.3.3(xii)

$$f : A \rightarrow X, \ g : X \times X \rightarrow X, \ m, n \in \mathbb{N}, \ m \leq n \vdash \\ H(n) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$$

1	1	$f: A \rightarrow X$	A	
2	2	$g: X \times X \rightarrow X$	A	
3	3	$m, n \in \mathbb{N}$	A	
4	4	$m \leq n$	A	
3,4	5	$\exists k \in \mathbb{N} (m + k = n)$	$Th. 10.4.5.6(3, 4)$	
6	6	$k \in \mathbb{N} \wedge m + k = n$	A	
1,2,3,6	7	$H(m + k) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$		$Th. 22.3.3.3(xi)(1, 2, 3, \wedge E1(6))$
1,2,3,6	8	$H(n) \upharpoonright_{\mathcal{T}_{A,m}} = H(m) = E(\wedge E2(6), 7)$		
1,2,3,4	9	$H(n) \upharpoonright_{\mathcal{T}_{A,m}} = H(m) \exists E(5, 6, 8)$		

Familie der Stufenträger

Th. 22.3.3.3(xiii)

$$D: \mathbb{N} \rightarrow \mathcal{P}(B), \forall n \in \mathbb{N} D(n) = \mathcal{T}_{A,n}$$

1	1	$n \in \mathbb{N}$	A	
1	2	$\mathcal{T}_{A,n} \subseteq B$	$Th. 22.3.2.8(1)$	
1	3	$\mathcal{T}_{A,n} \in \mathcal{P}(B)$	$Th. 3.16.2.1(2)$	
4	$\forall n \in \mathbb{N} \mathcal{T}_{A,n} \in \mathcal{P}(B)$	$\forall I(\rightarrow I(1, 3))$		
5	$\exists! J: \mathbb{N} \rightarrow \mathcal{P}(B) \forall n \in \mathbb{N} J(n) = \mathcal{T}_{A,n}$	$Th. 5.2.6.9(4)$		
6	$D: \mathbb{N} \rightarrow \mathcal{P}(B) \wedge \forall n \in \mathbb{N} D(n) = \mathcal{T}_{A,n}$	$Th. 2.10.9.7(5)$		
7	$D: \mathbb{N} \rightarrow \mathcal{P}(B)$	$\wedge E1(6)$		
8	$\forall n \in \mathbb{N} D(n) = \mathcal{T}_{A,n}$	$\wedge E2(6)$		

Vereinigung der Stufenträger

Th. 22.3.3.3(xiv)

$$\bigcup D[\mathbb{N}] = B$$

1	1	$Q \in \bigcup D[\mathbb{N}]$	A	
1	2	$\exists Y \in D[\mathbb{N}] (Q \in Y)$	$Th. 3.13.2.1(1)$	
3	3	$Y \in D[\mathbb{N}] \wedge Q \in Y$	A	
3	4	$\exists n \in \mathbb{N} (Y = D(n))$		$Th. 5.2.4.6(Th. 22.3.3.3(xiii), \wedge E1(3))$
5	5	$n \in \mathbb{N} \wedge Y = D(n)$	A	
3,5	6	$Q \in D(n)$		$= E(\wedge E2(5), \wedge E2(3))$
3,5	7	$Q \in \mathcal{T}_{A,n}$		$= E(\rightarrow E(\wedge E1(5), \forall E(Th. 22.3.3.3(xiii))), 6)$
3,5	8	$Q \in \Sigma_A^+$		$Th. 3.5.2.6(Th. 22.3.2.4(\wedge E1(5)), 7)$
3,5	9	$Q \in B$		$Def. 22.3.2.3(8, \exists I(\wedge I(\wedge E1(5), 7)))$
3	10	$Q \in B$	$\exists E(4, 5, 9)$	

1	11	$Q \in B$	$\exists E(2, 3, 10)$
	12	$Q \in \bigcup D[\mathbb{N}] \rightarrow Q \in B$	$\rightarrow I(1, 11)$
13	13	$Q \in B$	A
13	14	$\exists n \in \mathbb{N} (Q \in \mathcal{T}_{A,n})$	<i>Def.</i> 22.3.2.3(13)
15	15	$n \in \mathbb{N} \wedge Q \in \mathcal{T}_{A,n}$	A
15	16	$Q \in D(n)$	$= E(\rightarrow E(\wedge E1(15), \forall E(Th. 22.3.3.3(xiii))), \wedge E2(15))$
15	17	$D(n) \in D[\mathbb{N}]$	$Th. 5.2.4.6(Th. 22.3.3.3(xiii), \exists I(\wedge I(\wedge E1(15), = I)))$
15	18	$Q \in \bigcup D[\mathbb{N}]$	<i>Th.</i> 3.13.2.3(17, 16)
13	19	$Q \in \bigcup D[\mathbb{N}]$	$\exists E(14, 15, 18)$
	20	$Q \in B \rightarrow Q \in \bigcup D[\mathbb{N}]$	$\rightarrow I(13, 19)$
	21	$Q \in \bigcup D[\mathbb{N}] \leftrightarrow Q \in B$	$\leftrightarrow I(12, 20)$
	22	$\forall Q (Q \in \bigcup D[\mathbb{N}] \leftrightarrow Q \in B) \forall I(21)$	
	23	$\bigcup D[\mathbb{N}] = B$	<i>Ax.</i> 3.4.1.1(22)

Paarweise Stufenkohärenz

Th. 22.3.3.3(xv)

$$f: A \rightarrow X, g: X \times X \rightarrow X \vdash \\ \forall m, n \in \mathbb{N} \forall Q \in D(m) \cap D(n) H(m)(Q) = H(n)(Q)$$

1	1	$f: A \rightarrow X$	A
2	2	$g: X \times X \rightarrow X$	A
3	3	$m \in \mathbb{N}$	A
4	4	$n \in \mathbb{N}$	A
5	5	$Q \in D(m) \cap D(n)$	A
3,4,5	6	$Q \in \mathcal{T}_{A,m} \wedge Q \in \mathcal{T}_{A,n}$	<i>Th.</i> 3.10.2.3(5), <i>Th.</i> 22.3.3.3(xiii)(3, 4)
1,2,3	7	$H(m): \mathcal{T}_{A,m} \rightarrow X$	$\rightarrow E(3, \forall E(Th. 22.3.3.3(v)(1, 2)))$
1,2,4	8	$H(n): \mathcal{T}_{A,n} \rightarrow X$	$\rightarrow E(4, \forall E(Th. 22.3.3.3(v)(1, 2)))$
3,4	9	$m \leq n \vee n \leq m$	<i>Th.</i> 10.4.5.22(3, 4)
10	10	$m \leq n$	A
1,2,3,4,10	11	$H(n) \upharpoonright_{\mathcal{T}_{A,m}} = H(m)$	<i>Th.</i> 22.3.3.3(xii)(1, 2, 3, 4, 10)
3,4,10	12	$\mathcal{T}_{A,m} \subseteq \mathcal{T}_{A,n}$	<i>Th.</i> 22.3.2.7(3, 4, 10)
1,2,3,4,10	13	$\forall U \in \mathcal{T}_{A,m} H(n)(U) = H(m)(U)$	$\rightarrow E(11, \leftrightarrow E1(Th. 5.3.2.15(8, 7, 12)))$
1,2,3,4,5,10	14	$H(m)(Q) = H(n)(Q)$	<i>Th.</i> 2.9.1.1($\rightarrow E(\wedge E1(6), \forall E(13))$)
15	15	$n \leq m$	A

$$\begin{array}{ll}
1,2,3,4,15 & 16 \ H(m) \vdash_{\mathcal{T}_{A,n}} H(n) \\
& Th. \ 22.3.3.3(ii)(1, 2, 4, 3, 15) \\
3,4,15 & 17 \ \mathcal{T}_{A,n} \subseteq \mathcal{T}_{A,m} \\
& Th. \ 22.3.2.7(4, 3, 15) \\
1,2,3,4,15 & 18 \ \forall U \in \mathcal{T}_{A,n} \ H(m)(U) = H(n)(U) \\
& \rightarrow E(16, \leftrightarrow E1(Th. \ 5.3.2.15(7, 8, 17))) \\
1,2,3,4,5,15 & 19 \ H(m)(Q) = H(n)(Q) \\
& \rightarrow E(\wedge E2(6), \forall E(18)) \\
1,2,3,4,5 & 20 \ H(m)(Q) = H(n)(Q) \\
& \vee E(9, 10, 14, 15, 19) \\
1,2 & 21 \ \forall m, n \in \mathbb{N} \ \forall Q \in D(m) \cap D(n) \ H(m)(Q) = H(n)(Q) \\
& \forall I(\rightarrow I(3, \forall I(\rightarrow I(4, \forall I(\rightarrow I(5, 20)))))
\end{array}$$

Vereinigung der Stufenfunktionen Th. 22.3.3.3(xvi)

$$\begin{array}{l}
f: A \rightarrow X, \ g: X \times X \rightarrow X \vdash E: B \rightarrow X, \\
\forall n \in \mathbb{N} \ \forall Q \in \mathcal{T}_{A,n} \ E(Q) = H(n)(Q)
\end{array}$$

$$\begin{array}{ll}
1 & 1 \ f: A \rightarrow X \quad A \\
2 & 2 \ g: X \times X \rightarrow X \quad A \\
1,2 & 3 \ H: \mathbb{N} \rightarrow M \\
& Th. \ 22.3.3.3(ii)(1, 2) \\
4 & 4 \ D: \mathbb{N} \rightarrow \mathcal{P}(B) \quad Th. \ 22.3.3.3(iii) \\
5 & 5 \ n \in \mathbb{N} \quad A \\
1,2,5 & 6 \ H(n): D(n) \rightarrow X \\
& = E(\rightarrow E(5, \forall E(Th. \ 22.3.3.3(iii))), \rightarrow E(5, \forall E(Th. \ 22.3.3.3(v)(1, 2)))) \\
1,2 & 7 \ \forall n \in \mathbb{N} \ H(n): D(n) \rightarrow X \ \forall I(\rightarrow I(5, 6)) \\
1,2 & 8 \ \forall m, n \in \mathbb{N} \ \forall Q \in D(m) \cap D(n) \ H(m)(Q) = H(n)(Q) \\
& Th. \ 22.3.3.3(xv)(1, 2) \\
1,2 & 9 \ E: \bigcup D[\mathbb{N}] \rightarrow X \ \wedge \\
& \forall n \in \mathbb{N} \ \forall Q \in D(n) \ E(Q) = H(n)(Q) \\
& Th. \ 5.3.2.22(3, 4, 7, 8) \\
10 & 10 \ \bigcup D[\mathbb{N}] = B \quad Th. \ 22.3.3.3(xiv) \\
1,2 & 11 \ E: B \rightarrow X \quad = E(10, \wedge E1(9)) \\
1,2 & 12 \ \forall n \in \mathbb{N} \ \forall Q \in \mathcal{T}_{A,n} \ E(Q) = H(n)(Q) \\
& = E(Th. \ 22.3.3.3(iii), \wedge E2(9))
\end{array}$$

Rekursionsgleichungen der Vereinigung Th. 22.3.3.3(xvii)

$$\begin{array}{l}
f: A \rightarrow X, \ g: X \times X \rightarrow X \vdash \\
\forall a \in A \ E(\text{Blatt}_A(a)) = f(a), \\
\forall S, T \in B \ E(\text{Knoten}_A(S, T)) = g(E(S), E(T))
\end{array}$$

1	1	$f: A \rightarrow X$	A	
2	2	$g: X \times X \rightarrow X$	A	
3	3	$a \in A$	A	
	4	$0 \in \mathbb{N}$	$Ax. 10.3.1$	
	5	$\mathcal{T}_{A,0} \subseteq B$	$Th. 22.3.2.8(4)$	
	6	$\mathcal{T}_{A,0} \in \mathcal{P}(B)$	$Th. 3.16.2.1(5)$	
1,2	7	$H(0): \mathcal{T}_{A,0} \rightarrow X$		$\rightarrow E(4, \forall E(Th. 22.3.3.3(v)(1, 2)))$
3	8	$Blatt_A(a) \in \mathcal{C}_A(\mathcal{T}_{A,0})$		$Def. 22.3.2.1(Th. 22.3.2.4(4), 3)$
3	9	$Blatt_A(a) \in \mathcal{T}_{A, \text{succ}(0)}$	$Th. 22.3.2.3(4, 8)$	
	10	$\text{succ}(0) \in \mathbb{N}$	$Ax. 10.3.4(4)$	
1,2,3	11	$E(Blatt_A(a)) = H(\text{succ}(0))(Blatt_A(a))$		$Th. 22.3.3.3(xvi)(1, 2, 10, 9)$
1,2,3	12	$H(\text{succ}(0))(Blatt_A(a)) = \mathcal{R}_{f,g}(H(0))(Blatt_A(a))$		$= E(\rightarrow E(4, \forall E(Th. 22.3.3.3(ii)(1, 2))))$
1,2,3	13	$\mathcal{R}_{f,g}(H(0))(Blatt_A(a)) = f(a)$		$Th. 22.3.3.2(1, 2, 6, 7, 3)$
1,2,3	14	$E(Blatt_A(a)) = f(a)$	$= E(11, 12, 13)$	
1,2	15	$\forall a \in A E(Blatt_A(a)) = f(a)$	$\forall I(\rightarrow I(3, 14))$	
16	16	$S \in B$	A	
17	17	$T \in B$	A	
16,17	18	$\exists n \in \mathbb{N} (S, T \in \mathcal{T}_{A,n})$		$Th. 22.3.2.9(\wedge I(16, 17))$
19	19	$n \in \mathbb{N} \wedge S, T \in \mathcal{T}_{A,n}$	A	
19	20	$n \in \mathbb{N}$	$\wedge E1(19)$	
19	21	$S, T \in \mathcal{T}_{A,n}$	$\wedge E2(19)$	
19	22	$\mathcal{T}_{A,n} \subseteq B$	$Th. 22.3.2.8(20)$	
19	23	$\mathcal{T}_{A,n} \in \mathcal{P}(B)$	$Th. 3.16.2.1(22)$	
1,2,19	24	$H(n): \mathcal{T}_{A,n} \rightarrow X$		$\rightarrow E(20, \forall E(Th. 22.3.3.3(v)(1, 2)))$
1,2,19	25	$\mathcal{R}_{f,g}(H(n))(\text{Knoten}_A(S, T)) = g(H(n)(S), H(n)(T))$		$Th. 22.3.3.2(1, 2, 23, 24, 21)$
1,2,19	26	$H(\text{succ}(n))(\text{Knoten}_A(S, T)) = \mathcal{R}_{f,g}(H(n))(\text{Knoten}_A(S, T))$		$= E(\rightarrow E(20, \forall E(Th. 22.3.3.3(ii)(1, 2))))$
19	27	$\text{Knoten}_A(S, T) \in \mathcal{C}_A(\mathcal{T}_{A,n})$		$Def. 22.3.2.1(Th. 22.3.2.4(20), 21)$
19	28	$\text{Knoten}_A(S, T) \in \mathcal{T}_{A, \text{succ}(n)}$	$Th. 22.3.2.3(20, 27)$	
19	29	$\text{succ}(n) \in \mathbb{N}$	$Ax. 10.3.4(20)$	
1,2,19	30	$E(\text{Knoten}_A(S, T)) = H(\text{succ}(n))(\text{Knoten}_A(S, T))$		$Th. 22.3.3.3(xvi)(1, 2, 29, 28)$
1,2,19	31	$E(S) = H(n)(S) \wedge E(T) = H(n)(T)$		$Th. 22.3.3.3(xvi)(1, 2, 20, 21)$
1,2,16,17,19	32	$E(\text{Knoten}_A(S, T)) = g(E(S), E(T)) = E(30, 26, 25, 31)$		

		<i>Th. 22.3.3.3(xvi)(1, 2)</i>
1,2 4	$\forall a \in A \ E(\text{Blatt}_A(a)) = f(a)$	
		<i>Th. 22.3.3.3(xvii)(1, 2)</i>
1,2 5	$\forall S, T \in B \ E(\text{Knoten}_A(S, T)) = g(E(S), E(T))$	
		<i>Th. 22.3.3.3(xvii)(1, 2)</i>
1,2 6	$\exists ! E: B \rightarrow X \left(\forall a \in A \ E(\text{Blatt}_A(a)) = f(a) \wedge \right.$ $\left. \forall S, T \in B \ E(\text{Knoten}_A(S, T)) = g(E(S), E(T)) \right)$	
		$\exists ! I(3, 4, 5, \text{Th. 22.3.3.3(xviii)})$

□

Kapitel 4

Blattwörter, Klammerungen und Auswertung

4.1 Blattwort

Definition 22.4.1.1 (Einbuchstabenabbildung).

Mengen: A, a
Funktionen: ein_A
Neues Symbol:
Einbuchstabenabbildung: $\triangleright \text{ein}_A$

$$\begin{aligned} \text{ein}_A: A &\rightarrow A^+, \\ \forall a \in A \ (\text{ein}_A(a) &:= \langle a \rangle) \end{aligned}$$

$$\begin{array}{ll} 1 \ 1 \ a \in A & A \\ 1 \ 2 \ \langle a \rangle \in A^+ & Th. \ 22.2.2.4(1) \\ 3 \ \forall a \in A \ (\langle a \rangle \in A^+) & \forall I(\rightarrow I(1, 2)) \\ 4 \ \exists! F \left(F: A \rightarrow A^+ \wedge \forall a \in A \ F(a) = \langle a \rangle \right) & Th. \ 5.2.6.9(3) \end{array}$$

Theorem 22.4.1.1 (Die Konkatenation ist ein binärer Operator auf nichtleeren Wörtern).

Mengen: A, u, v
Funktionen: $\frown$

$$\text{BinOp}(A^+, \frown)$$

$$\begin{array}{ll} 1 \ 1 \ u, v \in A^+ & A \\ 1 \ 2 \ u \frown v \in A^+ & Th. \ 22.2.3.7(1) \end{array}$$

Theorem 22.4.1.2 (Existenz und Eindeutigkeit der Blattwortabbildung).

Mengen: A, a, S, T
Funktionen: $F, \text{ein}_A, \curvearrowright$

$$\exists! F: \mathcal{T}(A) \rightarrow A^+ \left(\begin{array}{l} \forall a \in A \ F(\text{Blatt}_A(a)) = \text{ein}_A(a) \ \wedge \\ \forall S, T \in \mathcal{T}(A) \ F(\text{Knoten}_A(S, T)) = \widehat{\curvearrowright}_{A^+}(F(S), F(T)) \end{array} \right)$$

1 $\text{ein}_A: A \rightarrow A^+$ Def. 22.4.1.1

2 BinOp($A^+, \curvearrowright$) Th. 22.4.1.1

3 $\widehat{\curvearrowright}_{A^+}: A^+ \times A^+ \rightarrow A^+$ Def. 22.2.5.1(2)

4 $\exists! F: \mathcal{T}(A) \rightarrow A^+ \left(\begin{array}{l} \forall a \in A \ F(\text{Blatt}_A(a)) = \text{ein}_A(a) \ \wedge \\ \forall S, T \in \mathcal{T}(A) \ F(\text{Knoten}_A(S, T)) = \widehat{\curvearrowright}_{A^+}(F(S), F(T)) \end{array} \right)$

Th. 22.3.3.3(1, 3)

Definition 22.4.1.2 (Blattwort eines Klammerungsbaums).

Mengen: A, a, S, T
Funktionen: wort_A
Neues Symbol:
Blattwort: $\triangleright \text{wort}_A$

$$\text{wort}_A := \iota F: \mathcal{T}(A) \rightarrow A^+ \left(\begin{array}{l} \forall a \in A \ F(\text{Blatt}_A(a)) = \text{ein}_A(a) \ \wedge \\ \forall S, T \in \mathcal{T}(A) \ F(\text{Knoten}_A(S, T)) = \widehat{\curvearrowright}_{A^+}(F(S), F(T)) \end{array} \right)$$

Theorem 22.4.1.3 (Typisierung und Rekursionsgleichungen des Blattwortes).

Mengen: A, a, S, T
Funktionen: wort_A

$$\begin{array}{l} \text{wort}_A: \mathcal{T}(A) \rightarrow A^+, \\ a \in A \vdash \text{wort}_A(\text{Blatt}_A(a)) = \langle a \rangle, \\ S, T \in \mathcal{T}(A) \vdash \text{wort}_A(\text{Knoten}_A(S, T)) = \text{wort}_A(S) \curvearrowright \text{wort}_A(T) \end{array}$$

1	$\text{wort}_A: \mathcal{T}(A) \rightarrow A^+ \wedge$	
	$\left(\forall a \in A \text{ wort}_A(\text{Blatt}_A(a)) = \text{ein}_A(a) \wedge \right.$	
	$\left. \forall S, T \in \mathcal{T}(A) \text{ wort}_A(\text{Knoten}_A(S, T)) = \widehat{}_{A^+}(\text{wort}_A(S), \text{wort}_A(T)) \right)$	<i>Th. 2.10.9.7(Th. 22.4.1.2), Def. 22.4.1.2</i>
2	$\text{wort}_A: \mathcal{T}(A) \rightarrow A^+$	$\wedge E1(1)$
3	$\forall a \in A \text{ wort}_A(\text{Blatt}_A(a)) = \text{ein}_A(a)$	$\wedge E1(\wedge E2(1))$
4	$\forall S, T \in \mathcal{T}(A) \text{ wort}_A(\text{Knoten}_A(S, T)) = \widehat{}_{A^+}(\text{wort}_A(S), \text{wort}_A(T))$	$\wedge E2(\wedge E2(1))$
5	$5 \ a \in A$	A
5	$6 \ \text{wort}_A(\text{Blatt}_A(a)) = \text{ein}_A(a)$	$\rightarrow E(\forall E(3), 5)$
5	$7 \ \text{ein}_A(a) = \langle a \rangle$	<i>Def. 22.4.1.1(5)</i>
5	$8 \ \text{wort}_A(\text{Blatt}_A(a)) = \langle a \rangle$	<i>Th. 2.9.1.2(6, 7)</i>
9	$9 \ S \in \mathcal{T}(A)$	A
10	$10 \ T \in \mathcal{T}(A)$	A
9	$11 \ \text{wort}_A(S) \in A^+$	<i>Th. 5.2.3.8(2, 9)</i>
10	$12 \ \text{wort}_A(T) \in A^+$	<i>Th. 5.2.3.8(2, 10)</i>
9,10	$13 \ \text{wort}_A(\text{Knoten}_A(S, T)) = \widehat{}_{A^+}(\text{wort}_A(S), \text{wort}_A(T))$	$\rightarrow E(10, \rightarrow E(9, \forall E(\forall E(4))))$
9,10	$14 \ \widehat{}_{A^+}(\text{wort}_A(S), \text{wort}_A(T)) = \text{wort}_A(S) \frown \text{wort}_A(T)$	<i>Def. 22.2.5.1(Th. 22.4.1.1, 11, 12)</i>
9,10	$15 \ \text{wort}_A(\text{Knoten}_A(S, T)) = \text{wort}_A(S) \frown \text{wort}_A(T)$	<i>Th. 2.9.1.2(13, 14)</i>

4.2 Klammerungen eines Wortes

Definition 22.4.2.1 (Klammerungsmenge eines Nichtleerwortes).

Mengen: A, w, T
Neues Symbol:
Klammerungsmenge: $\triangleright \text{Kl}_A(w)$

$$w \in A^+ \vdash \text{Kl}_A(w) := \{T \in \mathcal{T}(A) \mid \text{wort}_A(T) = w\}$$

Theorem 22.4.2.1 (Elementkriterium der Klammerungsmenge).

Mengen: A, w, T

$$w \in A^+ \vdash T \in \text{Kl}_A(w) \dashv\vdash T \in \mathcal{T}(A), \text{ wort}_A(T) = w$$

- | | | |
|---|---|----------------------------|
| 1 | $1 \ w \in A^+$ | A |
| 1 | $2 \ \text{Kl}_A(w) = \{S \in \mathcal{T}(A) \mid \text{wort}_A(S) = w\}$ | <i>Def. 22.4.2.1(1)</i> |
| 1 | $3 \ T \in \text{Kl}_A(w) \dashv\vdash T \in \mathcal{T}(A), \text{ wort}_A(T) = w$ | $E(2, \text{Th. 3.7.2.5})$ |

Theorem 22.4.2.2 (Existenz einer Klammerung).

Mengen: A, w, a, u, T

$$w \in A^+ \vdash \text{Kl}_A(w) \neq \emptyset$$

1	1	$a \in A$	A
1	2	$\langle a \rangle \in A^+$	<i>Th.</i> 22.2.2.4(1)
1	3	$\text{Blatt}_A(a) \in \mathcal{T}(A)$	<i>Th.</i> 22.3.2.10(1)
1	4	$\text{wort}_A(\text{Blatt}_A(a)) = \langle a \rangle$	<i>Th.</i> 22.4.1.3(1)
1	5	$\text{Blatt}_A(a) \in \mathcal{T}(A) \wedge \text{wort}_A(\text{Blatt}_A(a)) = \langle a \rangle$	$\wedge I(3, 4)$
1	6	$\text{Blatt}_A(a) \in \text{Kl}_A(\langle a \rangle)$	<i>Th.</i> 22.4.2.1(2, 5)
1	7	$\text{Kl}_A(\langle a \rangle) \neq \emptyset$	<i>Th.</i> 3.6.3.5(6)
	8	$\forall a \in A (\text{Kl}_A(\langle a \rangle) \neq \emptyset)$	$\forall I(\rightarrow I(1, 7))$
9	9	$u \in A^+$	A
10	10	$a \in A$	A
11	11	$\text{Kl}_A(u) \neq \emptyset$	A
11	12	$\exists T (T \in \text{Kl}_A(u))$	<i>Th.</i> 3.6.3.7(11)
13	13	$T \in \text{Kl}_A(u)$	A
9,13	14	$T \in \mathcal{T}(A) \wedge \text{wort}_A(T) = u$	<i>Th.</i> 22.4.2.1(9, 13)
9,13	15	$T \in \mathcal{T}(A)$	$\wedge E1(14)$
9,13	16	$\text{wort}_A(T) = u$	$\wedge E2(14)$
	10	17 $\text{Blatt}_A(a) \in \mathcal{T}(A)$	<i>Th.</i> 22.3.2.10(10)
9,10,13	18	$\text{Knoten}_A(T, \text{Blatt}_A(a)) \in \mathcal{T}(A)$	<i>Th.</i> 22.3.2.10(15, 17)
	10	19 $\text{wort}_A(\text{Blatt}_A(a)) = \langle a \rangle$	<i>Th.</i> 22.4.1.3(10)
9,10,13	20	$\text{wort}_A(\text{Knoten}_A(T, \text{Blatt}_A(a))) = \text{wort}_A(T) \frown \text{wort}_A(\text{Blatt}_A(a))$	<i>Th.</i> 22.4.1.3(15, 17)
9,10,13	21	$\text{wort}_A(\text{Knoten}_A(T, \text{Blatt}_A(a))) = u \frown \text{wort}_A(\text{Blatt}_A(a))$	$= E(16, 20)$
9,10,13	22	$\text{wort}_A(\text{Knoten}_A(T, \text{Blatt}_A(a))) = u \frown \langle a \rangle$	$= E(19, 21)$
	10	23 $\langle a \rangle \in A^+$	<i>Th.</i> 22.2.2.4(10)
9,10	24	$u \frown \langle a \rangle \in A^+$	<i>Th.</i> 22.2.3.7(9, 23)
9,10,13	25	$\text{Knoten}_A(T, \text{Blatt}_A(a)) \in \mathcal{T}(A) \wedge \text{wort}_A(\text{Knoten}_A(T, \text{Blatt}_A(a))) = u \frown \langle a \rangle$	$\wedge I(18, 22)$
9,10,13	26	$\text{Knoten}_A(T, \text{Blatt}_A(a)) \in \text{Kl}_A(u \frown \langle a \rangle)$	<i>Th.</i> 22.4.2.1(24, 25)
9,10,13	27	$\text{Kl}_A(u \frown \langle a \rangle) \neq \emptyset$	<i>Th.</i> 3.6.3.5(26)
9,10,11	28	$\text{Kl}_A(u \frown \langle a \rangle) \neq \emptyset$	$\exists E(12, 13, 27)$
9,10	29	$\text{Kl}_A(u) \neq \emptyset \rightarrow \text{Kl}_A(u \frown \langle a \rangle) \neq \emptyset$	$\rightarrow I(11, 28)$
	9	30 $\forall a \in A (\text{Kl}_A(u) \neq \emptyset \rightarrow \text{Kl}_A(u \frown \langle a \rangle) \neq \emptyset)$	$\forall I(\rightarrow I(10, 29))$
	31	$\forall u \in A^+ \forall a \in A (\text{Kl}_A(u) \neq \emptyset \rightarrow \text{Kl}_A(u \frown \langle a \rangle) \neq \emptyset)$	$\forall I(\rightarrow I(9, 30))$
32	32	$w \in A^+$	A

4.3 Auswertung eines Klammerungsbaums

Theorem 22.4.3.1 (Existenz und Eindeutigkeit der Baumauswertung).

Mengen: A, a, S, T
Funktionen: $\star, E, \text{id}_A$

$$\begin{aligned} \text{BinOp}(A, \star) \vdash \exists! E: \mathcal{T}(A) \rightarrow A \Big(\\ \forall a \in A \ E(\text{Blatt}_A(a)) = \text{id}_A(a) \ \wedge \\ \forall S, T \in \mathcal{T}(A) \ E(\text{Knoten}_A(S, T)) = \widehat{\star}_A(E(S), E(T)) \Big) \end{aligned}$$

$$\begin{aligned} 1 \quad & 1 \ \text{BinOp}(A, \star) \quad A \\ & 2 \ \text{id}_A: A \rightarrow A \quad \text{Th. 8.3.1.1} \\ 1 \quad & 3 \ \widehat{\star}_A: A \times A \rightarrow A \text{Def. 22.2.5.1(1)} \\ 1 \quad & 4 \ \exists! E: \mathcal{T}(A) \rightarrow A \Big(\\ & \quad \forall a \in A \ E(\text{Blatt}_A(a)) = \text{id}_A(a) \ \wedge \\ & \quad \forall S, T \in \mathcal{T}(A) \ E(\text{Knoten}_A(S, T)) = \widehat{\star}_A(E(S), E(T)) \Big) \end{aligned}$$

Th. 22.3.3.3(2, 3)

Definition 22.4.3.1 (Auswertung eines Klammerungsbaums).

Mengen: A, a, S, T
Funktionen: $\star, \text{ev}_\star$
Neues Symbol:
Baumauswertung: $\triangleright \text{ev}_\star$

$$\begin{aligned} \text{BinOp}(A, \star) \vdash \text{ev}_\star := \iota E: \mathcal{T}(A) \rightarrow A \Big(\\ \forall a \in A \ E(\text{Blatt}_A(a)) = \text{id}_A(a) \ \wedge \\ \forall S, T \in \mathcal{T}(A) \ E(\text{Knoten}_A(S, T)) = \widehat{\star}_A(E(S), E(T)) \Big) \end{aligned}$$

Theorem 22.4.3.2 (Typisierung und Rekursionsgleichungen der Baumauswertung).

Mengen: A, a, S, T
Funktionen: $\star, \text{ev}_\star$

$$\begin{aligned} \text{BinOp}(A, \star) \vdash \text{ev}_\star: \mathcal{T}(A) \rightarrow A, \\ \text{BinOp}(A, \star), \ a \in A \vdash \text{ev}_\star(\text{Blatt}_A(a)) = a, \\ \text{BinOp}(A, \star), \ S, T \in \mathcal{T}(A) \vdash \text{ev}_\star(\text{Knoten}_A(S, T)) = \text{ev}_\star(S) \star \text{ev}_\star(T) \end{aligned}$$

1	1	$\text{BinOp}(A, \star)$	A	
1	2	$\text{ev}_\star: \mathcal{T}(A) \rightarrow A \wedge$ $\left(\forall a \in A \text{ ev}_\star(\text{Blatt}_A(a)) = \text{id}_A(a) \wedge \right.$ $\left. \forall S, T \in \mathcal{T}(A) \text{ ev}_\star(\text{Knoten}_A(S, T)) = \widehat{\star}_A(\text{ev}_\star(S), \text{ev}_\star(T)) \right)$		$Th. 2.10.9.7(Th. 22.4.3.1(1)), Def. 22.4.3.1(1)$
	3	$\text{ev}_\star: \mathcal{T}(A) \rightarrow A$	$\wedge E1(2)$	
	4	$\forall a \in A \text{ ev}_\star(\text{Blatt}_A(a)) = \text{id}_A(a)$	$\wedge E1(\wedge E2(2))$	
	5	$\forall S, T \in \mathcal{T}(A) \text{ ev}_\star(\text{Knoten}_A(S, T)) = \widehat{\star}_A(\text{ev}_\star(S), \text{ev}_\star(T))$		$\wedge E2(\wedge E2(2))$
6	6	$a \in A$	A	
1,6	7	$\text{ev}_\star(\text{Blatt}_A(a)) = \text{id}_A(a)$	$\rightarrow E(\forall E(4), 6)$	
6	8	$\text{id}_A(a) = a$	$Th. 8.3.1.3(6)$	
1,6	9	$\text{ev}_\star(\text{Blatt}_A(a)) = a$	$Th. 2.9.1.2(7, 8)$	
10	10	$S \in \mathcal{T}(A)$	A	
11	11	$T \in \mathcal{T}(A)$	A	
1,10	12	$\text{ev}_\star(S) \in A$	$Th. 5.2.3.8(3, 10)$	
1,11	13	$\text{ev}_\star(T) \in A$	$Th. 5.2.3.8(3, 11)$	
1,10,11	14	$\text{ev}_\star(\text{Knoten}_A(S, T)) = \widehat{\star}_A(\text{ev}_\star(S), \text{ev}_\star(T))$		$\rightarrow E(11, \rightarrow E(10, \forall E(\forall E(5))))$
1,10,11	15	$\widehat{\star}_A(\text{ev}_\star(S), \text{ev}_\star(T)) = \text{ev}_\star(S) \star \text{ev}_\star(T)$		$Def. 22.2.5.1(1, 12, 13)$
1,10,11	16	$\text{ev}_\star(\text{Knoten}_A(S, T)) = \text{ev}_\star(S) \star \text{ev}_\star(T)$	$Th. 2.9.1.2(14, 15)$	